

SECTION 2 – ROOM LAYOUTS

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2-1 INTRODUCTION

When laying out a floor plan there are special considerations that must be taken into account due to the magnetic field effect on certain medical implants (including cardiac pacemakers, neurostimulators, and biostimulation devices) and the environmental effect (motors, steel, etc.) on the field homogeneity. The maximum magnetic field in which the equipment can be located is listed in Table 2-1. Selected magnetic shielding of some devices and equipment is possible but must be handled on an individual basis. Refer to Section 6, INTERCONNECTION DATA, for cable length considerations.

The RF shielded room (Magnet Room) is unique in that the room must be shielded from outside radio frequency interference. This is done by enclosing the room with metal walls, floors and ceiling. These shielding requirements impose special considerations which are addressed in Section 7, RF SHIELDED ROOM.

The Magnet Room can be magnetically shielded to reduce the magnet fringe field or to shield the magnet from the effects of the external environment. Refer to Section 3-4, MAGNETIC SHIELDING, for magnet shielding considerations.

TABLE 2-1
PROXIMITY LIMITS

GAUSS (mT) LIMIT See Notes 1 & 2	EQUIPMENT	
0.5 GAUSS (0.05mT) OR LESS See Note 4	• Nuclear cameras	
1 GAUSS (0.1mT) OR LESS See Note 4	• Positron Emission Tomography scanner • Linear Accelerator • Cyclotrons • Accurate Measuring scale • Image intensifiers • Color TV	• Video display (color, B/W, monochrome) • CT scanner • Ultrasound • Lithotripter • Electron microscope • Advantage Workstation with CRT Monitor
3 GAUSS (0.3mT) OR LESS See Note 4	• Power transformers • Main electrical distribution transformers • Moving steel equipment such as: – Vehicular traffic – Fork lift trucks – Dumb waiters – Electric transport carts	– Loading dock (truck traffic) – Elevators – Escalators – Helicopters (See Note 3)
5 GAUSS (0.5mT) OR LESS	• Cardiac pacemakers • Neurostimulators	• Biostimulation devices
10 GAUSS (1mT) OR LESS	• Magnetic tapes and floppy discs • Hard copy imagers • Line printers • Video Cassette Recorder (VCR) • Film processor • Credit cards, watches, and clocks • Telephone switching station • Water cooling equipment • HVAC equipment	• Major mechanical equipment room • Large steel equipment such as: – Emergency generators – Commercial laundry equipment – Food preparation area – Air conditioning chiller – Fuel storage tanks – Motors greater than 5 horsepower • X-ray tubes
30 GAUSS (3mT) OR LESS	• System Cabinet • Twin Accessory Cabinet	• TSCC equipment • TGWC equipment
50 GAUSS (5mT) OR LESS	• Octane Computer • Operator Workspace Cabinet • LCD Color Monitor for OW (See note 5) • 1.5T SRFD II Cabinet • ACGD/PDU Cabinet • Main Disconnect Device	• Magnet Monitor • Telephones • Metal detector for screening • LCD Color Monitor for Advantage Workstation • BrainWave Audio Console • BrainWave Cabinet
100 GAUSS (10mT) OR LESS	• S/C Power Supply • Service Tool Magnet Power Supply Cabinet • Service Tool Shim Power Supply Cabinet	• Shield/Cryo Cooler Compressor Cabinet • Oxygen Monitor • Pneumatic Patient Alert Control Box
200 GAUSS (20mT) OR LESS	• Penetration Panel • Blower Box	• Magnet Rundown Unit • Remote Oxygen Sensor Module
800 GAUSS (80 mT) OR LESS	• BrainWave Cart	
Note 1 Refer to SECTION 3, MAGNETIC FIELD CONSIDERATIONS, for magnet field plots. 2 Recommended limits given above are based on general MR site planning guidelines. Actual susceptibility of specific devices may vary significantly depending on electrical design orientation of the device relative to the magnetic field and the degree of interference considered unacceptable. 3 Verify operating limits with provider of helicopter service 4 The 0.5 gauss (0.05 mT), 1 gauss (0.1 mT), and 3 gauss (0.3 mT) or less proximity limits apply to ALL magnets. For additional Active Shield magnet proximity limits refer to Section 4-13 LCC (ACTIVE SHIELD) MAGNET CHANGING MAGNETIC ENVIRONMENT SPECIFICATIONS. 5 If gauss limit is more than indicated for OW with LCD Color Monitor then contact GE to determine impacts on OW equipment.		

2-2 ROOM SIZES

Table 2-2 contains minimum room dimensions necessary for an MR suite. Table 2-2 also contains issues which are created by reduction in service access, operator access, and equipment space.

TABLE 2-2
SIGNA TWINSPEED SYSTEM CONFIGURATION MINIMUM ROOM INSIDE CLEAR SPACE DIMENSIONS

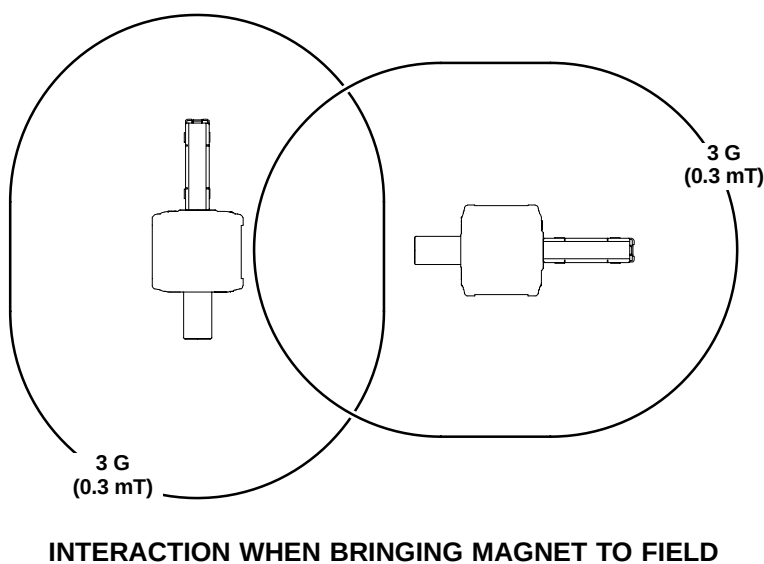
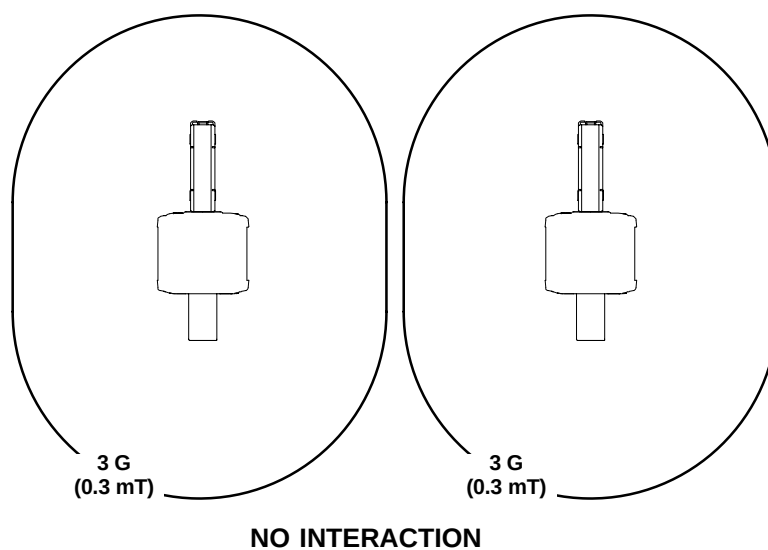
SYSTEM CONFIGURATION	EQUIPMENT ROOM MINIMUM VALUES See Note 6		MAGNET ROOM MINIMUM VALUES See Note 4			CONTROL ROOM MINIMUM VALUES See Note 6		TOTAL SYSTEM AREA ft ² (m ²)
	W x D ft-in. (m)	AREA ft ² (m ²)	W x D ft-in. (m)	AREA ft ² (m ²)	CEILING HEIGHT ft-in. (m)	W x D ft-in. (m)	AREA ft ² (m ²)	
System with Outdoor TSCC	8-9 x 13-0 (2.67 x 3.96)	113.75 (10.57)	10-11.27 X 19-7.1 (3.334 x 5.972) See Note 1, 2, 3, 5	214.3 (19.91)	8-9 (2.67) See Note 8	5* x 7 (1.52* x 2.13)	35 (3.24)	363.05 (33.72)
System with Indoor Water Cooled TSCC	8-9 x 18-0 (2.67 x 5.49)	157.50 (14.65)						406.80 (37.81)
System with Indoor Air Cooled TSCC	8-9 x 18-6 (2.67 x 5.64)	161.88 (15.06)						411.18 (38.21)
System with Outdoor TGWC	8-9 x 13-0 (2.67 x 3.96) See Note 7	113.75 (10.57) See Note 7						363.05 (33.72) See Note 7
System with Indoor Water Cooled TGWC	8-3 x 14-8 (2.51 x 4.47) See Note 7	121.00 (11.22) See Note 7						370.30 (34.37) See Note 7
System with Indoor Air Cooled TGWC	8-9 x 15-4 (2.67 x 4.67) See Note 7	134.17 (12.47) See Note 7						383.47 (35.62) See Note 7

- Note**
- 1 Must locate magnet in the side-to-side center of magnetic room shield.
 - 2 Minimum dimensions dependent on the magnetic field containment requested and dimensions of magnetic shield design.
 - 3 Room dimensions in front of LCC Magnet MUST allow for Gradient Coil Assembly and split bridge installation/servicing. LCC Magnets MUST USE special Gradient Coil Replacement Tool Kit for replacing the Gradient Coil Assembly. The Gradient Coil Replacement Tool Kit is shipped in a wooden crate on casters. Utilization of Gradient Coil Replacement Tool requires 107.75 in. (2737 mm) clear space in front of magnet. Note split bridge servicing requires 77.5 in. (1969 mm) clear space in front of the magnet.
 - 4 Absolute Minimum Magnet Room dimensions will result in limited operator clearances and increased Magnet Service time. Illustration 2-8 shows only 1.5 ft (0.46 m) clearance at end of Patient Transport.
 - 5 Room dimensions do not contain 5 gauss line to room.
 - 6 Minimum Equipment Room and Control Room dimensions do not permit placement of air conditioning units in room. Nor do they permit space for any optional equipment such as Advantage Workstation option, Laser Camera, etc.
 - 7 When a TGWC is selected then the customer site must still provide water cooling for the LCC Magnet Shield/Cryo Cooler Compressor Cabinet. Minimum Equipment Room for system with TGWC does not permit placement of equipment which will provide water cooling for Shield/Cryo Cooler Compressor.
 - 8 Magnet Room minimum ceiling height is 8 ft 2.5 in. (2.50 meter) when Low Ceiling Height Siting Option (M1060SR) is installed.
- * Width is dependent on Magnet Room door location and customer's approval of limited space available for operator.

2-3 MULTIPLE MR SYSTEMS SITE

2-3-1 Two Magnet Site Layout

For two magnet installations interaction can occur between the magnetic fields. For 2 magnets not to interact at all (including when bringing magnet to field) the 3 gauss lines of each magnet must not intersect. If the 3 gauss lines intersect but remain outside each magnet's cryostat there will be interaction between the magnets when bringing to field. The orientation of the magnets is irrelevant. Consult the MR Siting & Shielding group for closer proximity of magnets.



TWO MAGNET INSTALLATION
ILLUSTRATION 2-1

2-3-2 Equipment Room Shared By Multiple MR Systems

When the Equipment Room is shared by two or more MR systems of the same field strength there is a potential for cross-talk of RF energy between the MR systems. RF cross-talk may cause noise artifacts in images. Proper planning and installation of the multiple systems in the shared Equipment Room can reduce the potential for cross-talk.

- The RF Screen of the Magnet Room for each system needs to meet the RF Attenuation specifications in **Section 7-1 RF SHIELDED ROOM SPECIFICATION**.
- The Equipment Room design, layout, and installation must meet the requirements of **APPENDIX D – REQUIREMENTS FOR EQUIPMENT ROOM SHARED BY MULTIPLE MR SYSTEMS** for:
 - Relative location of certain equipment cabinets of one MR system relative to other MR system(s) equipment cabinets.
 - Separation of the Penetration Panel of one MR system relative to the Penetration Panel of other MR system(s).
 - Separation of interconnects of one MR system relative to interconnects other system(s) and the managing of any excess RF Receive and RF Transmit cables for each MR System.

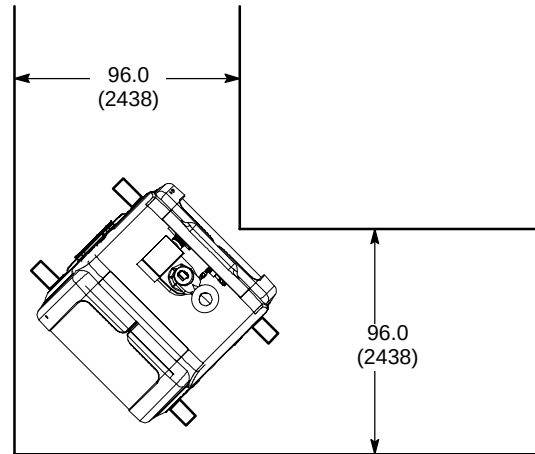
2-4 MINIMUM DOOR/HALLWAY SIZES & DELIVERY ROUTE WEIGHT CAPACITY

Table 2-3 lists minimum actual clearance opening dimensions for doors and hallways required by Signa equipment. Installation or replacement of components listed in Table 2-4 must be taken into consideration when determining hallway and door dimensions. Clearance for maneuvering around corners or turns must also be taken into consideration. Refer to SECTION 8, SHIPPING AND DELIVERY DATA, for Signa Component shipping dimensions.

TABLE 2-3
MINIMUM HALLWAY/DOOR DIMENSIONS

COMPONENT	MINIMUM HALLWAY/ DOOR WIDTH* in. (mm)	MINIMUM HALLWAY/ DOOR HEIGHT* in. (mm)	COMMENTS
Operator Workspace Table	32 (813)	80 (2032)	
Indoor Water Cooled TSCC [Model LC18MT/WC]	36 (914)	80 (2032)	
Indoor Air Cooled TSCC [Model LC20MT]	36 (914)	80 (2032)	
Equipment Cabinets	36 (914)	80 (2032)	Includes Indoor Water Cooled TGWC [Model LTL10] or Indoor Air Cooled TGWC [Model LC10M]
Cryogen delivery route and Storage Room	43 (1092)	80 (2032)	Width requirements due to size of 500 liter dewars. Width and height requirements vary dependent on the dewars used. Check with cryogen supplier.
Magnet	Refer to Note 1	Refer to Note 2	Refer to Table 2-4 for uncrated magnet dimensions.
RF Room Door	Refer to Section 7, RF SHIELDED ROOM	Refer to Section 7, RF SHIELDED ROOM	
Note * Minimum hallway and door dimensions are actual clearance openings. Width and height of rigging equipment is not included in above dimension. 1 Minimum width depends on access route to removable panels of RF shielded room wall. For straight path (i.e. no bends or turns) recommended to allow 6 in. (153 mm) on both sides of magnet. Appropriate calculations must be performed if turns exist along proposed magnet delivery route. Illustration 2-2 shows dimensions for 90° turn. 2 Final dimension is dependent on rigger equipment used, refer to SECTION 8, SHIPPING AND DELIVERY DATA.			

2-4 MINIMUM DOOR/HALLWAY SIZES & DELIVERY ROUTE WEIGHT CAPACITY(continued)



NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.

MAGNET MINIMUM DOOR/HALLWAY DIMENSIONS 90° TURN
ILLUSTRATION 2-2

TABLE 2-4
COMPONENT DIMENSIONS FOR INSTALLATION/REPLACEMENT

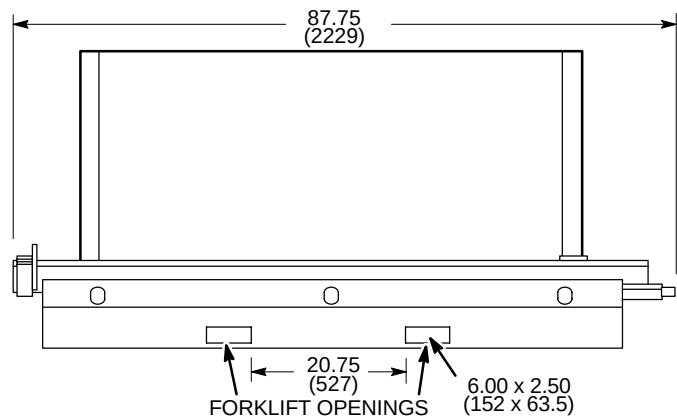
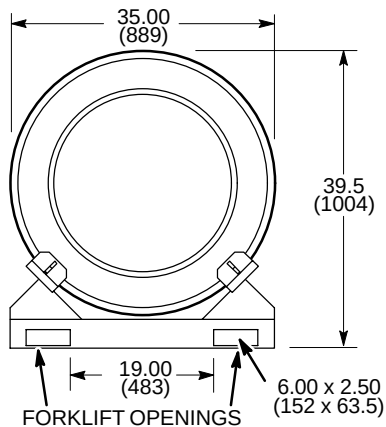
COMPONENT	APPROXIMATE WEIGHT lbs (kg)	OVERALL DIMENSIONS W x D x H in. (mm)	COMMENTS
Magnet (uncrated)	Refer to comments.	Refer to comments.	Refer to Section 8, SHIPPING AND DELIVERY DATA, for dimensions, illustrations and weights.
Split Bridge (LCC Magnet Enclosure)	40 (18)	21.5 x 77.3, x 7 (546 x 1969 x 18)	Room dimensions in front of LCC Magnet MUST allow for bridge installation/servicing and Gradient Coil Replacement, See Note 3.
Replacement RF Body Coil	155 (70)	30 x 30 x 60 (762 x 762 x 1524)	Replacement coil is shipped in a protective case. Weight & dimensions are for coil & case.
Replacement Gradient (TRM) Coil Assembly on a Shipping Cradle/Cart	See Note 1	35.5 x 99.84 x 55.88 (902 x 2536 x 1420) See Note 2 & 3	Initial TRM Gradient Coil Assembly is shipped installed in the Magnet. Shipping/installation cart is used to install replacement coil assembly only. Refer to Section 2-4-1, Gradient Coil Assembly, Cradle, & Cart Requirements.
Gradient (TRM) Coil Replacement Tool Kit Crate for LCC Magnets	750 (340)	30 x 86 x 28 (762 x 2184 x 711)	See Note 3.
Note 1 The replacement TRM Gradient Coil Assembly weight is approximately 2850 lbs (1293 kg), the shipping cradle is 132 lbs (60 kg), and the Gradient Coil Assembly shipping/installation cart weighs 855 lbs (389 kg). Therefore total shipping weight is 3837 lbs (1742 kg). The coil assembly outside diameter x length dimensions are 35.0 x 74.6 in. (888 x 1895 mm). 2 Gradient (TRM) Coil Assembly and shipping cart dimensions are with cart in lowest position. Cart can be adjusted to maximum height of 61.88 in. (1572 mm). 3 LCC Magnets MUST USE GE Service Tool Gradient Coil Replacement Kit for replacing the Gradient Coil Assembly. The Gradient Coil Replacement Tool Kit is shipped in a wooden crate on casters. Utilization of Gradient Coil Replacement Tool requires 107.75 in. (2737 mm) clear space in front of magnet.			

2-4-1 Gradient Coil Assembly, Cradle, & Cart Requirements

Initial TRM Gradient Coil Assembly is shipped installed in the Magnet. Shipping/installation cart is used to install replacement coil assembly only. The Gradient Coil Assembly will be delivered on an aluminium (re-useable) cradle. A forklift or crane/hoist rated for 4000 lbs (1818 kg) may be required to position the coil/cradle assembly onto the cart for installing the coil into the magnet, see Illustrations 2-3 and 2-4.

NOTE:

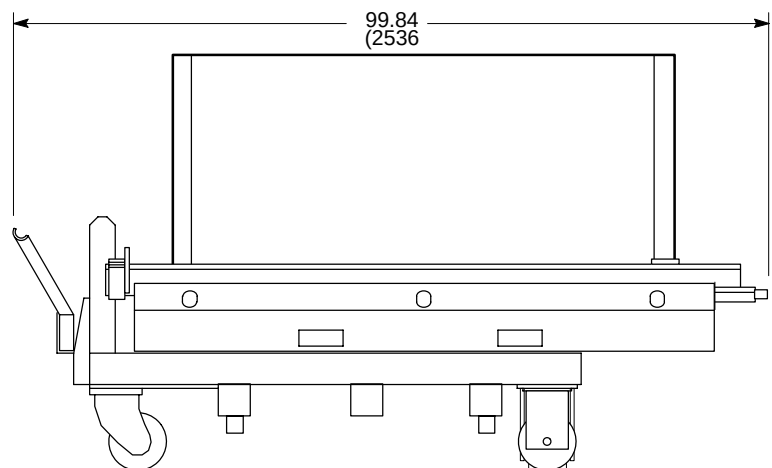
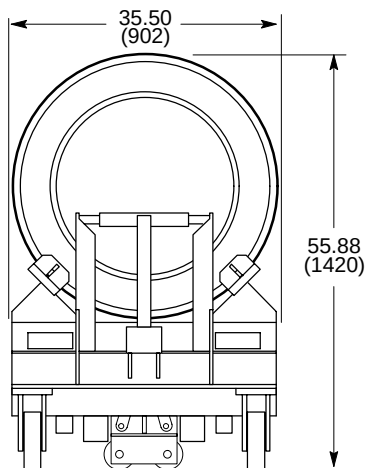
- ALL DIMENSIONS ARE IN INCHES ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT WITH TRM 2982 lbs (1353 kg)



REPLACEMENT GRADIENT (TRM) COIL ASSEMBLY & CRADLE
ILLUSTRATION 2-3

NOTE:

- ALL DIMENSIONS ARE IN INCHES ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT WITH TRM 3837 lbs (1742 kg)



SHORT ROOM REPLACEMENT GRADIENT (TRM) COIL ASSEMBLY & CRADLE ON CART
ILLUSTRATION 2-4

2-5 CABLING CONSIDERATIONS

Several different methods for running cables are listed below and the customer should carefully consider the advantages and disadvantages of each.

Care must be taken to protect interconnecting cords and cables from physical damage (including water). Branch circuit conductors must be enclosed in metal raceway or metal wireway when concealed or when installed under access flooring.

Consult local/national code for interconnects separation requirements (i.e. signal, power, water, etc.).

Note

Only non-magnetic metal material (e.g. aluminum) can be used when routing cables in the Magnet Room.

2-5-1 Floor Duct

Recessed floor duct has advantages when used within a single room or two adjacent rooms. Floor duct combines a neat functional appearance with accessibility and room for expansion. The disadvantage is the amount of work required to install it, which is generally prohibitive in old installations. Floor ducts can be used in the Magnet Room, however, they must meet the requirements in Section 7, RF SHIELDED ROOM.

2-5-2 Raceway

Raceways offer some unique advantages when routing cables. It is very practical to use in existing structures since it is surface-mounted. There is no problem with pre-terminated cables since the entire raceway system can be opened. Raceway systems are relatively easy to expand as compared to other means of routing cables. However, surface-mounted raceways are not recommended for routing cables within the Magnet Room due to the number/size of cables and the trip hazard of the raceway.

2-5-3 Access Flooring

Access flooring (i.e. removable panel flooring) is recommended for use in both the Equipment and Magnet Rooms due to the number and size of cables in the system. Cable accessibility and ease of alteration are just a few advantages of using access flooring. Floor duct with dividers placed above the Magnet Room floor but beneath the access flooring is a convenient method of separating electrical lines from water lines. However, if the area under the access flooring is used for an air plenum, cables may have to be in raceway depending on local and national codes. **Note**, the Signa system interconnecting cables in the Equipment room are FT4 or equivalent rated, not plenum rated.

2-5-4 Conduit

Conduit has some important restrictions when used with a MR system. The primary problem is that the majority of cables used are pre-terminated, which greatly simplifies interconnection, but makes cabling difficult because of the added dimensions of the connectors. As a consequence, conduit size must allow for the dimension of the connectors and the possibility of additional cables being added as the system is upgraded in the future. Always size the conduit to allow the cable to pass through with all other cables already in the conduit. Conduit should not be used for running the main GE MR system cables in the Magnet Room due to the number and size of conduits needed.

Note

MR personnel must have an unobstructed path from the patient table to the area directly behind the rear pedestal. Therefore cable routing methods must not interfere with this pathway.

Cable runs in the Magnet Room as well as throughout the system must be in accordance with local and national codes.

2-6 FLOORING

The finished floor in the Magnet Room should be water resistant and a type flooring to minimize the possibility of a static discharge. Use of an access floor with covering to minimize static discharge is recommended in the Equipment Room and the portion of the Magnet Room designated for cable routing. For safety purposes in the Magnet Room, it is required that the access flooring be made of aluminum. Depending on local and national codes, the area under the access floor may possibly be used as an air conditioning plenum. If the area under the access floor is to be used as an air plenum and for cable routing, 10 in. (254mm) of clear space from the underside of the access floor to the permanent floor is recommended. Cabling, plumbing (water lines), etc. routed under the access floor may affect air flow and needs to be considered if used as an air conditioning plenum. Also check local and national codes for fire protection requirements under access floor. **Note**, the Signa system interconnecting cables in the Equipment room are FT4 or equivalent rated, not plenum rated.

Ensure that the access flooring, if used, can support the equipment and any transport device needed to move the equipment.

If carpeting is used, it should either be anti-static carpeting or treated with an anti-static solution. Carpeting is not recommended in the Magnet Room or cryogen storage area as well as along the dewar delivery route due to possible problems with docking the patient table or moving the dewars.

Information on RF shielded room floor requirements can be found in Section 7, RF SHIELDED ROOM.

2-7 System Cooling Equipment

The Signa TwinSpeed system requires water cooling for the LCC Magnet Shield/Cryo Cooler Compressor Cabinet and the TRM Gradient Coil located in the Magnet bore. Refer to Section 1-2-1 System Cooling Equipment Selections for listing of catalog

- Twin System Cooling Cabinet (TSCC) which provides water cooling for the LCC Magnet Shield/Cryo Cooler Compressor Cabinet and the Gradient Coil located in the Magnet bore. Refer to Section 2-7-1 TSCC Siting Considerations for details.
- Twin Gradient Water Cooler (TGWC) which provides water cooling for ONLY the Gradient Coil located in the Magnet bore. When a TGWC is selected then the customer site must still provide water cooling for the LCC Magnet Shield/Cryo Cooler Compressor Cabinet. Refer to Section 2-7-2 TGWC Siting Considerations for details.

Note

True Mobile system water cooling for the Gradient Coil and the Shield/Cryo Cooler Compressor is provided by the Mobile Van equipment.

2-7-1 TSCC Configurations Siting Considerations



Continuous water cooling is critical for the Shield/Cryo Cooler Compressor and therefore MUST be available 24 hours per day / 7 days per week to maximize proper uninterrupted magnet operation. Water cooling is required immediately upon magnet arrival. Therefore the Twin System Cooling Cabinet (TSCC) and Main Disconnect Panel (MDP) must be installed and operational prior to magnet arrival.

Note

Consult local/national code for interconnects separation requirements (i.e. signal, power, water, etc.).

The TSCC is available in three fixed site configuration options:

- Outdoor Air Cooled TSCC [Model LC-20MT/O] has the TSCC unit located outside the building and a Remote Control Panel (RCP) located in the MR system Equipment Room; see Illustration 2-14 for Outdoor Air Cooled TSCC and Illustration 2-16 for the Remote Control Panel.
- Indoor Water Cooled TSCC [MODEL LC-18MT/WC]; see Illustration 2-17 and refer to Section 4-4-3 Water Cooled TSCC Requirements for water cooling requirements.
- Indoor Air Cooled TSCC [MODEL LC-20MT]; see Illustration 2-19.

Note

Customer provided temporary backup water cooling is recommended for the Shield/Cryo Cooler Compressor Cabinet. Refer to Section 4-4-2 Shield/Cryo Cooler Compressor Temporary Backup Water Cooling Recommendation.

2-7-1 TSCC Configurations Siting Considerations (Continued)**TSCC & System Interconnects/Separation Limitations**

Location of the Outdoor TSCC must meet the following limitations for the water lines (customer provided copper line and E&W/GE provided flexible tubing combined).

- Outdoor TSCC and the Gradient Coil located inside the magnet must not be separated by a distance greater than 200 ft (61 m) water line length.
- Outdoor TSCC and the Shield/Cryo Cooler Compressor Cabinet must not be separated by a distance greater than 160 ft (48.8 m) water line length.
- The Outdoor TSCC vertical separation from the Gradient Coil located inside the magnet and the Shield/Cryo Cooler Compressor must not exceed 100 ft (30.5 m) above or below the Gradient Coil and Shield/Cryo Cooler Compressor.
- The Outdoor TSCC and the RCP must not be separated by a distance greater than 100 ft (30.5 m) total interconnect length.

Location of the Indoor Water Cooled TSCC or Indoor Air Cooled TSCC must meet the following limitations for the water lines (E&W/GE provided flexible tubing).

- Indoor Water Cooled TSCC or Indoor Air Cooled TSCC and the Gradient Coil located inside the magnet must not be separated by a distance greater than 100 ft (30.5 m) water line length.
- Indoor Water Cooled TSCC or Indoor Air Cooled TSCC and the Shield/Cryo Cooler Compressor Cabinet must not be separated by a distance greater than 60 ft (18.3 m) water line length.
- The Indoor Water Cooled TSCC or Indoor Air Cooled TSCC vertical separation from the Gradient Coil located inside the magnet and the Shield/Cryo Cooler Compressor must not exceed 15 ft (4.6 m) above or below the Gradient Coil and Shield/Cryo Cooler Compressor.

The TSCC is powered from the system Main Disconnect Panel (MDP) via customer supplied wiring. Refer to **Section 6, INTERCONNECT DATA**, for additional interconnects information and requirements.

2-7-1 TSCC Configurations Siting Considerations (Continued)

Responsibility For Installation Tasks For TSCC Equipment

The TSCC subsystem equipment installation requires specific tasks to be performed by the Customer Contractor, GE Service, and Ellis & Watts Service. Table 2-5 lists the responsibility for the specific tasks. Refer to vendor manual for additional information concerning tasks.

TABLE 2-5
TSCC EQUIPMENT RESPONSIBILITY FOR INSTALLATION TASKS
PRIOR TO MAGNET DELIVERY & WHEN MAGNET IS DELIVERED/INSTALLED

APPLICABLE CONFIG			INSTALLATION TASKS FOR TSCC EQUIPMENT See Notes 1, 2, & 3	RESPONSIBILITY		
OUTDOOR (LC20MT/O)	INDOOR WATER COOLED (LC18MT/W)	INDOOR AIR COOLED (LC20MT)		CUSTOMER CONTRACTOR	GE SERVICE	E&W SERVICE PROVIDER
WHEN TSCC IS AT SITE & PRIOR TO MAGNET DELIVERY						
X	X	X	Unload TSCC from truck.	X		
X			Move TSCC to site outdoor location and mount. (Use seismic tie down brackets where required by law)	X		
X			Install Remote Control Panel in Equipment Room	X		
	X		Move TSCC to Equipment Room and mount. (Use seismic tie down brackets where required by law)	X		
		X	Move TSCC to Equipment Room and mount. (Use seismic tie down brackets & vibration isolators where required by law)	X		
X	X		Install customer water supply and return lines to TSCC	X		
X	X	X	Connect customer supplied power wiring from MDP to TSCC.	X		
X			Install power and control cable between TSCC & RCP	X		
WHEN MAGNET IS DELIVERED/INSTALLED						
X			Install water cooling lines from customer supplied/installed ball valves & hose barbs on copper lines from TSCC to 1) Gradient Coil and 2) Shield/Cryo Cooler Compressor. Refer to Section 6 – INTERCONNECT DATA for details of interconnects.		X	
	X	X	Install water cooling lines from TSCC to 1) Gradient Coil and 2) Shield/Cryo Cooler Compressor. Refer to Section 6 – INTERCONNECT DATA for details of interconnects.		X	
X	X	X	Fill TSCC with Propylene Glycol and startup unit		X	
X	X	X	See APPENDIX C Prerequisites for Scheduled Performance Verification of Chillers For Signa TwinSpeed and review for completeness. Send email request for initial performance verification to service@elliswatts.com . Email to include: System ID, Magnet Monitor System ID, Model and Serial Number of the SCC unit, and Customer location.		X	
X	X	X	Verify proper operation per E&W Chiller Startup Check List			X
Note 1 Refer to Ellis & Watts Technical Publication 470 Pre installation/Installation/Operating LC 20MT/O Chilled Water System for GE Signa TwinSpeed MRI Equipment for additional Outdoor TSCC information and details. 2 Refer to Ellis & Watts Technical Publication 471 Pre installation/Installation/Operating LC 18MT/ Chilled Water System for GE Signa TwinSpeed MRI Equipment for additional Indoor Water Cooled TSCC information and details. 3 Refer to Ellis & Watts Technical Publication 472 Pre installation/Installation/Operating LC 20MT Chilled Water System for GE Signa TwinSpeed MRI Equipment for additional Indoor Air Cooled TSCC information and details.						

2-7-2 TGWC Configurations Siting Considerations



When a TGWC is selected for the system then the customer site must still provide cooling for the LCC Magnet Shield/Cryo Cooler Compressor Cabinet; refer to Section 4-4-4 Shield/Cryo Cooler Compressor Water Cooling Requirements For Sites With TGWC

Note

Consult local/national code for interconnects separation requirements (i.e. signal, power, water, etc.).

The TGWC is available in three fixed site configuration options:

- Outdoor Air Cooled TGWC [MODEL LC-10MO] has the TGWC unit located outside the building and a Remote Control Panel (RCP) is located in the MR system Equipment Room; see Illustration 2-21 for Outdoor Air Cooled TGWC and Illustration 2-23 for the RCP.
- Indoor Water Cooled TGWC [MODEL LTL-10] has the TGWC unit and a Remote Control Panel both located in the MR system Equipment Room; see Illustration 2-24 and refer to **Section 4-4-5 Water Cooled TGWC Requirements** for water cooling requirements.
- Indoor Air Cooled TGWC [MODEL LC-10M]; see Illustration 2-26.

TGWC & System Interconnects/Separation Limitations

Location of the Outdoor TGWC must meet the following limitations for the water lines (customer provided copper line and GE provided flexible tubing combined).

- Outdoor TGWC and the Gradient Coil located inside the magnet must not be separated by a distance greater than 200 ft (61 m) water line length.
- The Outdoor TGWC vertical separation from the Gradient Coil located inside the magnet must not exceed 100 ft (30.5 m) above or below the Gradient Coil.
- The Outdoor TGWC and the RCP must not be separated by a distance greater than 100 ft (30.5 m) total interconnect length.

Location of the Indoor Water Cooled TGWC or Indoor Air Cooled TGWC must meet the following limitations for the water lines (GE provided flexible tubing).

- Indoor Water Cooled TGWC or Indoor Air Cooled TGWC and the Gradient Coil located inside the magnet must not be separated by a distance greater than 100 ft (30.5 m) water line length.
- The Indoor Water Cooled TGWC or Indoor Air Cooled TGWC vertical separation from the Gradient Coil located inside the magnet must not exceed 15 ft (4.6 m) above or below the Gradient Coil.

The Outdoor TGWC and Indoor Air Cooled TGWC are powered from the Main Disconnect Panel (MDP) via customer supplied wiring. The Indoor Water Cooled TGWC is powered from the MR system PDU. Refer to Section 6, INTERCONNECT DATA, for additional interconnects information and requirements.

2-7-2 TGWC Configurations Siting Considerations (Continued)**Responsibility For Installation Tasks For TGWC Equipment**

The TGWC subsystem equipment installation requires specific tasks to be performed by the Customer Contractor, GE Service, and Ellis & Watts Service. Table 2-6 lists the responsibility for the specific tasks. Refer to vendor manual for additional information concerning tasks.

TABLE 2-6
TGWC EQUIPMENT RESPONSIBILITY FOR INSTALLATION TASKS
PRIOR TO MAGNET DELIVERY & WHEN MAGNET IS DELIVERED/INSTALLED

APPLICABLE CONFIG			INSTALLATION TASKS FOR TGWC EQUIPMENT See Notes 1, 2, & 3	RESPONSIBILITY		
OUTDOOR (LC10MT/O)	INDOOR WATER COOLED (LC10)	INDOOR AIR COOLED (LC10MT)		CUSTOMER CONTRACTOR	GE SERVICE	E&W SERVICE PROVIDER
WHEN TGWC IS AT SITE & PRIOR TO MAGNET DELIVERY						
X	X	X	Unload TGWC from truck.	X		
X			Move TGWC to site outdoor location and mount. (Use seismic tie down brackets where required by law)	X		
X			Install Remote Control Panel in Equipment Room	X		
	X		Move TGWC and Remote Control Panel to Equipment Room and mount. (Use seismic tie down brackets where required by law)	X		
		X	Move TGWC to Equipment Room and mount. (Use seismic tie down brackets where required by law)	X		
X	X		Install customer water supply and return lines to TGWC	X		
X		X	Connect customer supplied power wiring from MDP to TGWC.	X		
X	X		Install power and control cable between TGWC & RCP	X		
WHEN MAGNET IS DELIVERED/INSTALLED						
X			Install water cooling lines from customer supplied/installed ball valves & hose barbs on copper lines from TGWC to Gradient Coil. Refer to Section 6 – INTERCONNECT DATA for details of interconnects.		X	
	X	X	Install water cooling lines from TGWC to Gradient Coil. Refer to Section 6 – INTERCONNECT DATA for details of interconnects.		X	
X		X	Fill TGWC with Propylene Glycol and startup		X	
X		X	Verify proper operation per E&W Chiller Startup Check List			X
Note 1 Refer to Ellis & Watts Technical Publication 474 Pre installation/Installation/Operating LC 10MT/O Chilled Water System for GE Signa TwinSpeed MRI Equipment for additional Outdoor TGWC information and details. 2 Refer to Ellis & Watts Technical Publication 469 Pre installation/Installation/Operating LTL10 Chilled Water System for GE Twin Magnet MRI Equipment for additional Indoor Water Cooled TGWC information and details. 3 Refer to Ellis & Watts Technical Publication 473 Pre installation/Installation/Operating LC 10MT Chilled Water System for GE Signa TwinSpeed MRI Equipment for additional Indoor Air Cooled TGWC information and details.						
(Continued)						

2-7-2 TGWC Configurations Siting Considerations (Continued)

TABLE 2-6 (Continued)
**TGWC EQUIPMENT RESPONSIBILITY FOR INSTALLATION TASKS
 PRIOR TO MAGNET DELIVERY & WHEN MAGNET IS DELIVERED/INSTALLED**

APPLICABLE CONFIG			INSTALLATION TASKS FOR TGWC EQUIPMENT See Notes 1, 2, & 3	RESPONSIBILITY		
OUTDOOR (LC10MT/O)	INDOOR WATER COOLED (LC10)	INDOOR AIR COOLED (LC10MT)		CUSTOMER CONTRACTOR	GE SERVICE	E&W SERVICE PROVIDER
WHEN MR PDU IS DELIVERED/INSTALLED						
	X		Connect E&W supplied power cable from PDU to TGWC.		X	
	X		Fill TGWC with Propylene Glycol, startup and verify operation		X	
X	X	X	See APPENDIX C Prerequisites for Scheduled Performance Verification of Chillers For Signa TwinSpeed and review for completeness. Send email request for initial performance verification to service@elliswatts.com . Email to include: System ID, Magnet Monitor System ID, Model and Serial Number of the SCC unit, and Customer location.		X	
Note 1 Refer to Ellis & Watts <i>Technical Publication 474 Pre installation/Installation/Operating LC 10MT/O Chilled Water System for GE Signa TwinSpeed MRI Equipment</i> for additional Outdoor TGWC information and details. 2 Refer to Ellis & Watts <i>Technical Publication 469 Pre installation/Installation/Operating LTL – 10 Chilled Water System for GE Twin Magnet MRI Equipment</i> for additional Indoor Water Cooled TGWC information and details. 3 Refer to Ellis & Watts <i>Technical Publication 473 Pre installation/Installation/Operating LC 10MT Chilled Water System for GE Signa TwinSpeed MRI Equipment</i> for additional Indoor Air Cooled TGWC information and details.						

2-8 SPECIAL SITING CONSIDERATIONS

The following system accessories or options have special siting concerns which need to be considered.

2-8-1 Blower Box (MG6)

The Blower Box (MG6) provides cooling air for the Patient Comfort Module in the Magnet Enclosure. The Blower Box will be mounted within the RF Shielded Room and connect to the Patient Comfort Module by 4.0 inch (101.6 mm) OD flexible vinyl air ducting. **The Blower Box contains magnetic material and must be securely mounted to the floor of the Magnet Room or a support shelf on the Magnet Room wall or ceiling with support provided under the box. RF Shield integrity must be maintained for mounting the Blower Box within the Magnet Room.** The flexible vinyl air duct routes from the Blower Box through the Magnet Enclosure Rear Pedestal cable access and connects to the Patient Comfort Module in the Magnet Enclosure. Refer to Illustrations 2-27 for Blower Box and mounting dimensions.

Note

Blower Box mounting requires customer supplied hardware (ie. lag bolts, screws, etc.) appropriate for the surface on which the box will be mounted.

2-8-2 Pneumatic Patient Alert (PA1)

The Pneumatic Patient Alert system is a stand alone system that will allow the Patient to contact the Operator even when the intercom volume is turned down. The Control Box is to be located near the Operator Workspace. The Control Box audible and visual alarm will be activated by the patient squeeze bulb which is located on the Magnet Enclosure and connected by pneumatic tubing through the Penetration Panel to the Control Box. The Control Box should be mounted with consideration for ease of use by operator, remaining within sight of operator, and within 5 ft (1.5 m) of an electrical outlet. The Control Box can be powered from an outlet on the Operator Workspace. Refer to Illustration 2-42 for Control Box mounting dimensions.

2-8-3 SPT Phantom Set Shipping/Storage Cabinet

System Performance Test provides the customer and GE Service with a means to quickly verify whether critical parameters affecting image quality are within specifications. The test uses a set of phantoms and a nesting plate for proper positioning of the phantoms on the Patient Table. The phantom set and nesting plate are provided in a cabinet which protects the pieces during shipment and storage at site. The cabinet is not magnetic therefore it can be stored inside the Magnet Room if so desired and moved to the Patient Table for ease of positioning the phantoms. See Illustration 2-44 for dimensions information.

2-8-4 Oxygen Monitor (OM1) (Optional)

The optional Oxygen Monitor should be mounted near the Operator Workspace. The Oxygen Monitor alarm will be activated by the remote oxygen sensor located in the Magnet Room. All cellular telephones, even if not in use, should be kept at least 20 feet (6.1 meters) away from the Oxygen Monitor to prevent possible false trips of Oxygen Monitor alarms.

2-8-5 Spectroscopy for 1.5T Magnet Only (Optional)

Multi-Nuclear Spectroscopy (MNS)

The MNS option provides the Spectroscopy Acquisition Option which consists of the MNS Cabinet (contains the Broadband RF Amplifier) and several module kits to modify the SRF Cabinet, Penetration Panel in the Magnet Room, and system cabling. Refer to **Section 6-8, SPECTROSCOPY INTERCONNECTIONS** cable length information and other cable data.

Hydrogen Only (PROBE SV/Q) Spectroscopy

No additional siting impacts.

2-8-6 Advantage Workstation Option Additional Hardware Features

The Advantage Workstation (AW) option can be utilized with cardiac and neurovascular software offerings. Refer to *Direction 2261309-100 Advantage Workstation Ultra60 Pre-Installation* for more information.

The customer must determine the desired location for the AW relative to the imaging system. The following factors should be considered in determining AW location.

- If the customer is purchasing RTIP (neurovascular) option, then the customer may want to observe the experiment real time while operating the Signa system versus reviewing the data at a later time. If the former, then consideration should be made to co-locate the AW near the system Operator Workspace (OW).

Note

LCD monitor is recommended for the AW to be located near the Signa system Operator Workspace due to gauss field proximity limits.

- If the customer is going to perform cardiac studies only, then co-location is not necessary at this time. However, co-location of the AW may be recommended for future applications.

2-8-7 Magnet Monitor

The Magnet Monitor performs the functions of a cryogen meter and LCC Magnet pressure control with readout display capability on the unit. The Magnet Monitor also monitors vacuum level of the Quiet Technology Magnet Enclosure and the TSCC/TGWC operation parameters. The Magnet Monitor with its modem allows for remote monitoring during system warranty period or available as part of a GE Service contract. A dedicated direct-distance-dialing voice-grade telephone line with access located near the Magnet Monitor in the Equipment Room for remote monitoring is required, refer to **Section 2-8-8 System Monitoring & Support Connectivity Requirements** for additional information.

Magnet Monitor remote monitoring operation is valuable to maximize proper uninterrupted magnet operation. The Magnet Monitor is powered from the Main Disconnect Panel (MDP) and must be mounted near the MDP, see Illustration 2-5. The Magnet Monitor should be mounted approximately 60 in. (1524 mm) above the floor in the Equipment Room but outside the 10 gauss zone. Refer to **Section 6 – INTERCONNECT DATA** for details of Magnet Monitor, optional UPS, and system interconnects.

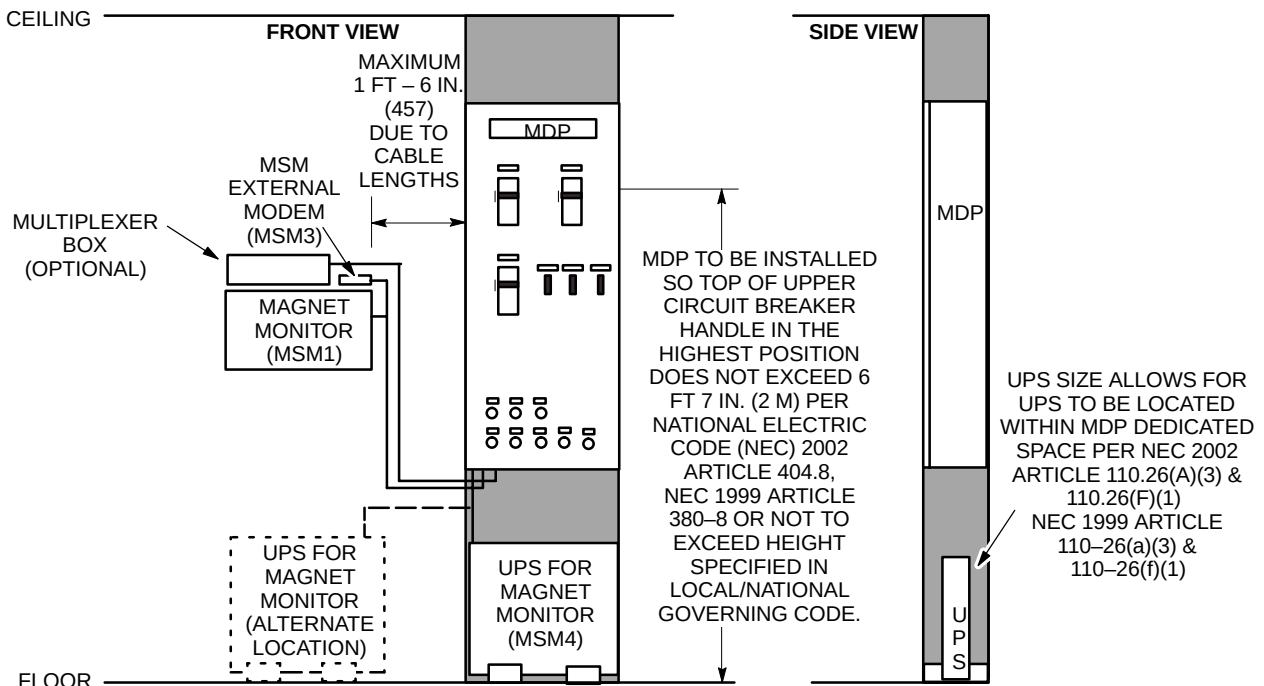
An optional small Uninterruptible Power System (UPS) is available for the Magnet Monitor. This optional UPS provides power to the Magnet Monitor equipment via the MDP. In the event of facility power outage, the UPS will maintain Magnet Monitor operation for sufficient time to communicate status via the remote monitoring.

Note

The UPS for the Magnet Monitor equipment is provided as part of the Magnet subsystem for True Mobile configurations.

NOTE:

SPACE ABOUT ELECTRICAL EQUIPMENT PER NATIONAL ELECTRICAL CODE (NEC) 2002 ARTICLE 110.26 OR NEC 1999 ARTICLE 110-26.



MAGNET MONITOR (MSM) & ASSOCIATED EQUIPMENT, UPS FOR MAGNET MONITOR, AND MDP LOCATIONS

ILLUSTRATION 2-5

2-8-8 System Monitoring & Support Connectivity Requirements

One of the system monitoring and support connectivity configurations listed in Table 2-7 must be provided for system installation and serviceability purposes. The network connection (if selected configuration) and telephone lines are to be provided and paid for by the customer.

TABLE 2-7
SYSTEM MONITORING & SUPPORT CONNECTIVITY REQUIREMENTS

CONFIGURATION	PHONE LINE	USE/LOCATION
Network connection & telephone lines	One <u>voice-grade</u> telephone line (voice line)	Available for Service Personnel use, located in the Control Room
	Two Network connection with a static IP Addresses See Note 1	One access located near the Operator Workspace (OW) in the Control Room.
		One access located near the Magnet Monitor (MSM) in the Equipment Room for remote monitoring. This Ethernet connection must not lose power when the MR system is shutdown.
Multiple telephone lines	One <u>voice-grade</u> telephone line (voice line)	Available for Service Personnel use, located in the Control Room
	One line must be a dedicated direct-distance-dialing <u>voice-grade</u> line (data line)	Access located near the Operator Workspace (OW) in the Control Room. See Note 2 & 3
	One line must be a dedicated direct-distance-dialing <u>voice-grade</u> line (data line)	Access located near the Magnet Monitor (MSM) in the Equipment Room for remote monitoring. See Note 2 & 3
Note 1 For North America & South America: One Internet accessible Virtual Private Network Connection with a static IP Address is required. For Europe: One ISDN Network Connection with a static IP Address is required. 2 A dedicated direct-distance-dialing <u>voice-grade</u> telephone line can be shared for Operator Workspace (OW) and Magnet Monitor (MSM) requirement through the use of a multiplexer box. The following multiplexer boxes are available for customer purchase. 46-328475P1 4 Line Phone Multiplexer box; 115 VAC input power 46-328475P3 4 Line Phone Multiplexer box; 220 VAC input power If the customer chooses not to purchase the multiplexer box then the customer must provide an additional line for each requirement as stated in this table. 3 If a Multiplexer Box is used then the Magnet Monitor MUST be Channel 1 to allow for call out after a power outage.		

2-8-9 BrainWave Option For Signa 1.5T TwinSpeed with EXCITE Technology Systems ONLY

The BrainWave option provides an integrated set of software and hardware stimulus components for the Signa/AW RTIP application to produce brain activation images from MR Blood Oxygen Level Dependent (BOLD) scan data. Stimulus devices include audio (headphones), visual (goggles), and push-button response box held by patient. Stimulus software resides on the AW (AW4.0 or later required) used by RTIP and on a PC in the BrainWave Cabinet located in the Equipment Room.

The BrainWave option consists of the following major items:

- **BrainWave Audio Console**
Used to select the sound media and communicate to/from the patient. The Audio Console is to be located on the Signa system Operator Workspace Table.
- **BrainWave Cabinet**
Located in the Equipment Room, the BrainWave Cabinet contains the following BrainWave equipment: Stimulus PC, LCD Display, keyboard, and mouse for the Stimulus PC, Video Interface, Graphic (audio) Equalizer, and Response Interface Module.
- **BrainWave Cart**
Located in the Magnet Room, the BrainWave Cart holds the Visual Display Assembly consisting of the Visual Projector, Image Guides, and Glasses which are factory connected together as one assembly. The color LCD Visual Projector receives video signals from the Video Interface in the BrainWave Cabinet, converts the signals to optical images, and then optically transmits the images to the left/right Patient Glasses via two flexible Image Guides. The Glasses are mounted to the inside of the Signa Head Coil with the Glasses Mount Assembly that provides for proper viewing of images by patients.
- **Interconnect cables**
The Audio Console is powered from the Operator Workspace. The BrainWave Cabinet is powered from the Signa System Cabinet. There are interconnects from Audio Console to the Signa System Cabinet and Penetration Panel, from System Cabinet to the Penetration Panel to the BrainWave Cart via the Magnet Enclosure. Refer to **Section 6, INTERCONNECT DATA**, for additional interconnects information and requirements.

2-9 ARCHITECTURAL REMINDERS

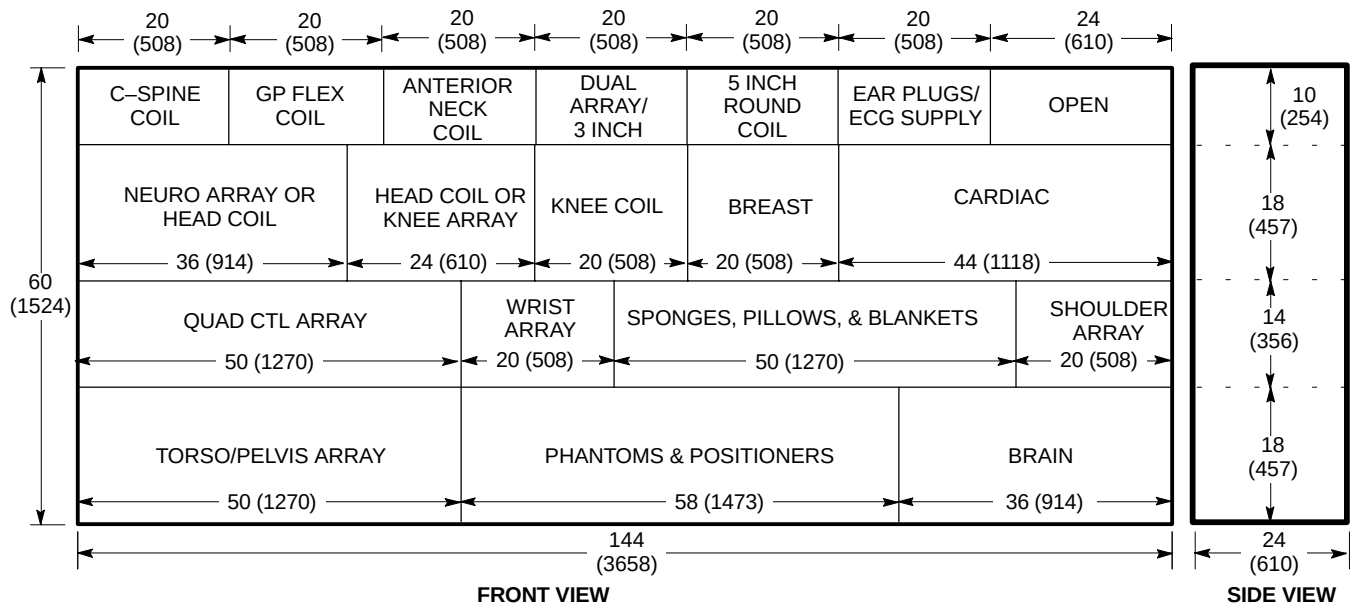
1. Pay attention to isogauss limits, not only for placement of equipment in rooms, but also for isogauss limits with respect to outside environment.
2. The customer is responsible for establishing protocols to warn persons with cardiac pacemakers, neurostimulators, and biostimulation devices of the potential danger of entering magnetic fields greater than 5 gauss (exclusion zone).
3. Due to the periodic cryogen servicing of the magnet, consideration must be given to the delivery route of the cryogens to the Magnet Room. The service route should be level and therefore steps or steep ramps must be avoided. Maximum acceptable incline along dewar delivery route is 1:12 (5°).
- 3A. Cryogen dewars must not be stored within the Magnet Room due to the safety issues of seismic considerations, spillage effects, fire hazards and explosive effects of compressed gas. Also the magnetic field inhomogeneity is affected by the physical shape of the non-magnetic dewars. If the magnet is shimmed with no dewars in the Magnet Room, the magnetic field inhomogeneity will slightly change once dewars are moved into the room (or vice versa). All dewars must be stored outside of the Magnet Room and more than 10 ft (3.05 m) from isocenter of the magnet in all directions.
- 3B. Means must be provided to secure gas cylinders used for cryogen transfills in an upright position using a removable chain or strap. This is to prevent the cylinders from falling, which may cause injury or damage.
4. If elevators are to be used along cryogen delivery route, verify that elevator dimensions and weight capacity is sufficient to handle the cryogen dewars. Also, elevator must be dedicated with restricted access during cryogen transport (will not allow stops between initial start and final floor destination).
5. The operator seated at the Operator Workspace should have an unobstructed view of the patient on the transport table when table is docked to the magnet.
6. It is recommended that the Magnet Room viewing window be of fine mesh screening material (as opposed to a "honeycomb-type pattern") for better visibility of the patient from the Operator Workspace.
7. Operators in Magnet Room must have easy access to the scan control switches located on both front side panels of the magnet enclosure.
8. A patient preparation/emergency area should be located near the Magnet Room and direct patient access must be available from the Magnet Room to the patient preparation/emergency area.
9. Customer provided and paid for telephone lines must be supplied for system installation and serviceability purposes per **Section 2-8-8 System Monitoring & Support Connectivity Requirements**.
10. A lockable storage cabinet can be provided and maintained by GE Medical Systems Service for storage of GE Medical Systems service documentation/tools. Cabinet to be approximately 36 in. (914 mm) wide, 18 in. (457 mm) depth, and 72 in. (1829 mm) high.

2-9 ARCHITECTURAL REMINDERS (Continued)

11. Corrosive chemicals must not be stored or used in the Equipment Room. These include chemicals used for film processor storage tanks, processor chemical recovery systems, etc. Such chemicals can contribute to increased equipment failures, increased system downtime, and decreased reliability. Film processor equipment installation must meet the manufacturer's requirements (e.g. ventilation specifications) and all applicable national and local codes. Also, consideration should be given to the location of this equipment and chemical fumes relative to human contact as it relates to locating this equipment and chemicals in the control area.
12. Storage space for system accessories and supplies should be planned for and included in room layout drawings. Illustrations 2-6 and 2-7 show a suggestion for a shelf/cabinet arrangement.
13. Recommend protecting floors while moving heavy pieces of equipment (e.g. ACGD/PDU Cabinet, TRM Body Coil Assembly, TSCC, TGWC).

NOTE:

- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- PERIPHERAL VASCULAR ARRAY IS NOT INCLUDED IN THE BELOW STORAGE SHELF/CABINET. CONTACT GE MEDICAL SYSTEMS ACCESSORIES FOR PV ARRAY STORAGE RACK OPTION AND DIMENSION INFORMATION FOR THE STORAGE RACK.

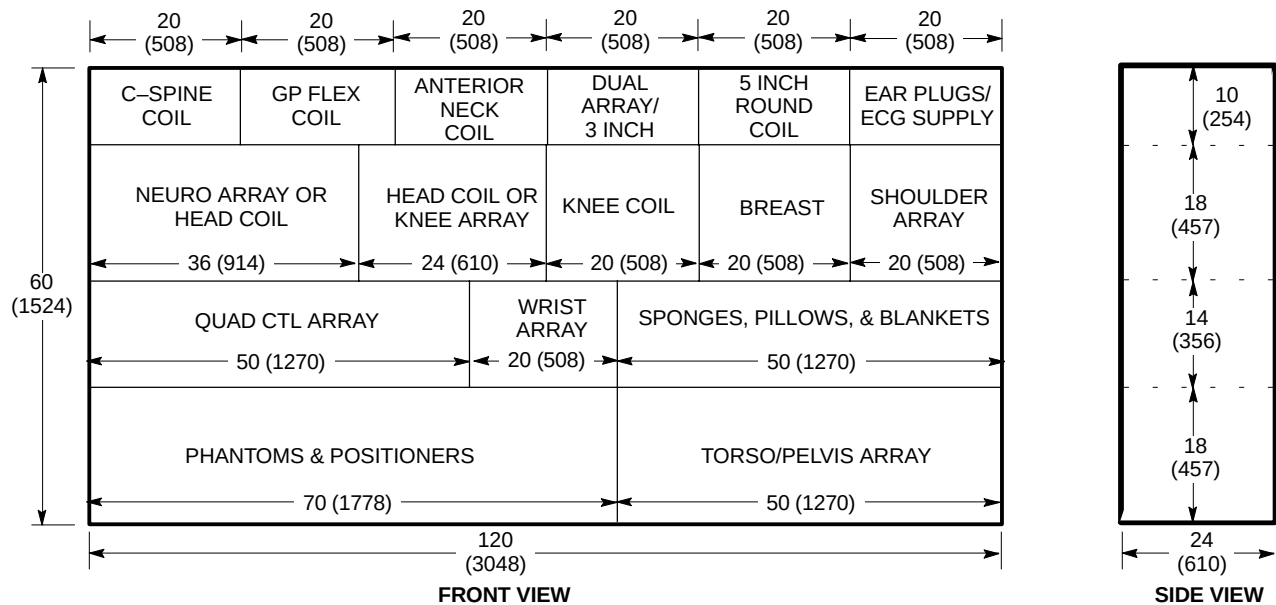


SUGGESTION FOR ACCESSORIES & SUPPLIES STORAGE SHELF/CABINET FOR SIGNA TWINSPEED WITH EXCITE TECHNOLOGY
ILLUSTRATION 2-6

2-9 ARCHITECTURAL REMINDERS (Continued)

NOTE:

- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- PERIPHERAL VASCULAR ARRAY IS NOT INCLUDED IN THE BELOW STORAGE SHELF/CABINET. CONTACT GE MEDICAL SYSTEMS ACCESSORIES FOR PV ARRAY STORAGE RACK OPTION AND DIMENSION INFORMATION FOR THE STORAGE RACK.



SUGGESTION FOR ACCESSORIES & SUPPLIES STORAGE SHELF/CABINET FOR SIGNA TWINSPEED
ILLUSTRATION 2-7

2-10 FLOOR LOADING AND WEIGHTS

This section contains loading considerations for the MR system. Listed in Table 2-8 are the weights, floor loading, and normal mounting methods for MR components.

TABLE 2-8
FLOOR LOADING

COMPONENT	NET WT lbs (kg) See Note 2	OVERALL DIMENSIONS W x D x H in. (mm)	LOAD PATTERN in. (mm)	NORMAL MOUNTING METHOD
LCC Magnet, Quiet Technology Enclosure, and Coils	See Notes 1 & 3 & Refer to Section 2-10-1	83.27 x 117.67 x 101.96 (2115 x 2989 x 2587.21)	See Illustration 2-11 See Illustration 7-1.	Magnet & Table Dock Asm. Resting on base, bolted to floor with RF Screen Room vendor supplied/ installed bolts & anchors.
Patient Table (See Note 4)	630 (286)	27 x 89 x 38 (686 x 2261 x 965)	Rectangular base 23 x 65 (584 x 1651). Four casters 8 (203) dia. x 1.5 (38) wide.	Mobile.
Blower Box	20 (9)	14.5 x 14 x 13.7 (368 x 356 x 348)	Square base 14.5 x 14 (368 x 356)	Anchor to floor, shelf, or wall.
1.5T SRFD II Cabinet	495 (225)	23.46 x 45.5 x 71.06 (596 x 1156 x 1805)	Rectangular base 22 x 33.5 (559 x 851). Four casters each 2.5 (63.5) dia. x 1.75 (44.5) wide.	Set on floor on casters.
Signa 1.5 TwinSpeed with EXCITE Technology System Cabinet	468 (213)	23.25 x 42 x 76.5 (591 x 1067 x 1943)	Rectangular base 22 x 30 (584 x 762). Four casters each 2.5 (63.5) dia. x 1.75 (44.5) wide.	Set on floor on casters.
Signa 1.5 TwinSpeed System Cabinet	494 (225)	23.25 x 42 x 76.5 (591 x 1067 x 1943)	Rectangular base 22 x 30 (584 x 762). Four casters each 2.5 (63.5) dia. x 1.75 (44.5) wide.	Set on floor on casters.
ACGD/PDU Cabinet	2136 (969)	23.5 x 37.25 x 85.6 (597 x 927 x 2174) See Note 5	Rectangular base 33.5 x 22 (851 x 559). Four casters each 3.0 (76.2) dia. x 1.82 (46.2) wide.	Set on floor on casters.
Twin Accessory Cabinet	600 (273)	23.32 x 38.75 x 50.13 (592 x 984 x 1273)	Rectangular base 22 x 30 (584 x 762). Four casters 2.5 (63.5) dia. x 1.75 (44.5) wide.	Set on floor and rest on casters.
Shield/Cryo Cooler Compressor Cabinet	275 (125)	17.72 x 19.7 x 26.93 (450 x 500 x 684)	Rectangular base 17.72 x 19.7 (450 x 500). Four casters 3 (76) dia. x .75 (19) wide.	Set on floor and rest on casters.
Twin System Cooling Cabinet (TSCC) See Note 6				
Outdoor Air Cooled	1200 (544)	74.83 x 45.5 x 66.88 (1901 x 1156 x 1699)	See Illustration 2-14.	Bolted to mounting pad external to building.
Indoor Water Cooled	1000 (454)	55.4 x 33.59 x 73 (1407 x 358 x 1854)	See Illustration 2-17.	Set on floor and rest on casters.
Indoor Air Cooled	1410 (640)	65 x 32.5 x 71.26 (1651 x 825.5 x 1810)	See Illustration 2-19.	Set on floor and rest on casters.
Twin Gradient Water Cooler (TGWC) See Note 6				
Outdoor Air Cooled	780 (354)	49.5 x 34 x 76.2 (1257 x 864 x 1936)	See Illustration 2-21.	Bolted to mounting pad external to building.
Indoor Water Cooled	175 (79)	33.94 x 19.66 x 23.67 (86.21 x 49.94 x 60.13)	See Illustration 2-24.	Set on floor and rest on 4 casters.
Indoor Air Cooled	1290 (585)	28.4 x 33 x 73 (721 x 838 x 1854)	See Illustration 2-26.	Set on floor and rest on casters & front pads on anti-tip legs
(Continued)				

2-10 FLOOR LOADING AND WEIGHTS (Continued)

TABLE 2-8 (Continued)
FLOOR LOADING

COMPONENT	NET WT lbs (kg) See Note 2	OVERALL DIMENSIONS W x D x H in. (mm)	LOAD PATTERN in. (mm)	NORMAL MOUNTING METHOD
Operator Workspace Table with LCD Color Monitor (See Note 7)	175 (80)	54 x 43 x 52 (1372 x 1092 x 1321)	See Illustration 2-37.	Set on floor: Table on leveling pads & Cabinet rest on casters. Anchor Table to floor per Section 2-10-3.
Operator Workspace Cabinet (See Note 7)	192 (87)	18.5 x 29 x 26 (470 x 737 x 660)	See Illustration 2-38.	Set on floor and rest on casters.
UPS for Magnet Monitor*	34 (15)	19.4 x 3.5 x 17 (494 x 89 x 432)	See Illustration 2-45	Set on floor on 2 mounting brackets.
Advantage Workstation*	Refer to Direction 2261309-100 <i>Advantage Workstation 3.1 Ultra60 Pre-Installation</i> . Also refer to Section 2-8-6 Advantage Workstation Option Additional Hardware Features to determine location of workstation.			
BrainWave Cabinet*	387 (176)	23.5 x 23 x 69 (597 x 584 x 1753)	Rectangular base 23 x 22 (584 x 559). Four leveling pads each 1.5 (38) dia.	Casters for location. Set on floor on leveling pads.
BrainWave Cart*	71 (32)	18 x 36 x 44.5 (457 x 914 x 1130)	Rectangular base 16.5 x 22.5 (419 x 572). Four casters each 2.875 (73) dia. x 2.0 (50.8) wide.	Set on floor on casters.
MNS Cabinet*	430 (195)	23.56 x 42 x 70.7 (598 x 1067 x 1797)	Rectangular base 33.5 x 22 (851 x 559). Four leveling pads each 1.5 (38) dia.	Casters for location. Set on floor on four leveling pads.
500 Liter Helium Dewar**	812 (369)	42 dia. x 67.5 high (1067 x 1715)	Mobile.	Four casters.
250 Liter Helium Dewar**	519 (236)	36 dia. x 64.0 high (914 x 1626)	Mobile.	Four casters.
Note - Weight of 1.5T LCC (Active Shield) magnet with Quiet Technology Enclosure, RF/Gradient (TRM) Body Coil, and cryogenics is 12,826 lbs (5818 kg). - Consult a structural engineer on method of calculating proper weight/unit area for floor loading. - Refer to Section 2-4 MINIMUM DOOR/HALLWAY SIZES & DELIVERY ROUTE WEIGHT CAPACITY for Gradient Coil Assembly replacement weight and dimension requirements. 630 lbs (286 kg) includes 350 lbs (159 kg) patient. - ACGD/PDU Cabinet height with Fan Module removed is 74.8 inches (1900 mm). - System Cooling Cabinet is available in the following configuration options (must choose one): TSCC (Provides water cooling for Shield/Cryo Cooler Compressor and Gradient Coil). - Outdoor Air Cooled TSCC installed external to building with Remote Control Panel (RCP) wall mounted in the Signa system Equipment Room - Indoor Water Cooled TSCC located in or near the Signa system Equipment Room - Indoor Air Cooled TSCC located in or near the Signa system Equipment Room. - True Mobile system equipment, contact Van Manufacturer for details. TGWC (Provides water cooling for Gradient Coil ONLY. Customer/site provided water cooling is required for Shield/Cryo Cooler Compressor, refer to Section 4-4-4 Shield/Cryo Cooler Compressor Water Cooling Requirements For Sites With TGWC. - Outdoor Air Cooled TGWC installed external to building with Remote Control Panel (RCP) wall mounted in the Signa system Equipment Room. - Indoor Water Cooled TGWC and Remote Control Panel (RCP) wall mounted both located in the Signa system Equipment Room - Indoor Air Cooled TGWC located in or near the Signa system Equipment Room. - The Operator Workspace Cabinet minimum area location is under the Workspace Table, see in Illustration 2-37. An alternate location is possible with the Operator Workspace Cabinet located against the right side of the Workspace Table. * Optional Equipment. ** Dewar specifications may vary. Check with cryogen supplier for exact weight and dimensions.				

2-10-1 Magnet Loading Considerations

In addition to the weight of the riggers equipment, special consideration must be given to the weight of the magnet along the delivery route. Refer to SECTION 8, SHIPPING AND DELIVERY DATA, for the shipping weight of the magnet (i.e. magnet shipped with liquid cryogen and RF/Gradient coil inside magnet bore and with partial enclosure installed). Structural reinforcement may be required along the magnet delivery route. It is required that a structural engineering analysis be performed on the Magnet Room floor and delivery route to determine its load bearing capacity.

During magnet installation, leveling plates or shims **MUST** be installed between magnet feet and the floor to compensate for variation within floor levelness. Refer to Section 7-6-4 FLOORS for levelness requirements. For LCC (Active Shield) Magnet, GE Medical Systems supplied aluminum shims are installed to complete contact between entire surface of each of the four magnet feet and the floor.

2-10-2 Anchoring And Seismic Considerations

It is the responsibility of the customer to obtain any and all approvals necessary for the construction of equipment support and seismic anchoring.

The center of gravity for MR system components are given for use in seismic calculations. If the MR cabinets are required by code to be anchored, refer to seismic drawings available on request from your local GEMS Installation Specialist. For the TSCC and TGWC units refer to Table 2-9 for mounting information.

**TABLE 2-9
SYSTEM COOLING EQUIPMENT MOUNTING INFORMATION**

SYSTEM COOLING EQUIPMENT	MOUNTING ILLUSTRATION	COMMENTS (See Note 1)
Outdoor TSCC unit	2-14 & 2-15	Unit has pre-drilled holes that should be used for mounting which should be used to aid in preventing the cabinet from shifting during a seismic event.
Indoor Water Cooled TSCC	2-17 & 2-18	Unit has tie down brackets supplied with unit which should be used to aid in preventing the cabinet from shifting during a seismic event. Site Architects and Engineers need to take into consideration the access floor framing layout in order to have substantial support for the tie down brackets.
Indoor Air Cooled TSCC	2-19 & 2-20	Unit has tie down brackets supplied with unit which should be used to aid in preventing the cabinet from shifting during a seismic event. Site Architects and Engineers need to take into consideration the access floor framing layout in order to have substantial support for the tie down brackets.
Outdoor TGWC	2-21 & 2-22	Unit has tie down brackets supplied with unit which should be used to aid in preventing the cabinet from shifting during a seismic event.
Indoor Water Cooled TGWC	2-24	Unit has tie down brackets supplied with unit which should be used to aid in preventing the cabinet from shifting during a seismic event. Site Architects and Engineers need to take into consideration the access floor framing layout in order to have substantial support for the tie down brackets.
Indoor Air Cooled TGWC	2-26	Unit has tie down brackets supplied with unit which should be used to aid in preventing the cabinet from shifting during a seismic event. Site Architects and Engineers need to take into consideration the access floor framing layout in order to have substantial support for the tie down brackets.
Note 1 Sites requiring seismic anchoring by code should have the site architect and engineer review the response spectra and/or Uniform Builders Code (UBC) for their location and then contact Ellis & Watts International for assistance in seismic planning of the site. Contact: Ellis & Watts International Inc. 4400 Glen Willow Lake Lane Batavia, OH 45103 USA Telephone: 513-752-9000 FAX: 513-752-4983		

2-10-2 Anchoring And Seismic Considerations (Continued)**Magnet Mounting**

It is the customer's responsibility to coordinate magnet mounting methods with the RF shielded room vendor to prevent RF leaks and secondary grounding problems. Refer to Table 2-10 for magnet mounting information and **Section 7-7 ANCHOR SPECIFICATION FOR MR EQUIPMENT INSIDE RF SHIELDED ROOM.**

TABLE 2-10
MAGNET MOUNTING INFORMATION

MAGNET	MOUNTING ILLUSTRATION	COMMENTS
1.5T LCC (Active Shield)	2-11	The magnet must be bolted to the floor, refer to Section 4-15, Vibration. Bolt hole openings in the magnet base are to be used to anchor the magnet.

2-10-3 Operator Workspace Mounting Requirements

The Operator Workspace Table sets on the floor with the Workspace Cabinet positioned under the Table towards the right side, see Illustration 2-37. Note, the Workspace Table with the Color Host Monitor must be bolted to the floor for safe use of the table and equipment. The Color Host Monitor can be bolted to the Workspace Table to prevent the monitor being moved within the magnetic field.

2-11 COMPONENT DIMENSIONS

To assist in completing your room layout, refer to Table 2-11 for list of component Illustrations.

TABLE 2-11
MR SYSTEM COMPONENT ILLUSTRATIONS LIST

ILLUSTRATION NAME	ILLUSTRATION NUMBER
SIGNA TWINSPEED 1.5T LCC (ACTIVE SHIELD) MAGNET (MINIMUM SERVICE AREA)	2-8
SIGNA TWINSPEED 1.5T LCC (ACTIVE SHIELD) MAGNET ENCLOSURE (FRONT AND REAR VIEWS)	2-9
SIGNA TWINSPEED 1.5T LCC (ACTIVE SHIELD) MAGNET CABLE ACCESS	2-10
1.5T LCC (ACTIVE SHIELD) MAGNET LOAD PATTERN	2-11
SHIELD/CRYO COOLER COMPRESSOR CABINET (MS5)	2-12
SIGNA TWINSPEED MAIN DISCONNECT PANEL (MDP)	2-13
OUTDOOR AIR COOLED TWIN SYSTEM COOLING CABINET [MODEL LC 20MT/O] (TSCC) FOR SHIELD/CRYO COOLER COMPRESSOR & GRADIENT COIL COOLING WATER	2-14
OUTDOOR AIR COOLED TSCC [MODEL LC 20MT/O] MOUNTING	2-15
REMOTE CONTROL PANEL (RCP) FOR OUTDOOR AIR COOLED TSCC [MODEL LC 20MT/O]	2-16
INDOOR WATER COOLED TWIN SYSTEM COOLING CABINET [MODEL LC-18MT/WC] (TSCC) FOR SHIELD/CRYO COOLER COMPRESSOR & GRADIENT COIL COOLING WATER	2-17
INDOOR WATER COOLED TSCC [MODEL LC 18MT/WC] MOUNTING	2-18
INDOOR AIR COOLED TWIN SYSTEM COOLING CABINET [MODEL LC 20MT] (TSCC) FOR SHIELD/CRYO COOLER COMPRESSOR & GRADIENT COIL COOLING WATER	2-19
INDOOR AIR COOLED TSCC MOUNTING	2-20
OUTDOOR AIR COOLED TWIN GRADIENT WATER COOLER [MODEL LC 10MTO] (TGWC) FOR GRADIENT COIL COOLING WATER ONLY	2-21
OUTDOOR AIR COOLED TGWC [MODEL LC 10MTO] MOUNTING	2-22
REMOTE CONTROL PANEL (RCP) FOR OUTDOOR AIR COOLED TGWC [MODEL LC 10MTO]	2-23
INDOOR WATER COOLED TWIN GRADIENT WATER COOLER [MODEL LTL 10] (TGWC) FOR GRADIENT COIL COOLING WATER ONLY	2-24
REMOTE CONTROL PANEL (RCP) FOR INDOOR WATER COOLED TGWC [MODEL LTL 10]	2-25
INDOOR AIR COOLED TWIN GRADIENT WATER COOLER [MODEL LC 10M] (TGWC) FOR GRADIENT COIL COOLING WATER ONLY	2-26
BLOWER BOX (MG4)	2-27
1.5T SRFD II CABINET (MR1)	2-28
ACGD/PDU CABINET (MR3)	2-29
SYSTEM CABINET (MR2) FOR SIGNA TWINSPEED WITH EXCITE TECHNOLOGY	2-30
SYSTEM CABINET (MR2) FOR SIGNA TWINSPEED	2-31
TWIN ACCESSORY CABINET (TAC)	2-32
MAGNET MONITOR (MSM1)	2-33
PATIENT TRANSPORT TABLE (PT1)	2-34
PENETRATION PANEL (PP1)	2-35
PENETRATION PANEL COVER	2-36
OPERATOR WORKSPACE (OW1)	2-37
OPERATOR WORKSPACE CABINET (OW1 A2)	2-38
OPERATOR WORKSPACE COMPONENTS POSITIONED ON TABLE TOP – LCD COLOR MONITOR OPTION	2-39
(Continued)	

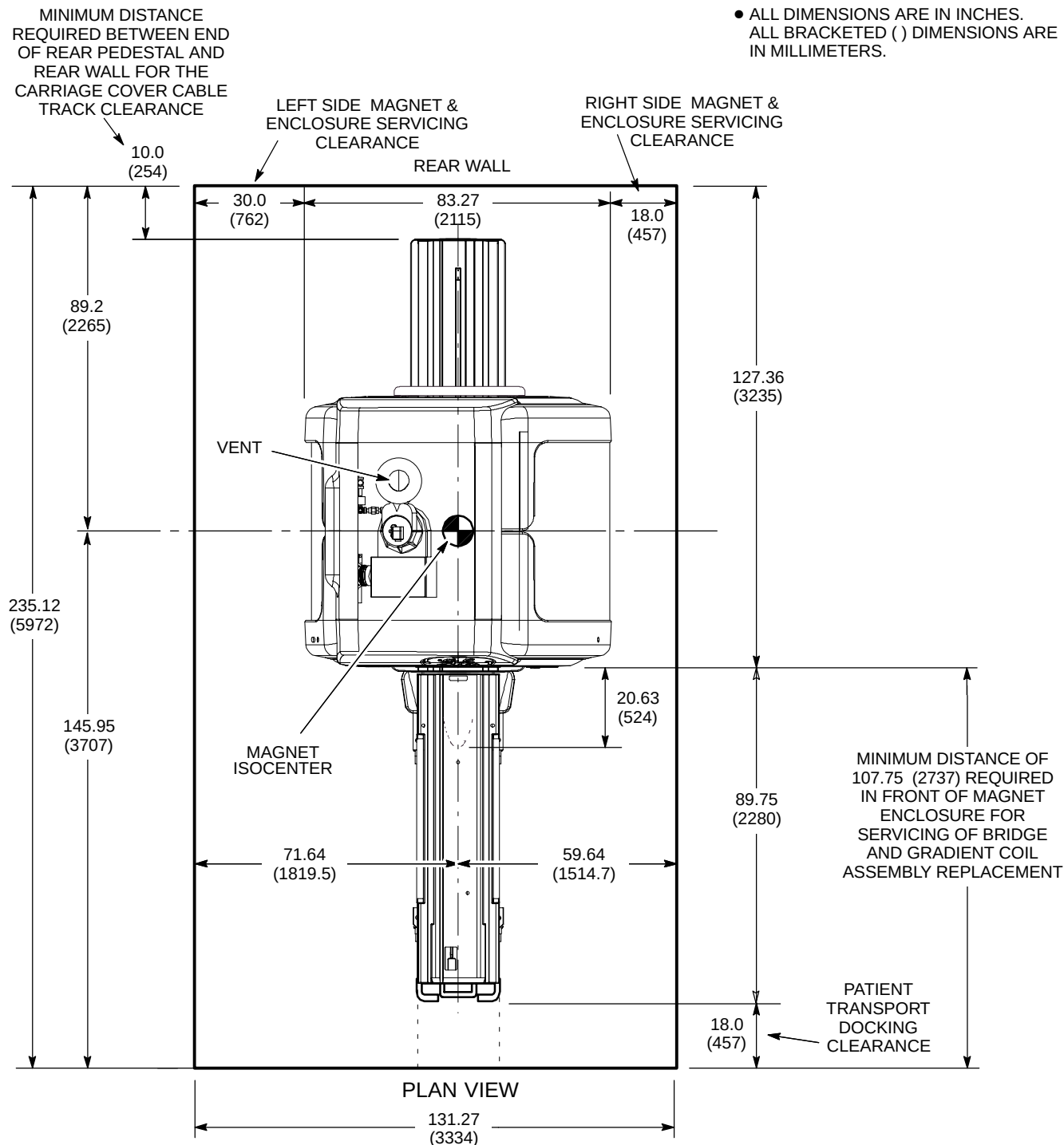
2-11 COMPONENT DIMENSIONS (Continued)

TABLE 2-11
MR SYSTEM COMPONENT ILLUSTRATIONS LIST (Continued)

ILLUSTRATION NAME	ILLUSTRATION NUMBER
OPERATOR WORKSPACE COMPONENTS POSITIONED ON TABLE TOP – OCTANE COMPUTER	2-40
OPERATOR WORKSPACE COMPONENTS POSITIONED ON TABLE TOP – KEYBOARD	2-41
PNEUMATIC PATIENT ALERT CONTROL BOX (PA1)	2-42
MAGNET RUNDOWN UNIT (MS4)	2-43
SPT PHANTOM SET SHIPPING/STORAGE CART	2-44
UPS FOR MAGNET MONITOR – STANDARD FOR TRUE MOBILE CONFIGURATION, OPTIONAL FOR OTHER CONFIGURATIONS	2-45
DC LIGHTING CONTROLLER – OPTIONAL	2-46
OXYGEN MONITOR (OM1) – OPTIONAL	2-47
REMOTE OXYGEN SENSOR MODULE (OM3) – OPTIONAL	2-48
MNS CABINET (MNS) – OPTIONAL	2-49
BRAINWAVE AUDIO CONSOLE – BRAINWAVE OPTION	2-50
BRAINWAVE CABINET – BRAINWAVE OPTION	2-51
BRAINWAVE CART – BRAINWAVE OPTION	2-52

NOTE:

- ALL DIMENSIONS ARE IN INCHES.
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.

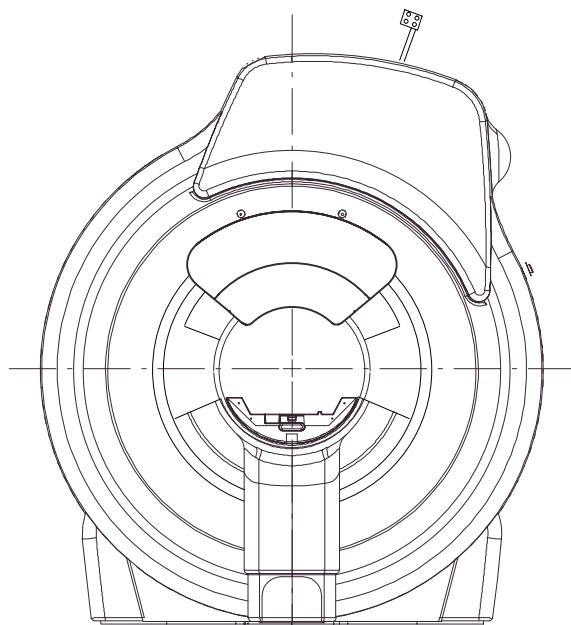


SIGNA TWINSPEED 1.5T LCC (ACTIVE SHIELD) MAGNET (MINIMUM SERVICE AREA)

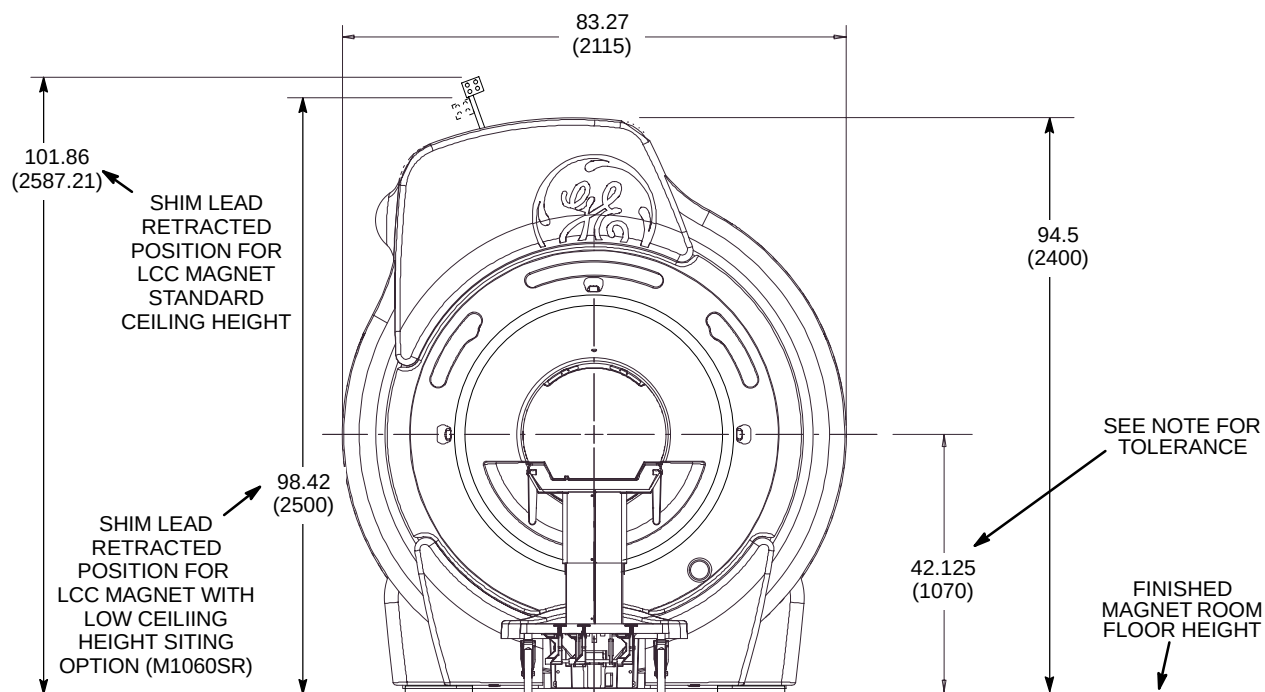
ILLUSTRATION 2-8

NOTE:

- ALL DIMENSIONS ARE IN INCHES.
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- FINISHED FLOOR TO MAGNET CENTER LINE HEIGHT DIMENSION MUST BE 42.125 ± 0.25 in. (1070 ± 6.35 mm) TO ALLOW PATIENT TABLE TO PROPERLY DOCK TO THE ENCLOSURE.



REAR VIEW



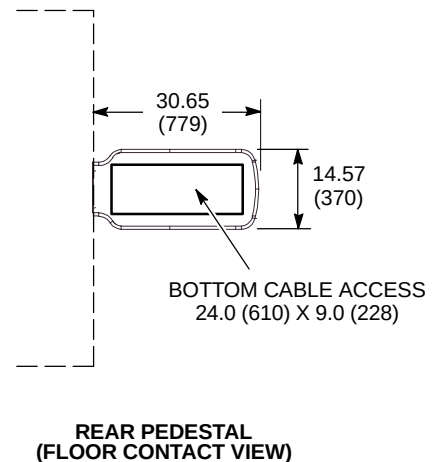
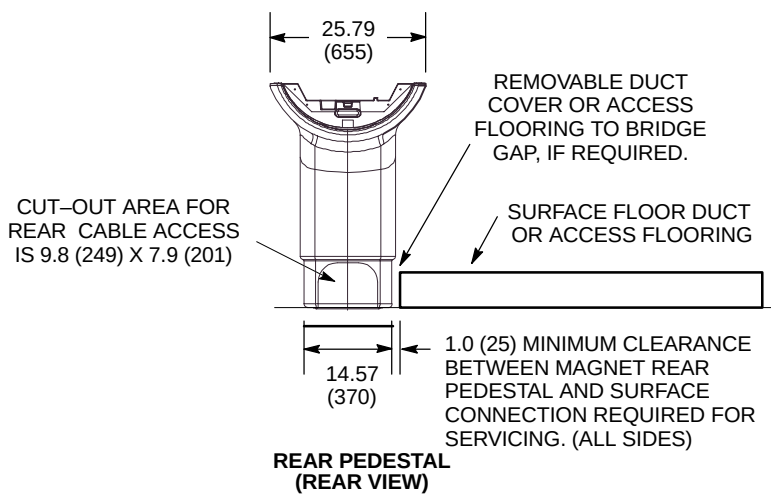
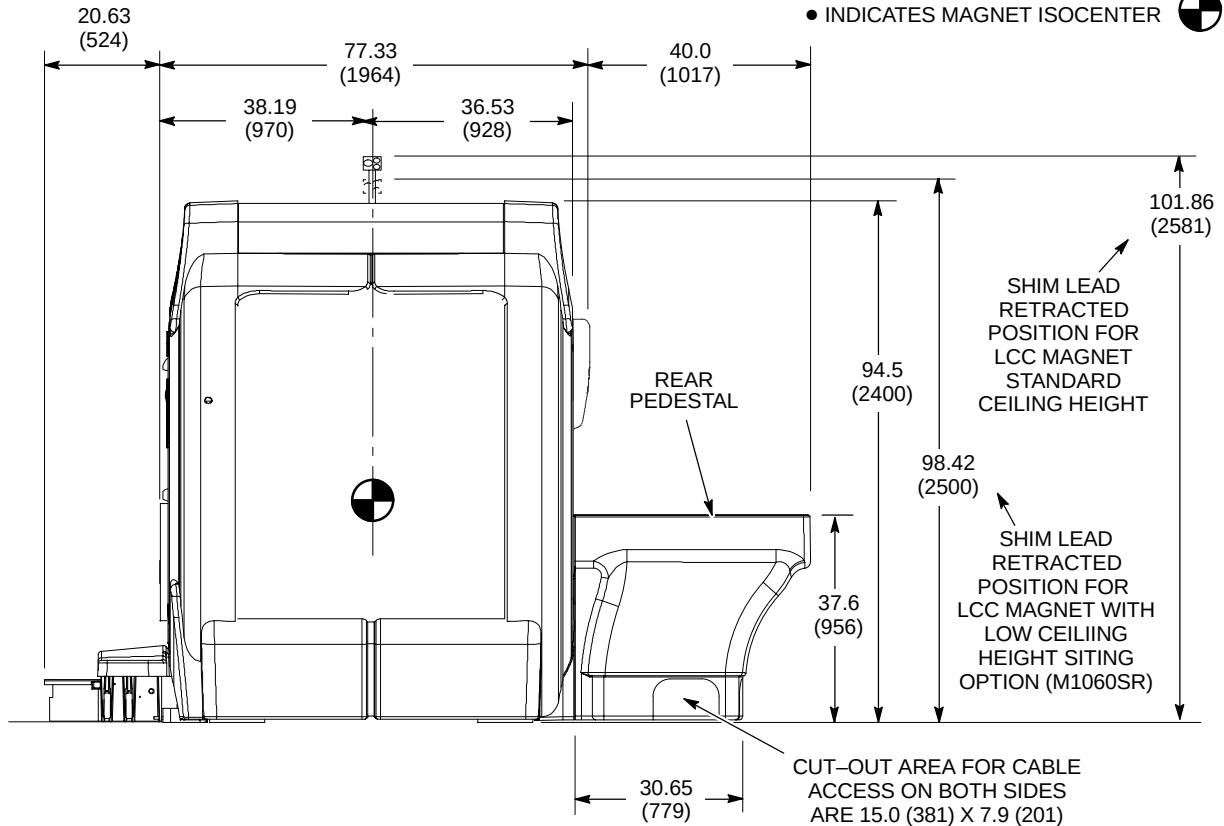
FRONT VIEW

SIGNA TWINSPEED 1.5T LCC (ACTIVE SHIELD) MAGNET ENCLOSURE (FRONT AND REAR VIEWS)

ILLUSTRATION 2-9

NOTE:

- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- INDICATES MAGNET ISOCENTER

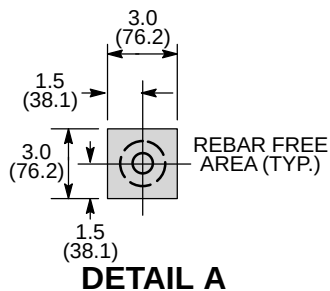


SIGNA TWINSPEED 1.5T LCC (ACTIVE SHIELD) MAGNET CABLE ACCESS
ILLUSTRATION 2-10

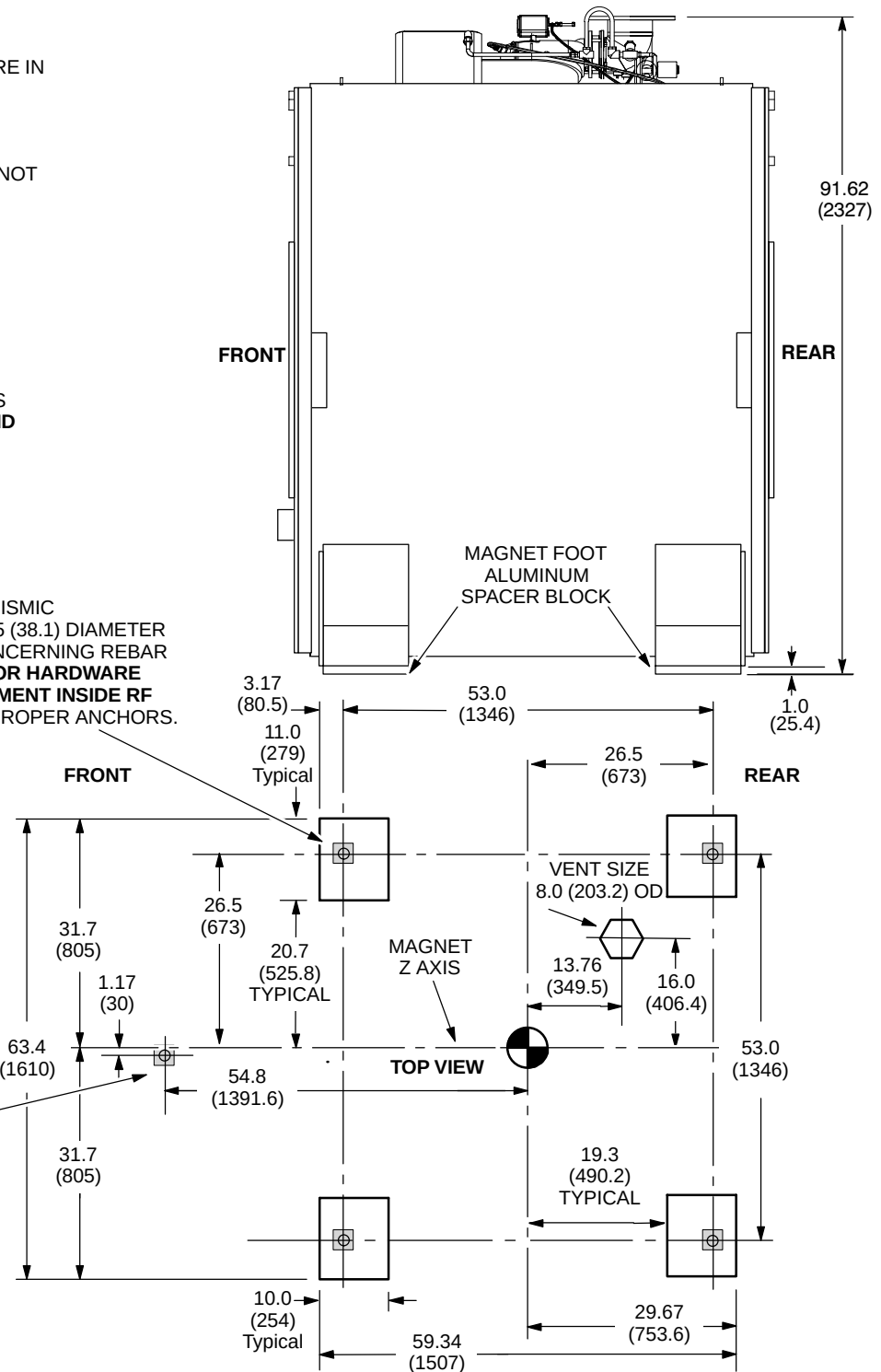
NOTE:

- ALL DIMENSIONS ARE IN INCHES.
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- LIFTING/JACKING BEAMS ARE ATTACHED TO SIDES OF MAGNET. REFER TO ILLUSTRATION 8-1.
- ALUMINUM LEVELING PLATES ARE NOT SHOWN.
- APPROX. WEIGHT OF MAGNET, CRYOGENS, COILS, AND ENCLOSURE WITH TRM 12,826 lbs (5818 kg)
- STEEL REBAR MUST NOT BE POSITIONED IN SHADED AREAS TO PREVENT INTERFERENCE WITH MOUNTING BOLTS.
- FOR MAGNET MOVING DIMENSIONS REFER TO **SECTION 8 SHIPPING AND DELIVERY DATA**

SEISMIC AND NON-SEISMIC
MAGNET MOUNTING HOLES (4) 1.5 (38.1) DIAMETER
SEE NOTE ABOVE & DETAIL A CONCERNING REBAR
REFER TO **SECTION 7-7 ANCHOR HARDWARE REQUIREMENTS FOR MR EQUIPMENT INSIDE RF SHIELDED ROOM** TO DETERMINE PROPER ANCHORS.



M10 ANCHOR FOR
TABLE DOCK MOUNTING,
LOCATED OFFSET FROM
MAGNET CENTER LINE.
FOR ADDITIONAL
MOUNTING INFORMATION
REFER TO **SECTION 7-7 ANCHOR HARDWARE REQUIREMENTS FOR MR EQUIPMENT INSIDE RF SHIELDED ROOM.**

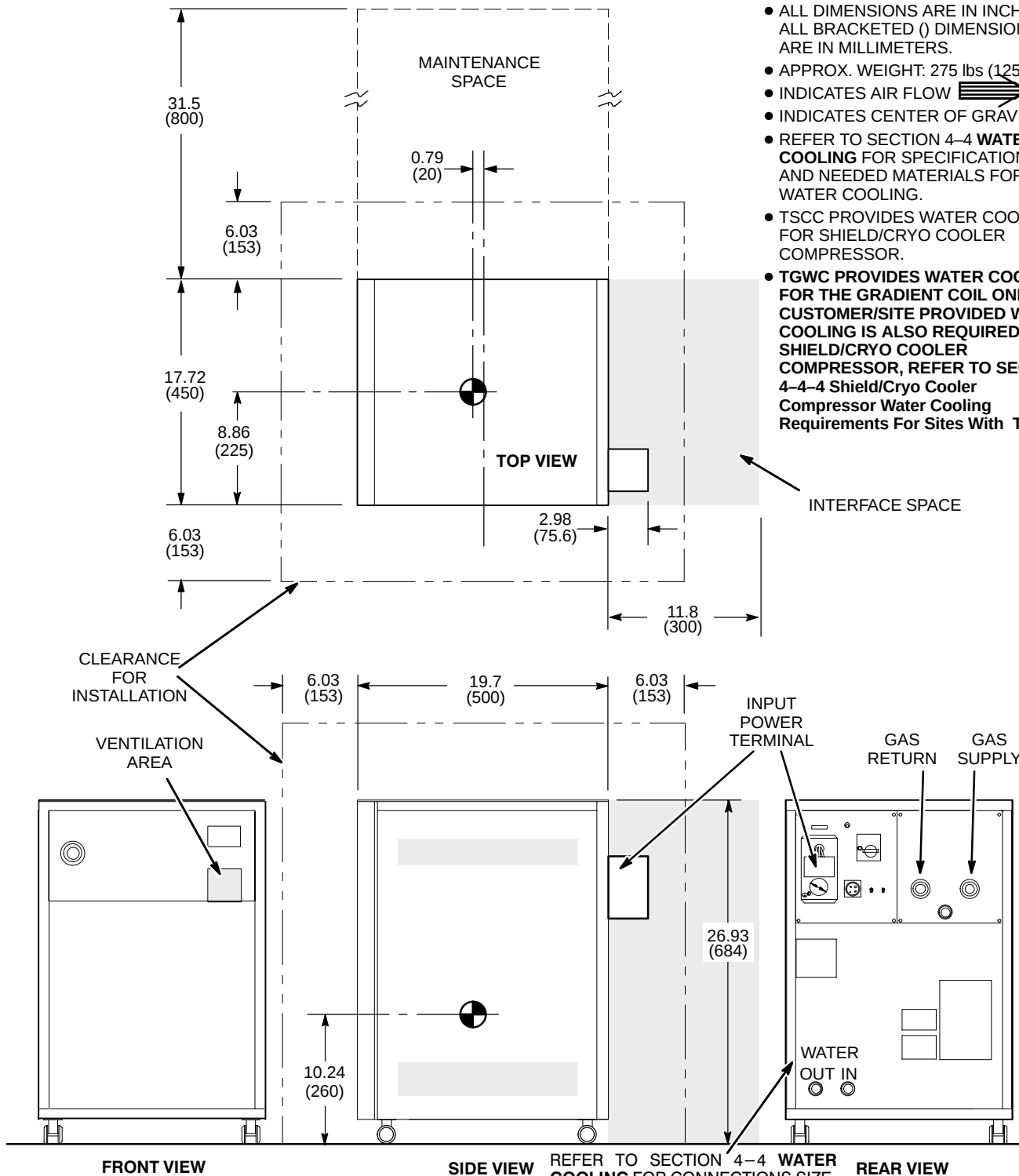


1.5T LCC (ACTIVE SHIELD) MAGNET LOAD PATTERN

ILLUSTRATION 2-11

NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS
ARE IN MILLIMETERS.
- APPROX. WEIGHT: 275 lbs (125 kg)
- INDICATES AIR FLOW 
- INDICATES CENTER OF GRAVITY 
- REFER TO SECTION 4-4 **WATER COOLING** FOR SPECIFICATIONS AND NEEDED MATERIALS FOR WATER COOLING.
- TSCC PROVIDES WATER COOLING FOR SHIELD/CRYO COOLER COMPRESSOR.
- TGWC PROVIDES WATER COOLING FOR THE GRADIENT COIL ONLY. CUSTOMER/SITE PROVIDED WATER COOLING IS ALSO REQUIRED FOR SHIELD/CRYO COOLER COMPRESSOR, REFER TO SECTION 4-4-4 Shield/Cryo Cooler Compressor Water Cooling Requirements For Sites With TGWC.

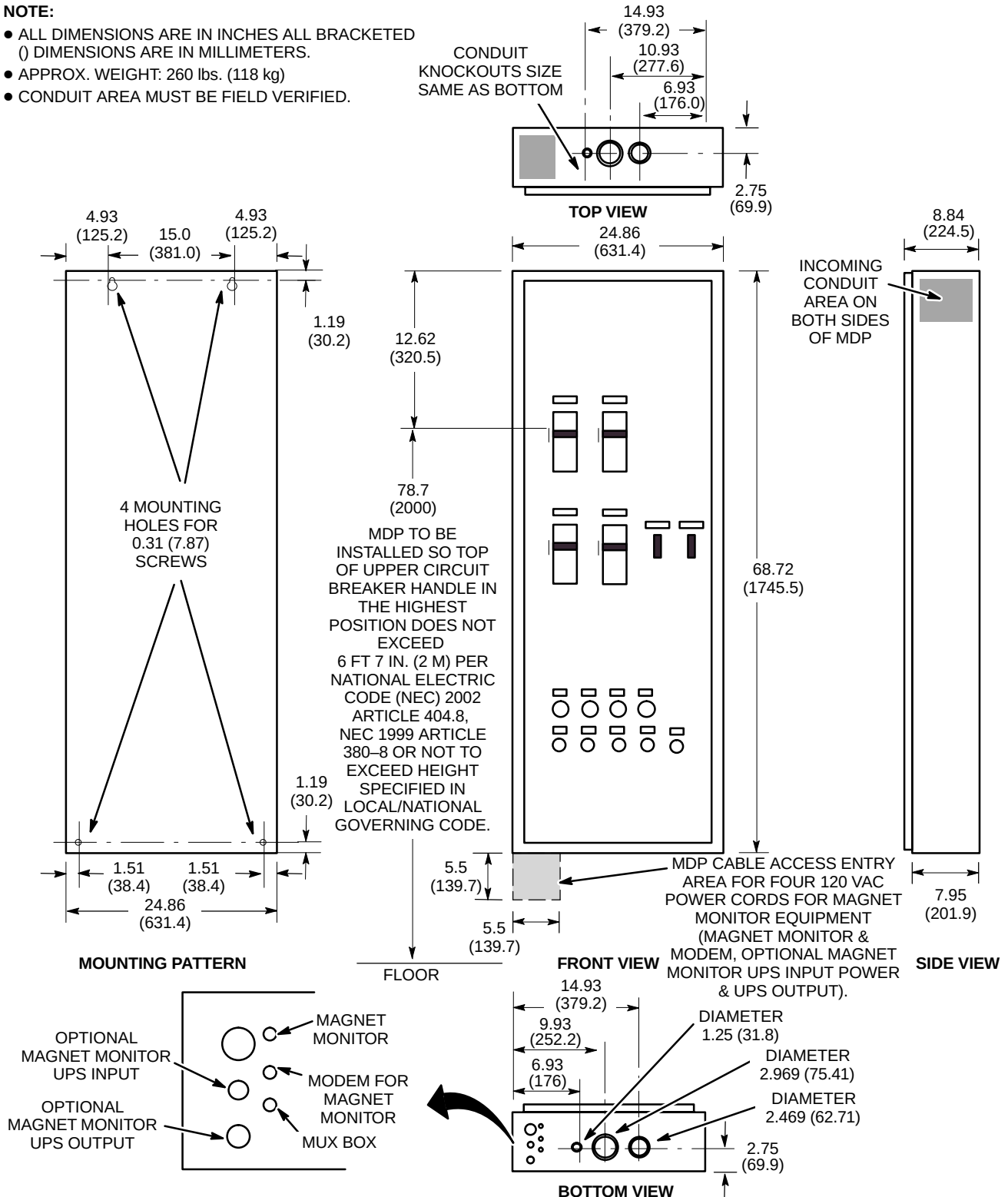


SHIELD/CRYO COOLER COMPRESSOR CABINET (MS5)

ILLUSTRATION 2-12

NOTE:

- ALL DIMENSIONS ARE IN INCHES ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 260 lbs. (118 kg)
- CONDUIT AREA MUST BE FIELD VERIFIED.

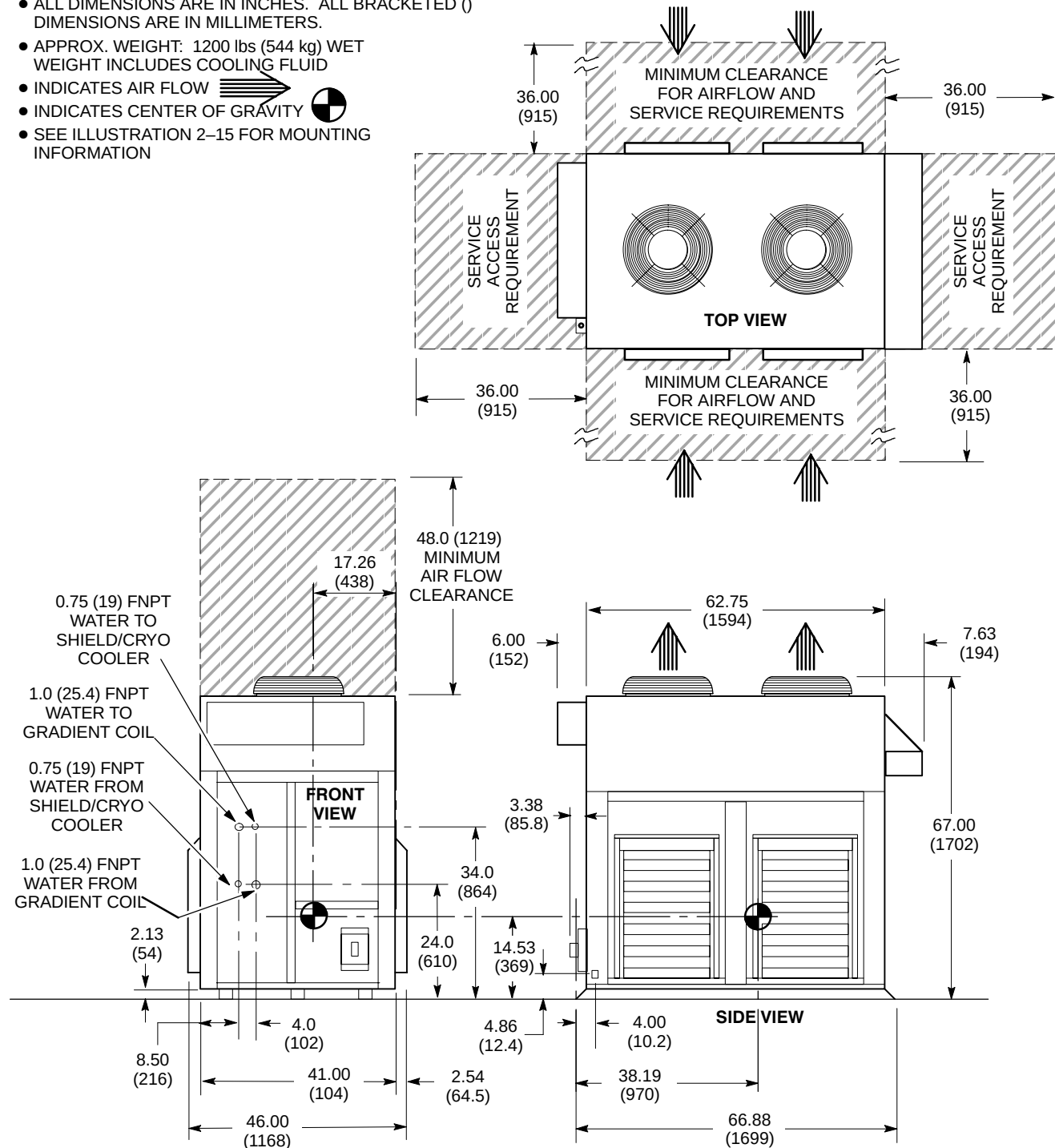


SIGNA TWINSPEED MAIN DISCONNECT PANEL (MDP) M3088TH

ILLUSTRATION 2-13

NOTE:

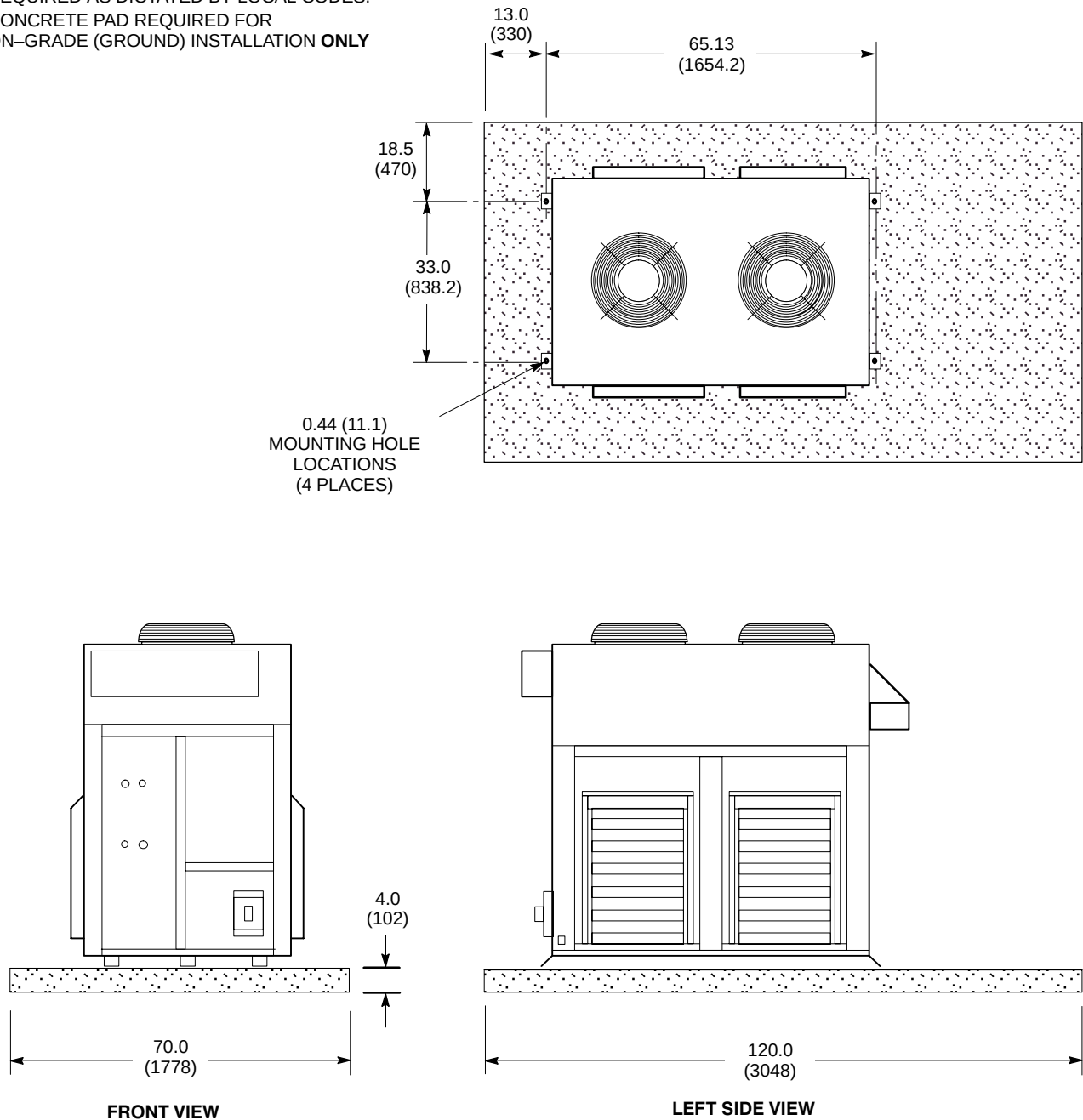
- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 1200 lbs (544 kg) WET WEIGHT INCLUDES COOLING FLUID
- INDICATES AIR FLOW 
- INDICATES CENTER OF GRAVITY 
- SEE ILLUSTRATION 2-15 FOR MOUNTING INFORMATION



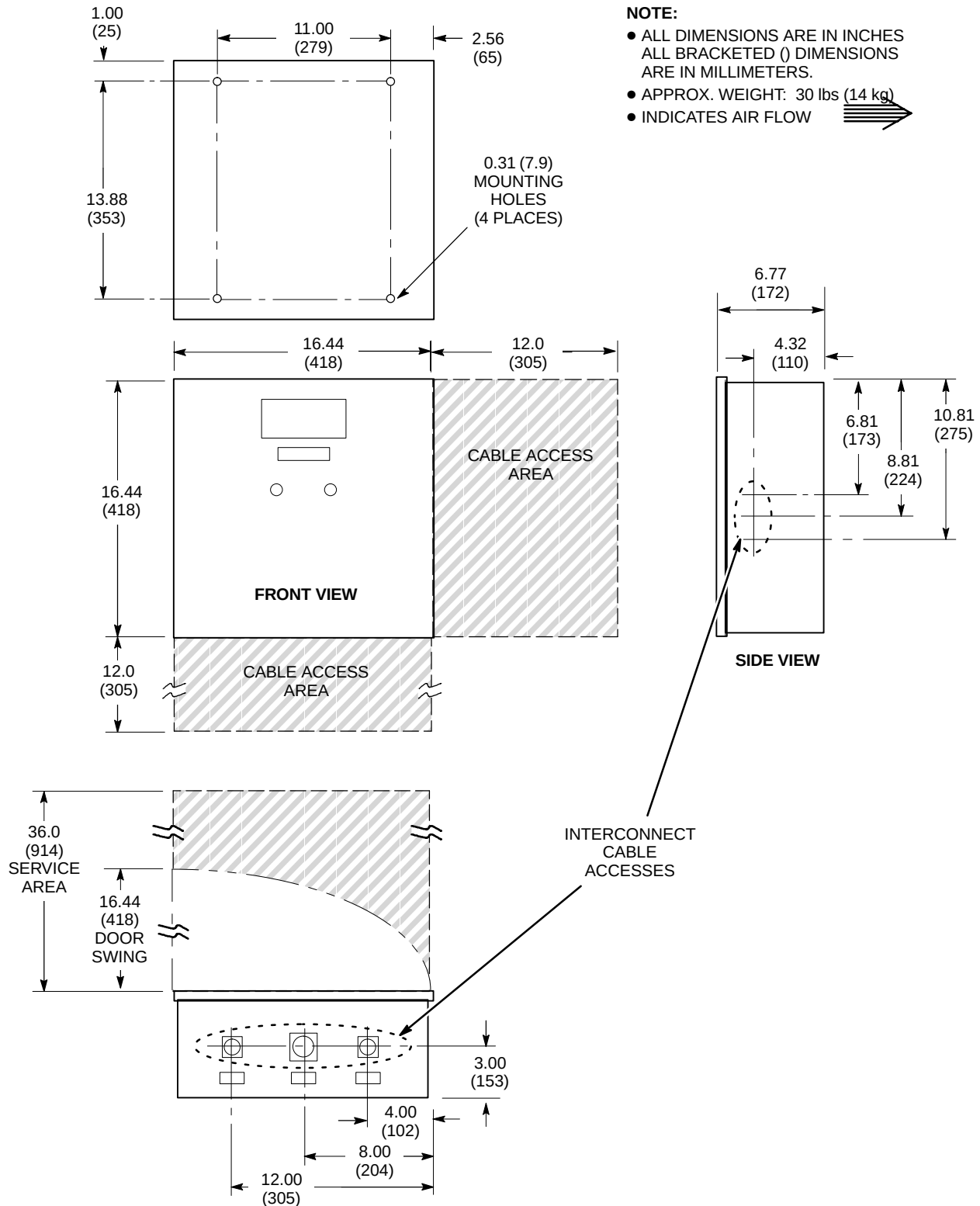
OUTDOOR AIR COOLED TWIN SYSTEM COOLING CABINET [MODEL LC 20MT/O] (TSCC)
FOR SHIELD/CRYO COOLER COMPRESSOR & GRADIENT COIL COOLING WATER
ILLUSTRATION 2-14

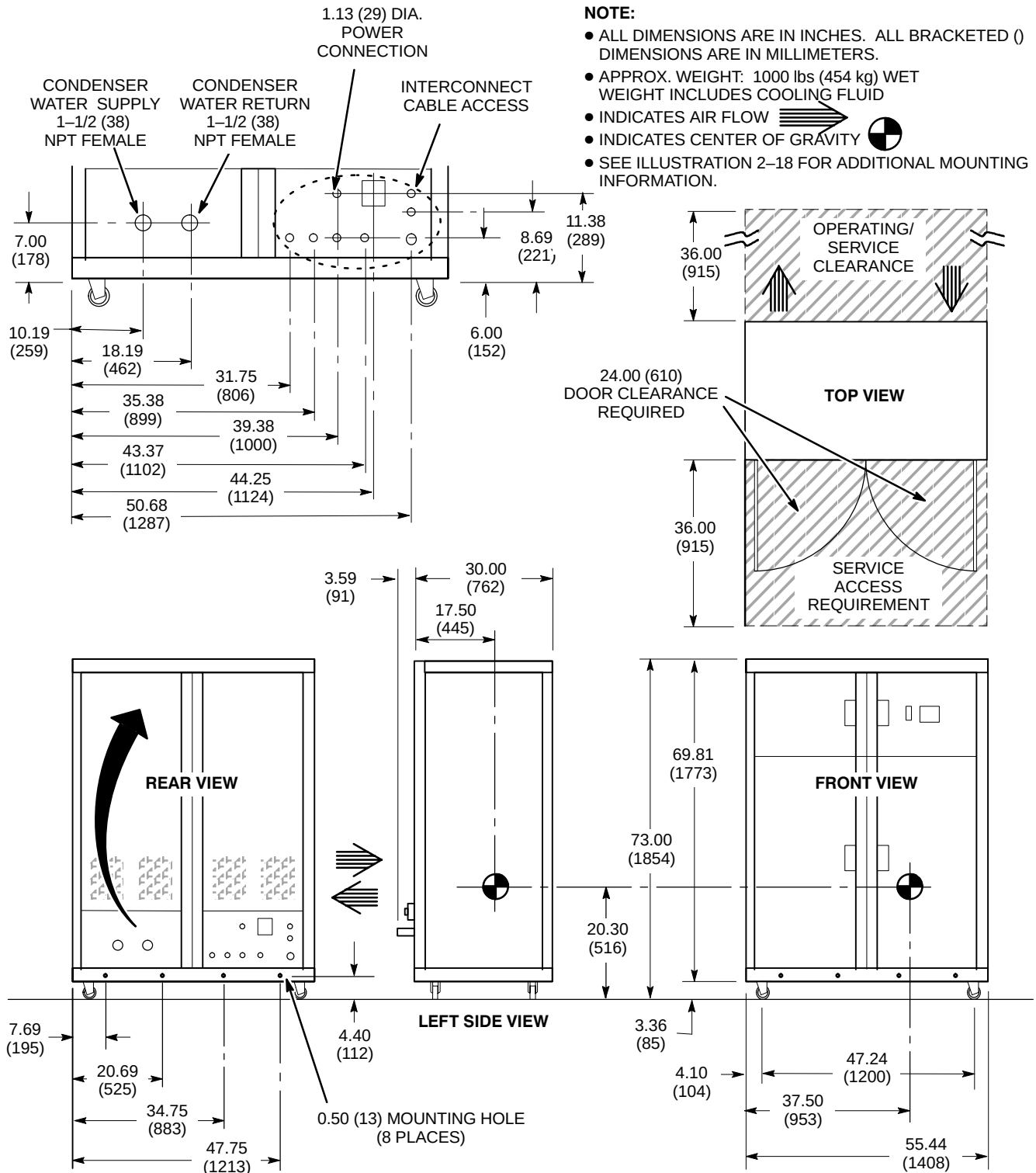
NOTE:

- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- MINIMUM 4 INCH (102 MM) DEEP CONCRETE PAD OF 2500 PSI CONCRETE REQUIRED. ADDITIONAL CONCRETE DEPTH MAY BE REQUIRED AS DICTATED BY LOCAL CODES.
- CONCRETE PAD REQUIRED FOR ON-GRADE (GROUND) INSTALLATION **ONLY**



OUTDOOR AIR COOLED TSCC [MODEL LC 20MT/O] MOUNTING
ILLUSTRATION 2-15

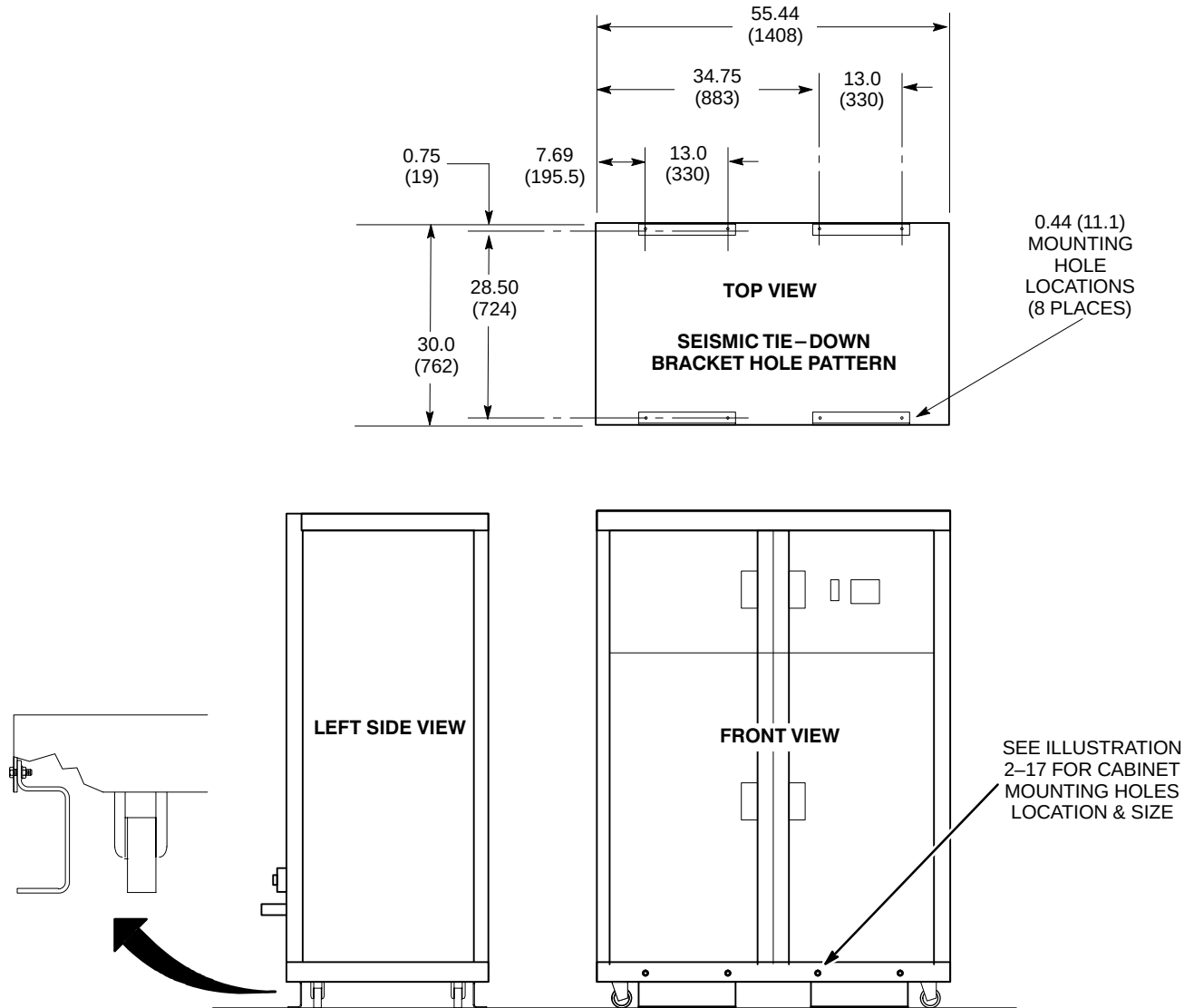




INDOOR WATER COOLED TWIN SYSTEM COOLING CABINET [MODEL LC 18MT/WC] (TSCC)
FOR SHIELD/CRYO COOLER COMPRESSOR & GRADIENT COIL COOLING WATER
ILLUSTRATION 2-17

NOTE:

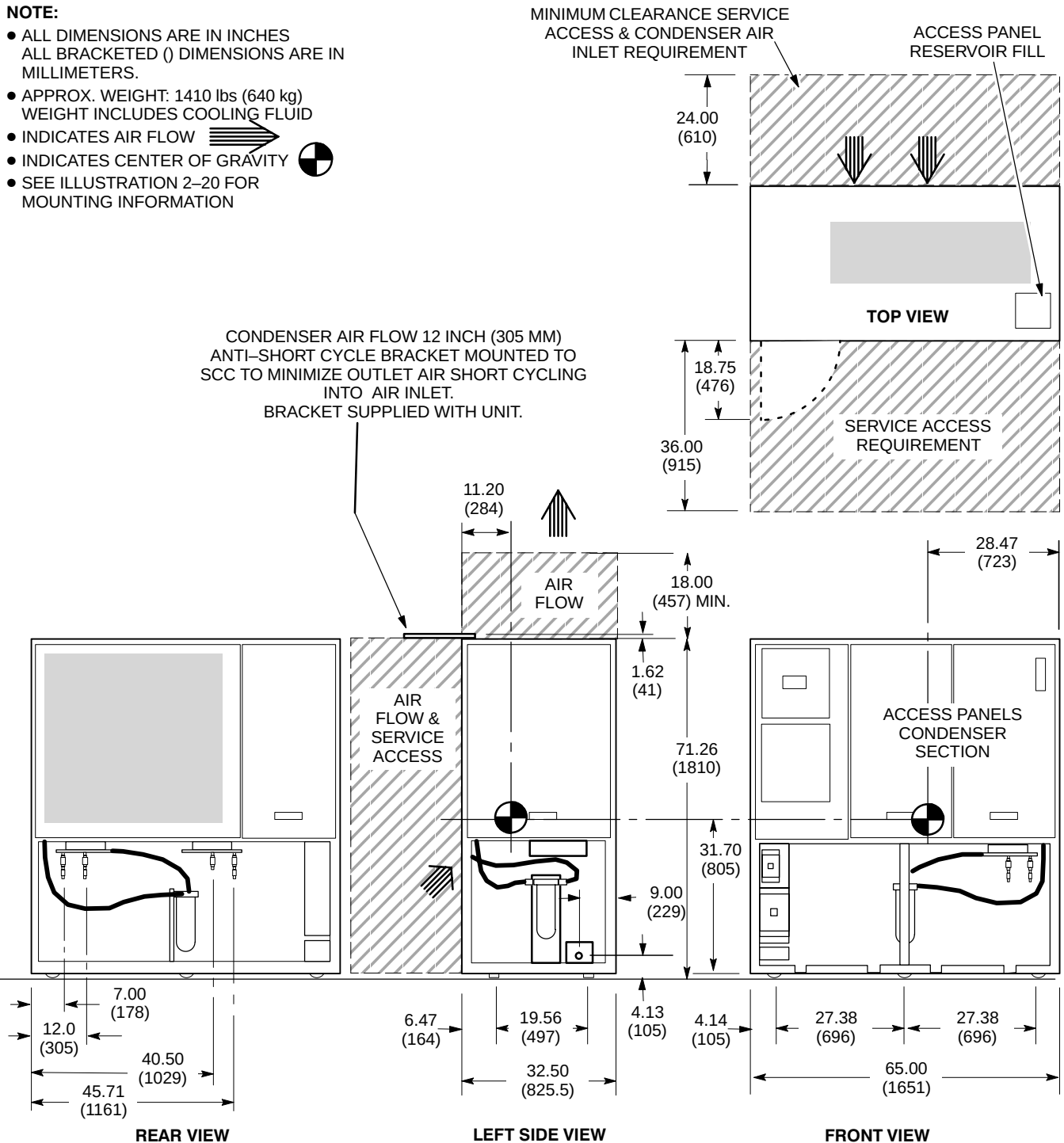
- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.



INDOOR WATER COOLED TSCC [MODEL LC 18MT/WC] MOUNTING
ILLUSTRATION 2-18

NOTE:

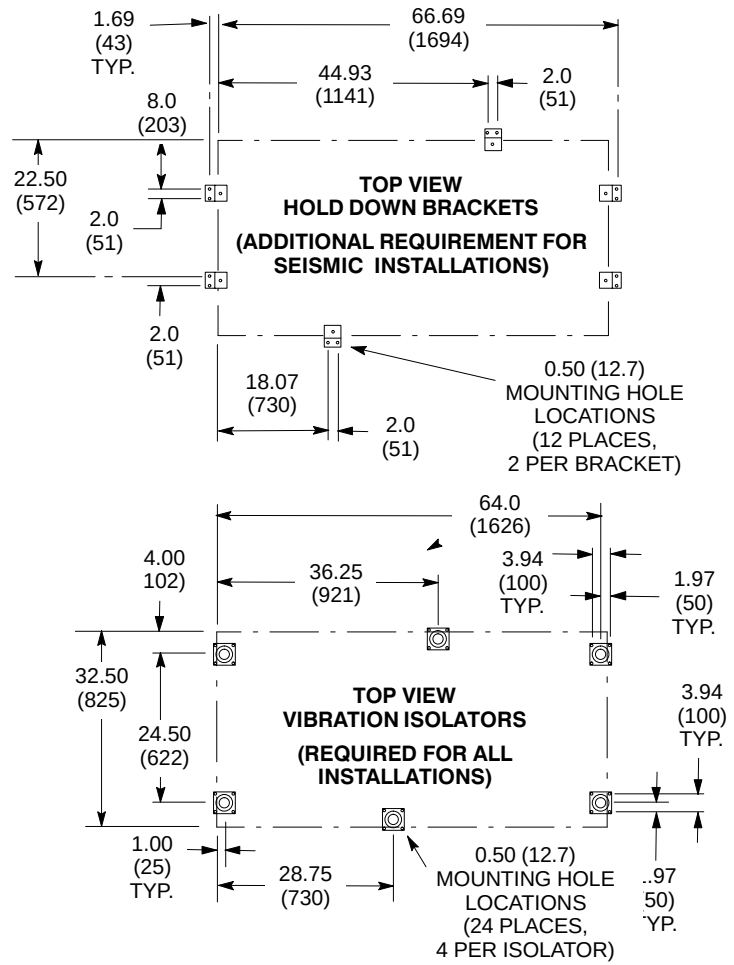
- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 1410 lbs (640 kg)
WEIGHT INCLUDES COOLING FLUID
- INDICATES AIR FLOW 
- INDICATES CENTER OF GRAVITY 
- SEE ILLUSTRATION 2-20 FOR MOUNTING INFORMATION



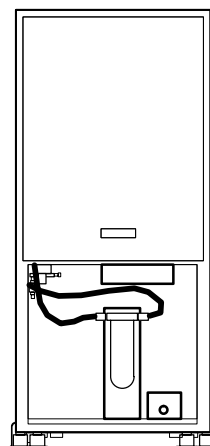
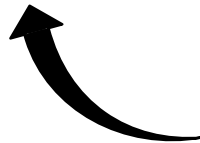
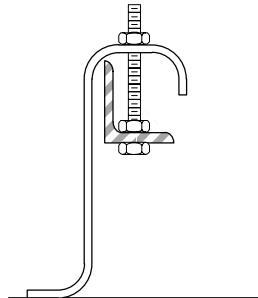
INDOOR AIR COOLED TWIN SYSTEM COOLING CABINET [MODEL LC 20MT] (TSCC)
FOR SHIELD/CRYO COOLER COMPRESSOR & GRADIENT COIL COOLING WATER
ILLUSTRATION 2-19

NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.

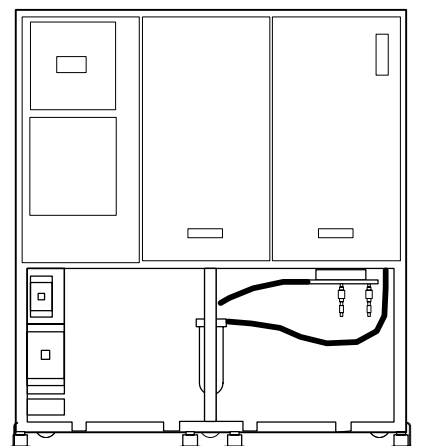


**HOLD DOWN BRACKET
DETAIL**



LEFT SIDE VIEW

3.8
(96.5)



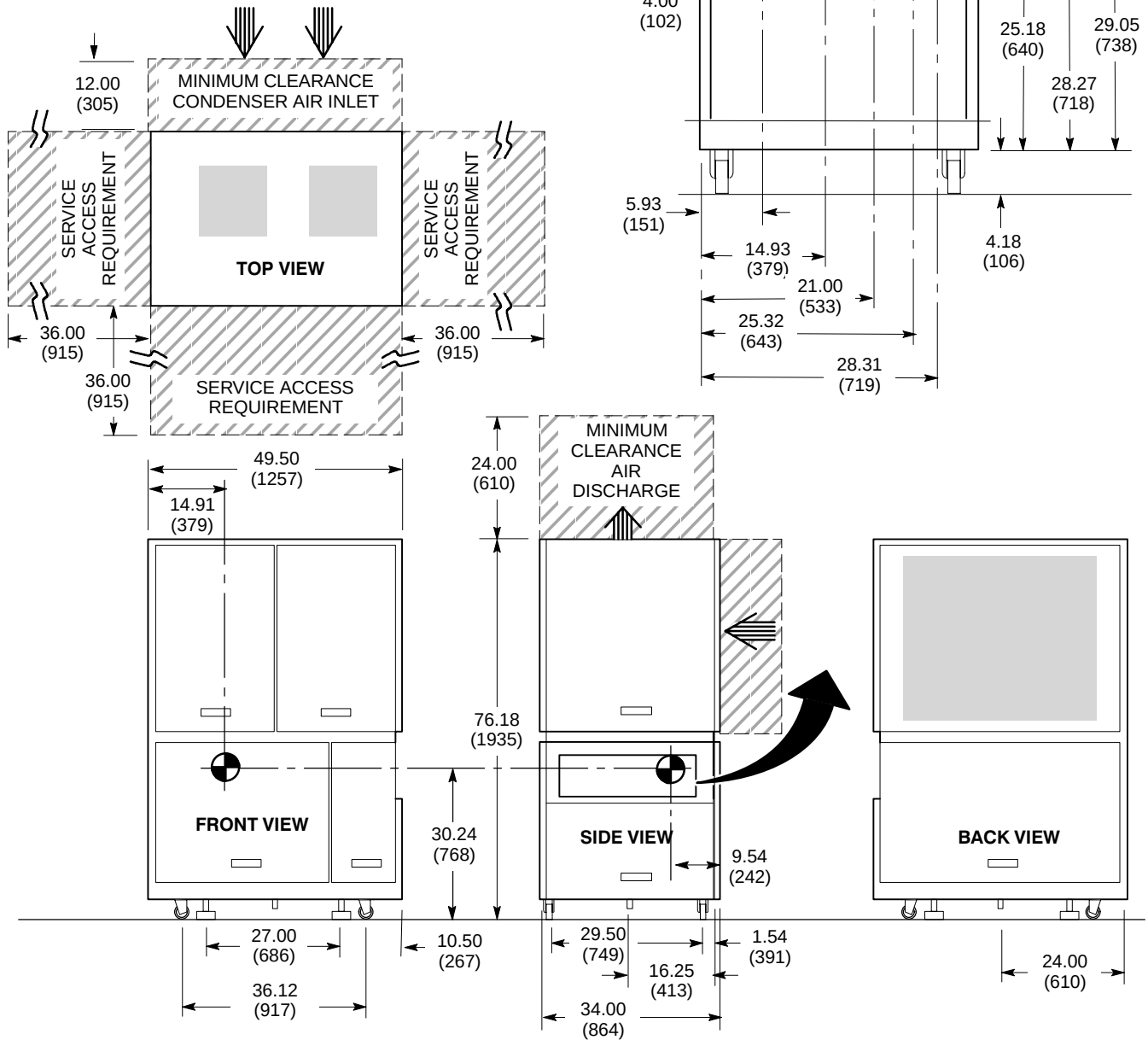
FRONT VIEW

INDOOR AIR COOLED TSCC MOUNTING

ILLUSTRATION 2-20

NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 780 lbs (345 kg) WEIGHT INCLUDES COOLING FLUID
- INDICATES AIR FLOW 
- INDICATES CENTER OF GRAVITY 
- SEE ILLUSTRATION 2-22 FOR MOUNTING INFORMATION
- TGWC PROVIDES WATER COOLING FOR THE GRADIENT COIL ONLY. CUSTOMER/SITE PROVIDED WATER COOLING IS ALSO REQUIRED FOR SHIELD/CRYO COOLER COMPRESSOR, REFER TO SECTION 4-4-4 Shield/Cryo Cooler Compressor Water Cooling Requirements For Sites With TGWC.

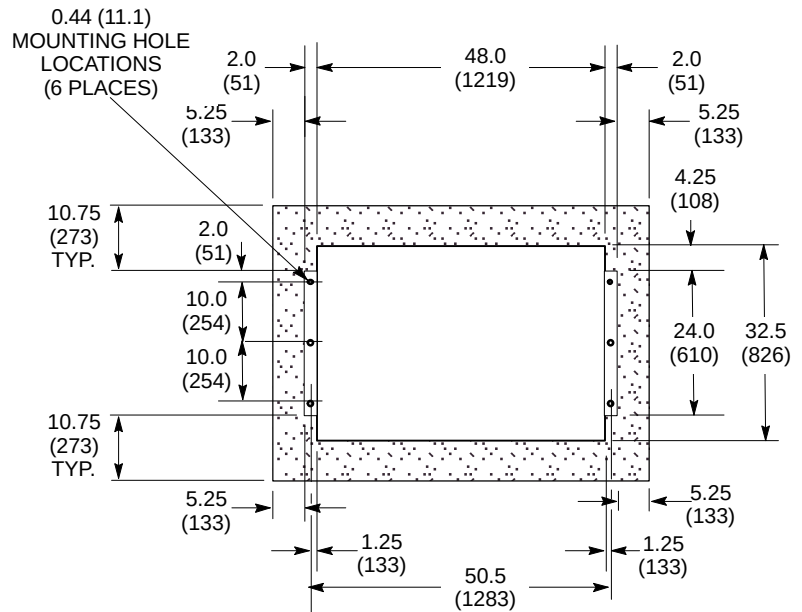


**OUTDOOR AIR COOLED TWIN GRADIENT WATER COOLER [MODEL LC 10MTO] (TGWC)
FOR GRADIENT COIL COOLING WATER ONLY**

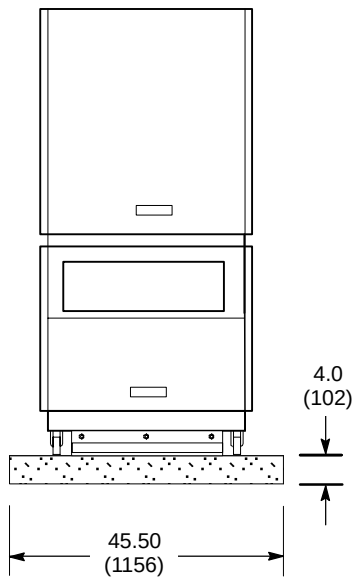
ILLUSTRATION 2-21

NOTE:

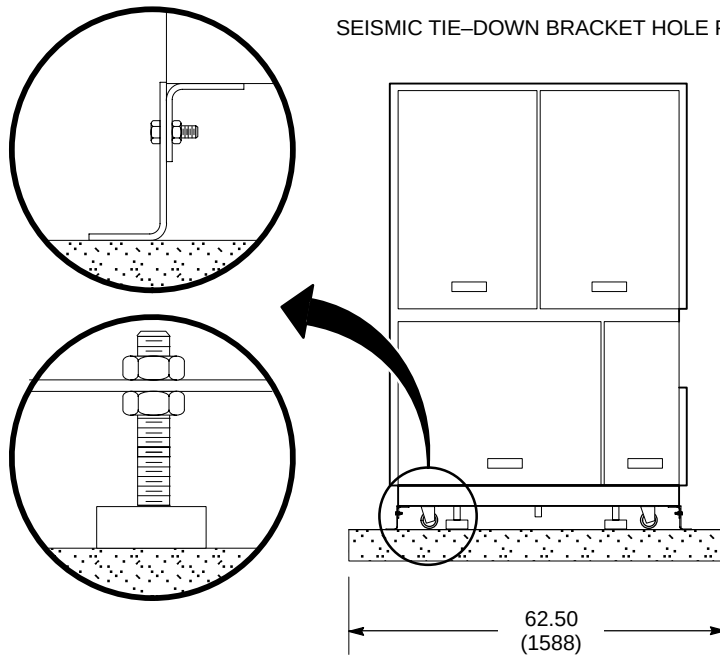
- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- MINIMUM 4 INCH (102 MM) DEEP CONCRETE PAD OF 2500 PSI CONCRETE REQUIRED. ADDITIONAL CONCRETE DEPTH MAY BE REQUIRED AS DICTATED BY LOCAL CODES.
- CONCRETE PAD REQUIRED FOR ON-GRADE (GROUND) INSTALLATION **ONLY**



SEISMIC TIE-DOWN BRACKET HOLE PATTERN

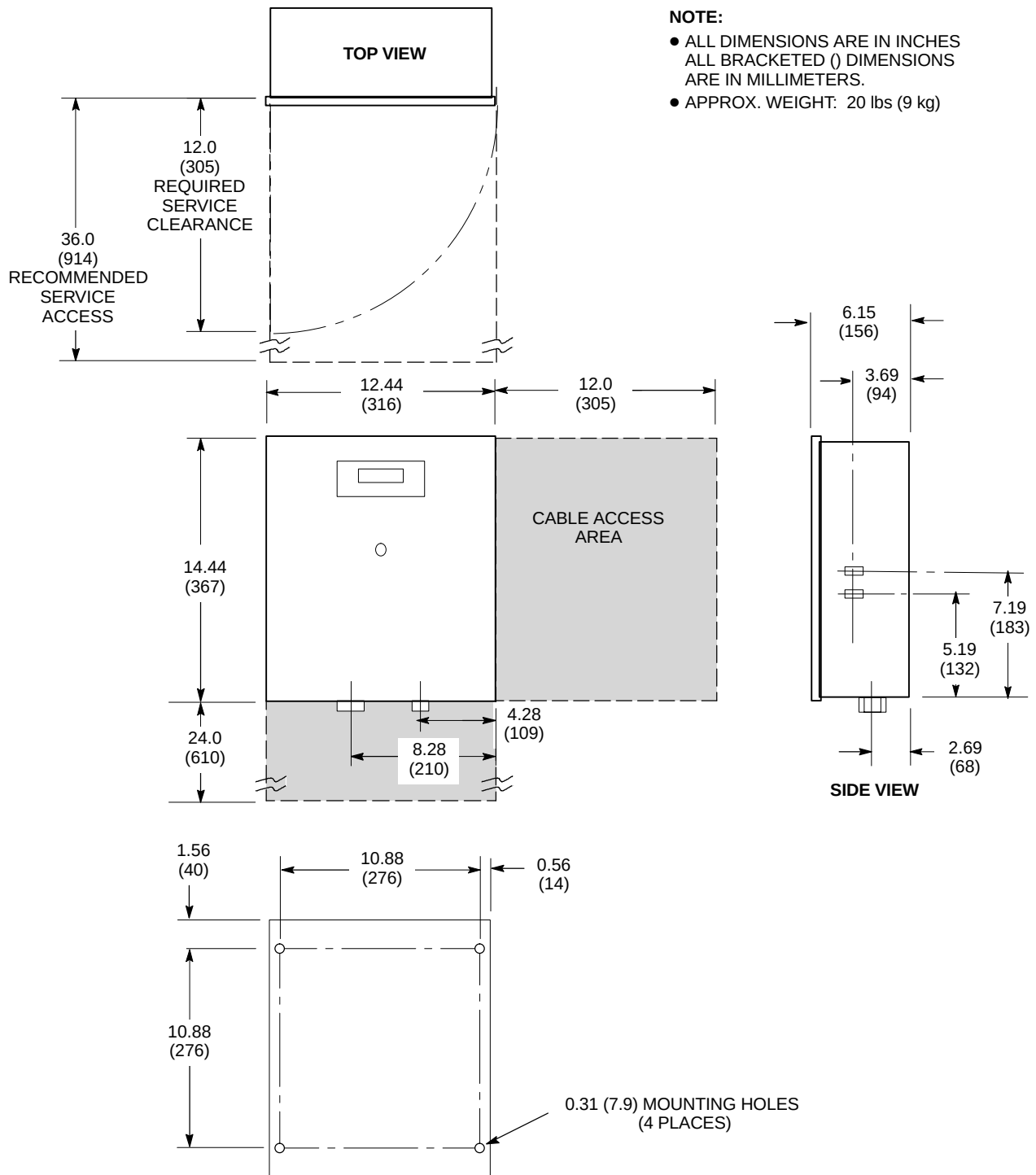


RIGHT SIDE VIEW



FRONT VIEW

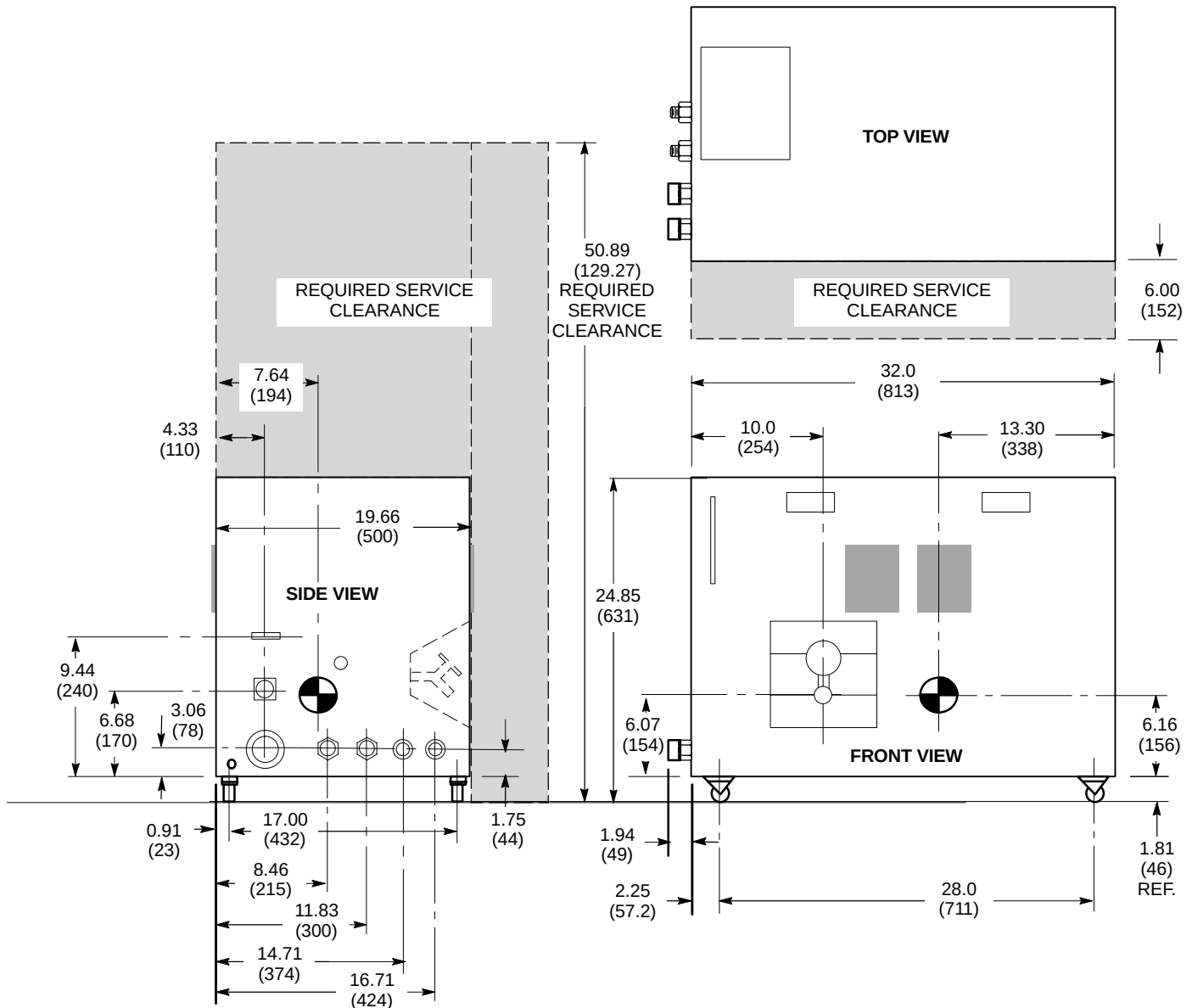
OUTDOOR AIR COOLED TGWC [MODEL LC 10MTO] MOUNTING
ILLUSTRATION 2-22



REMOTE CONTROL PANEL (RCP) FOR OUTDOOR AIR COOLED TGWC [MODEL LC 10MTO]
ILLUSTRATION 2-23

NOTE:

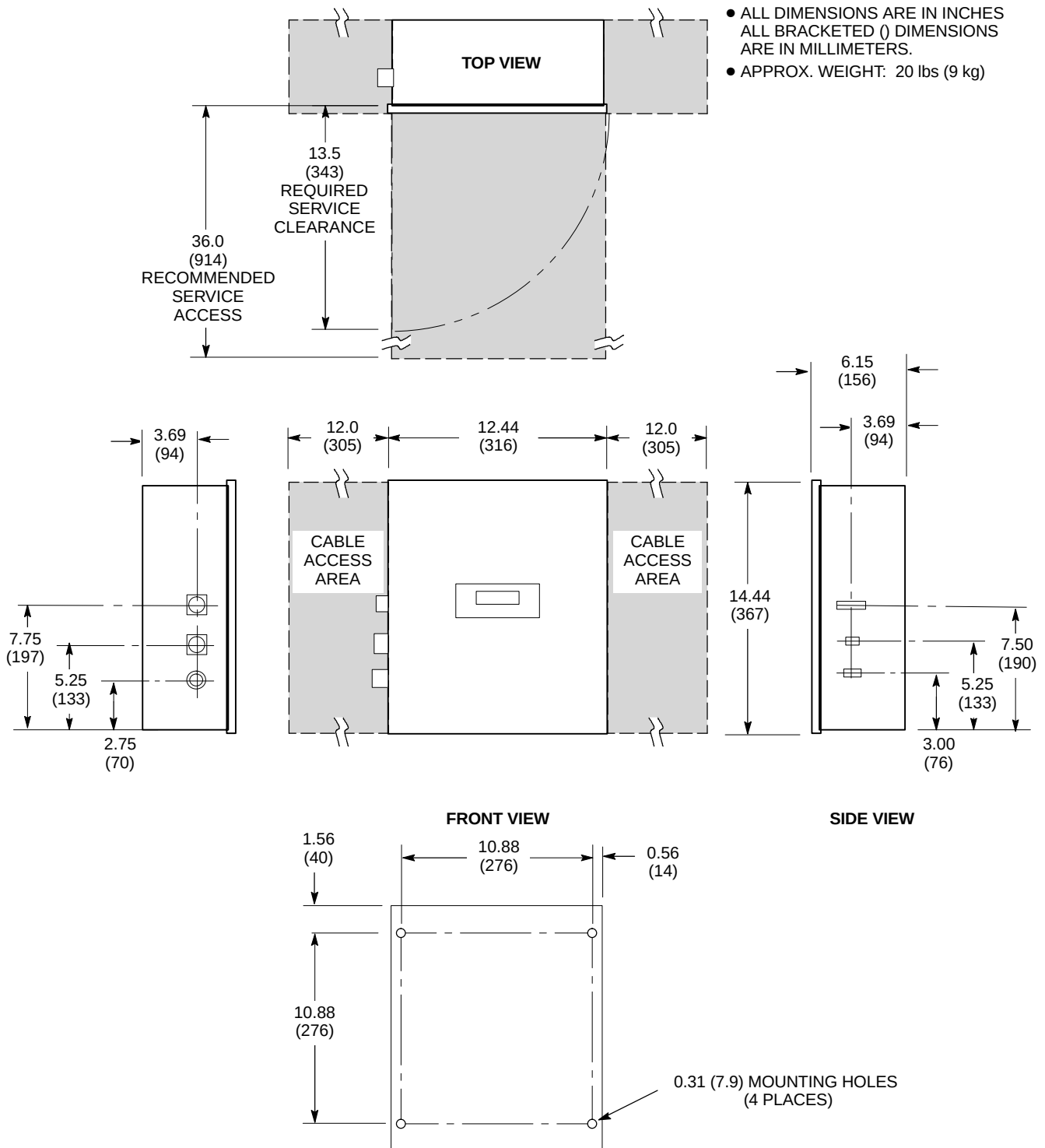
- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: TGWC 175 lbs (79 kg) WEIGHT INCLUDES COOLING FLUID
- INDICATES AIR FLOW
- INDICATES CENTER OF GRAVITY
- **TGWC PROVIDES WATER COOLING FOR THE GRADIENT COIL ONLY. CUSTOMER/SITE PROVIDED WATER COOLING IS ALSO REQUIRED FOR SHIELD/CRYO COOLER COMPRESSOR, REFER TO SECTION 4-4-4 Shield/Cryo Cooler Compressor Water Cooling Requirements For Sites With TGWC.**



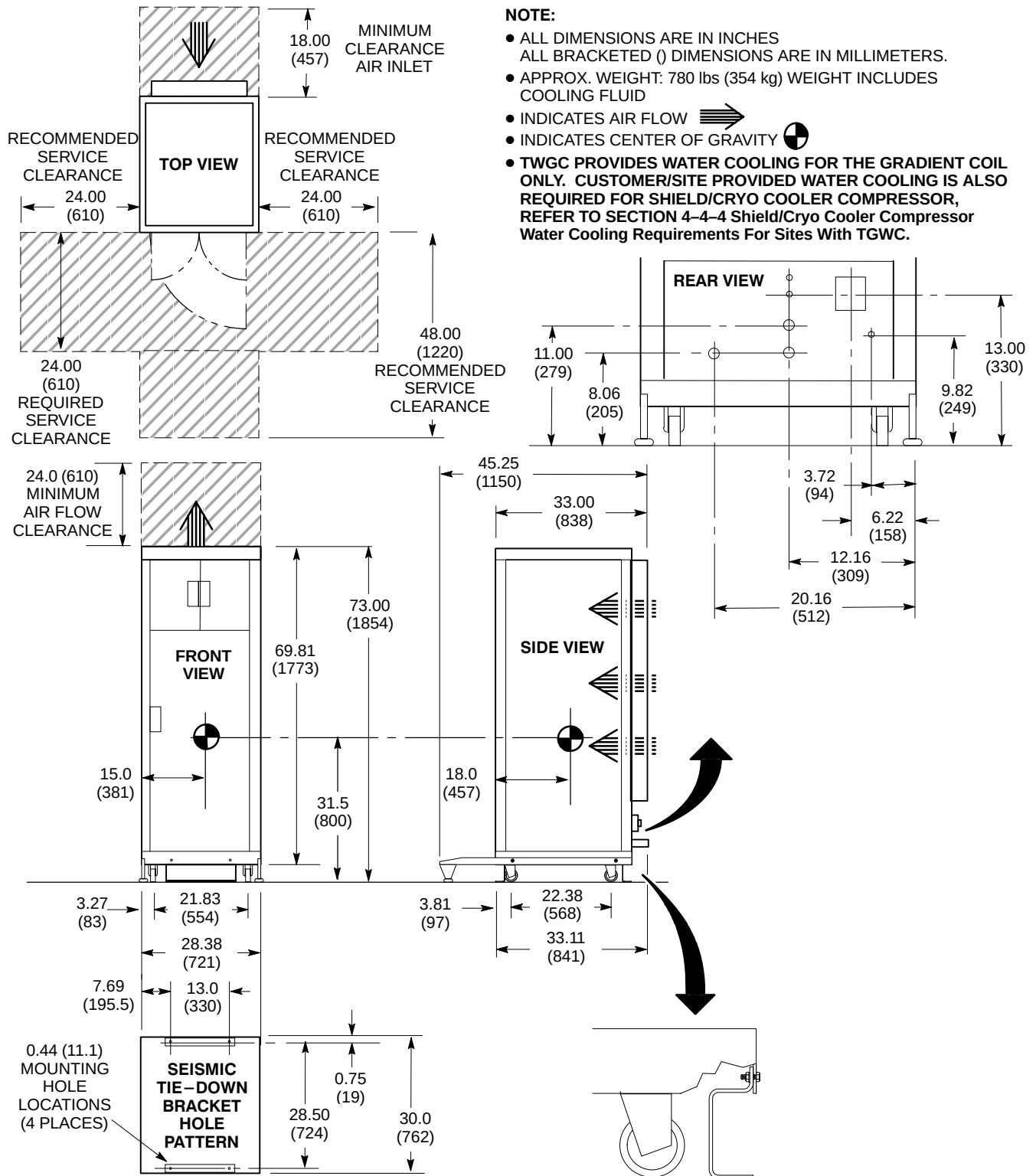
INDOOR WATER COOLED TWIN GRADIENT WATER COOLER [MODEL LTL 10] (TGWC)
FOR GRADIENT COIL COOLING WATER ONLY
ILLUSTRATION 2-24

NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS
ARE IN MILLIMETERS.
- APPROX. WEIGHT: 20 lbs (9 kg)



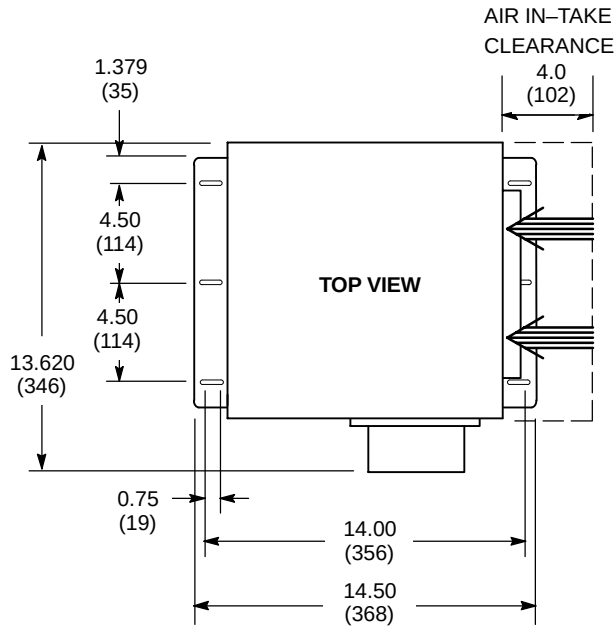
REMOTE CONTROL PANEL (RCP) FOR INDOOR WATER COOLED TGWC [MODEL LTL 10]
ILLUSTRATION 2-25



INDOOR AIR COOLED TWIN GRADIENT WATER COOLER [MODEL LC 10M] (TGWC)

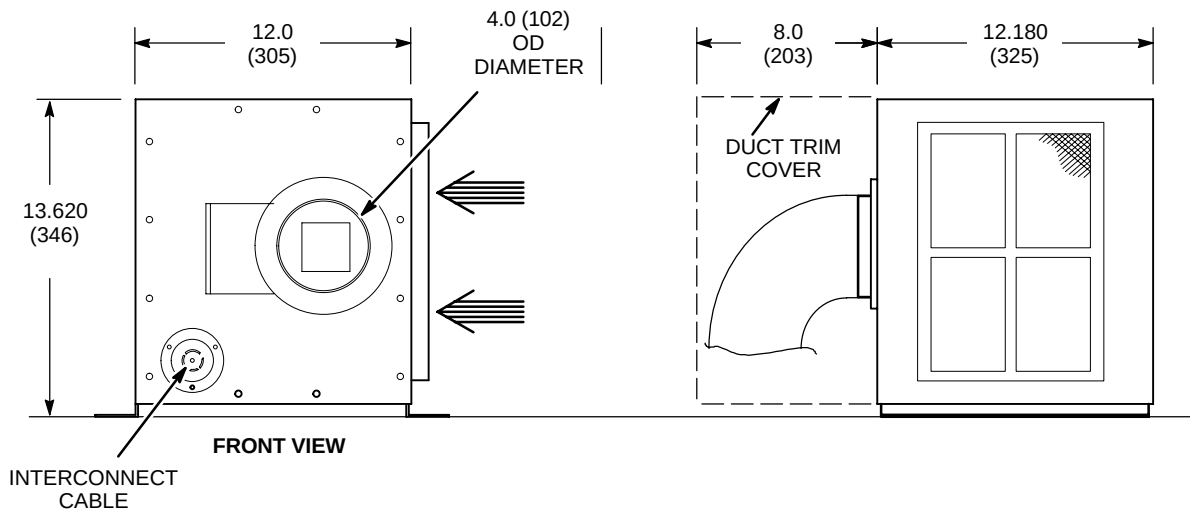
FOR GRADIENT COIL COOLING WATER ONLY

ILLUSTRATION 2-26



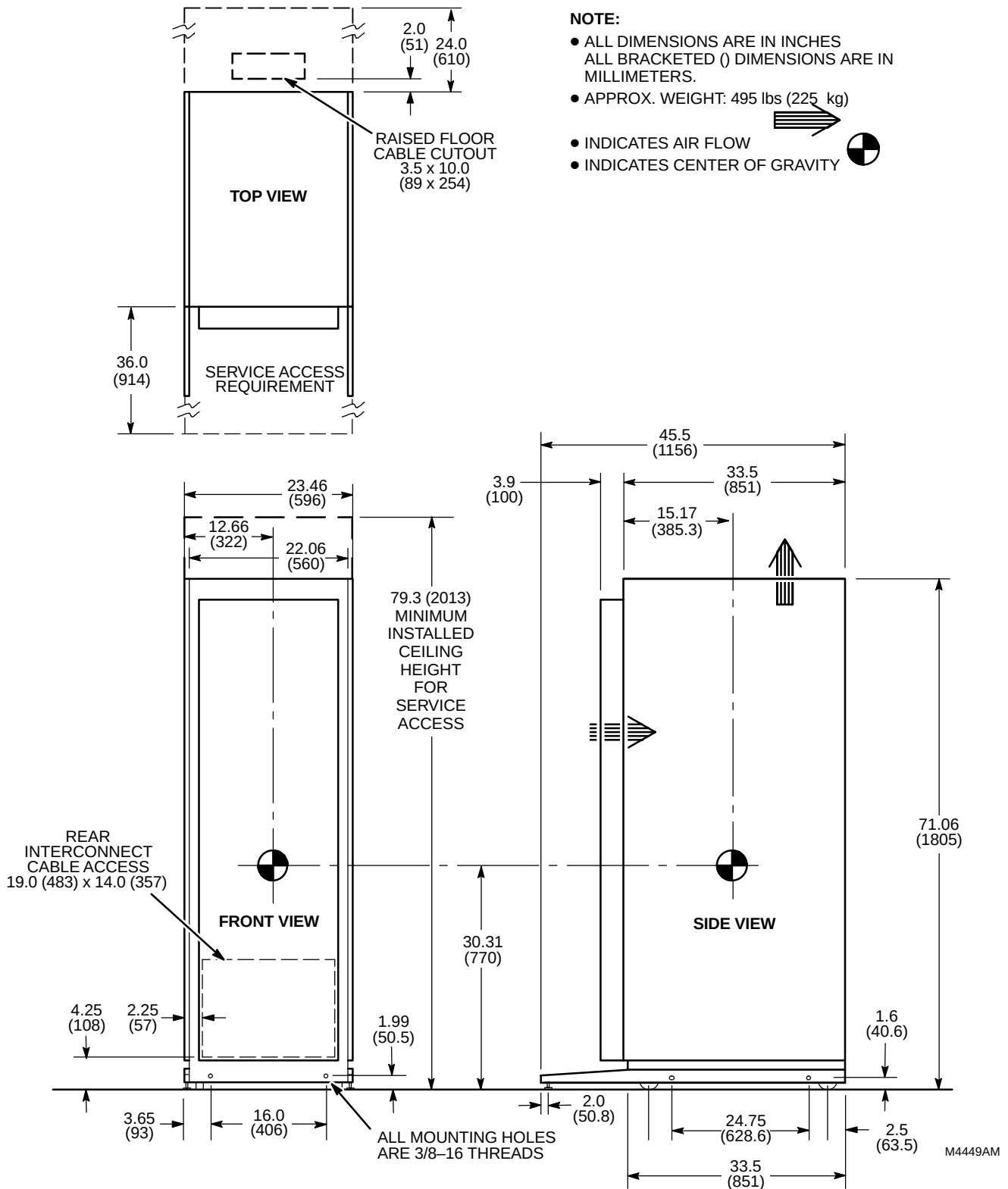
NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 20 lbs (9 kg)
- BLOWER BOX MUST BE ANCHORED DUE FERROUS COMPONENTS.
- INDICATES AIR FLOW ➡



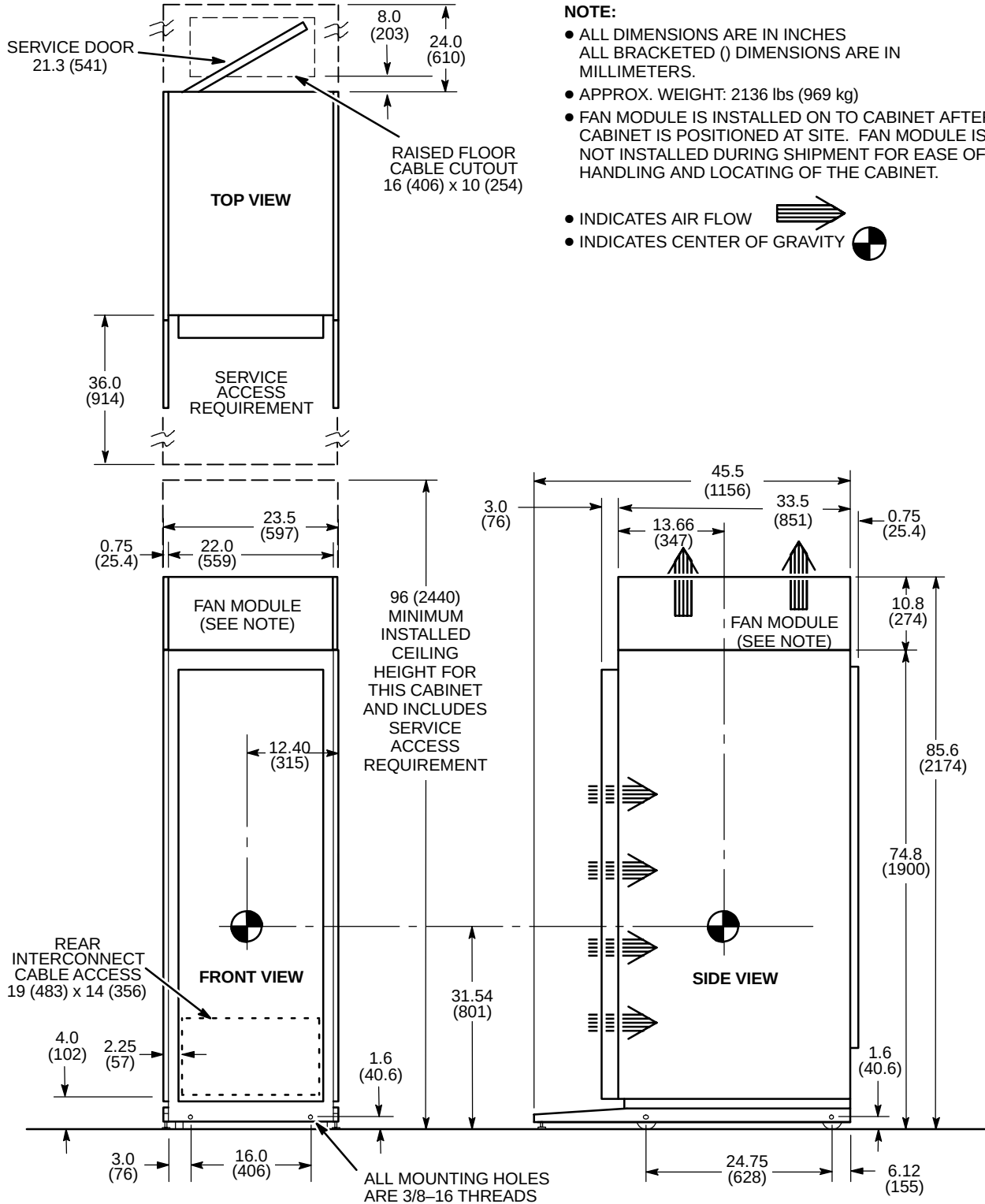
BLOWER BOX (MG6)

ILLUSTRATION 2-27



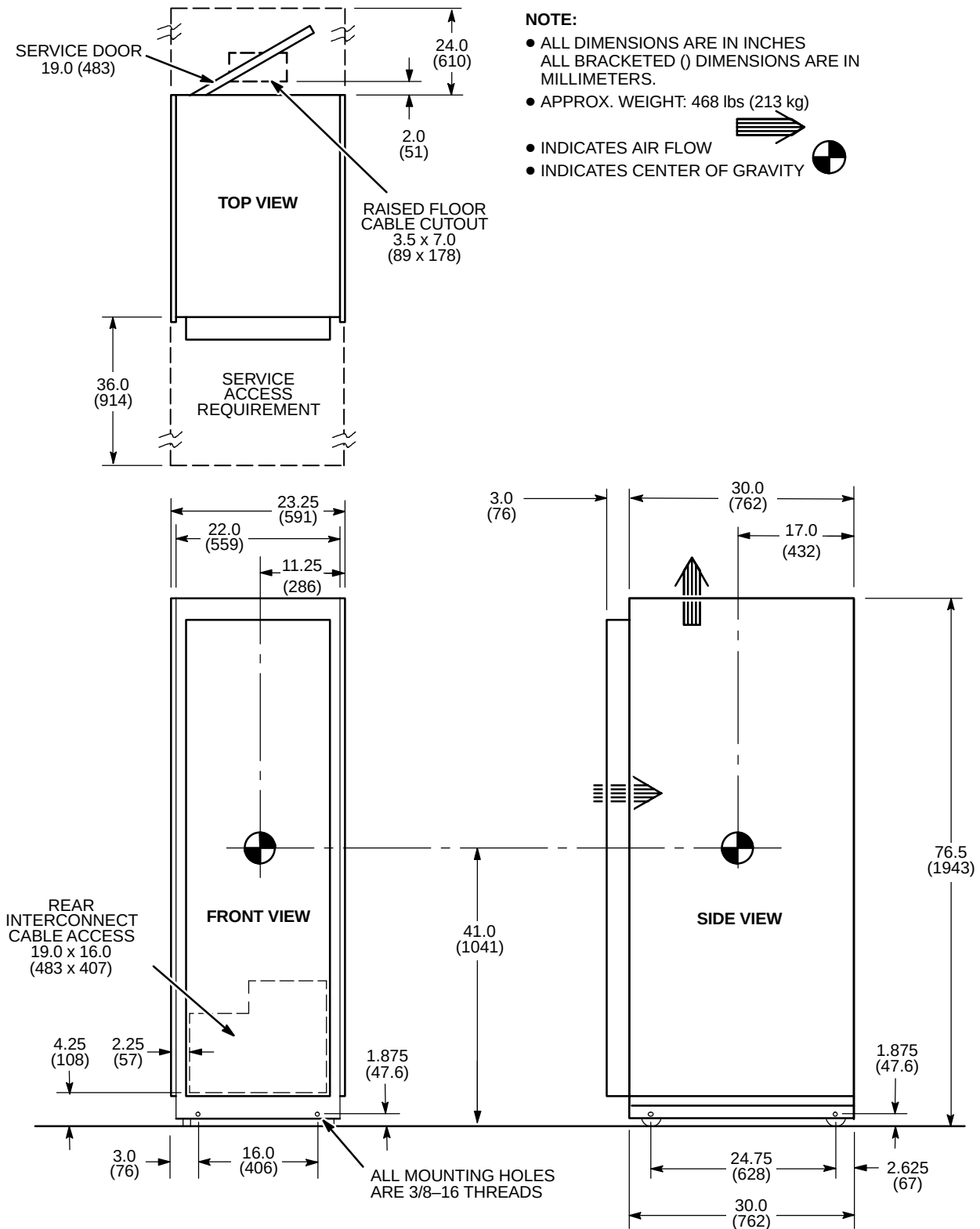
1.5T SRFD II CABINET (MR1)

ILLUSTRATION 2-28

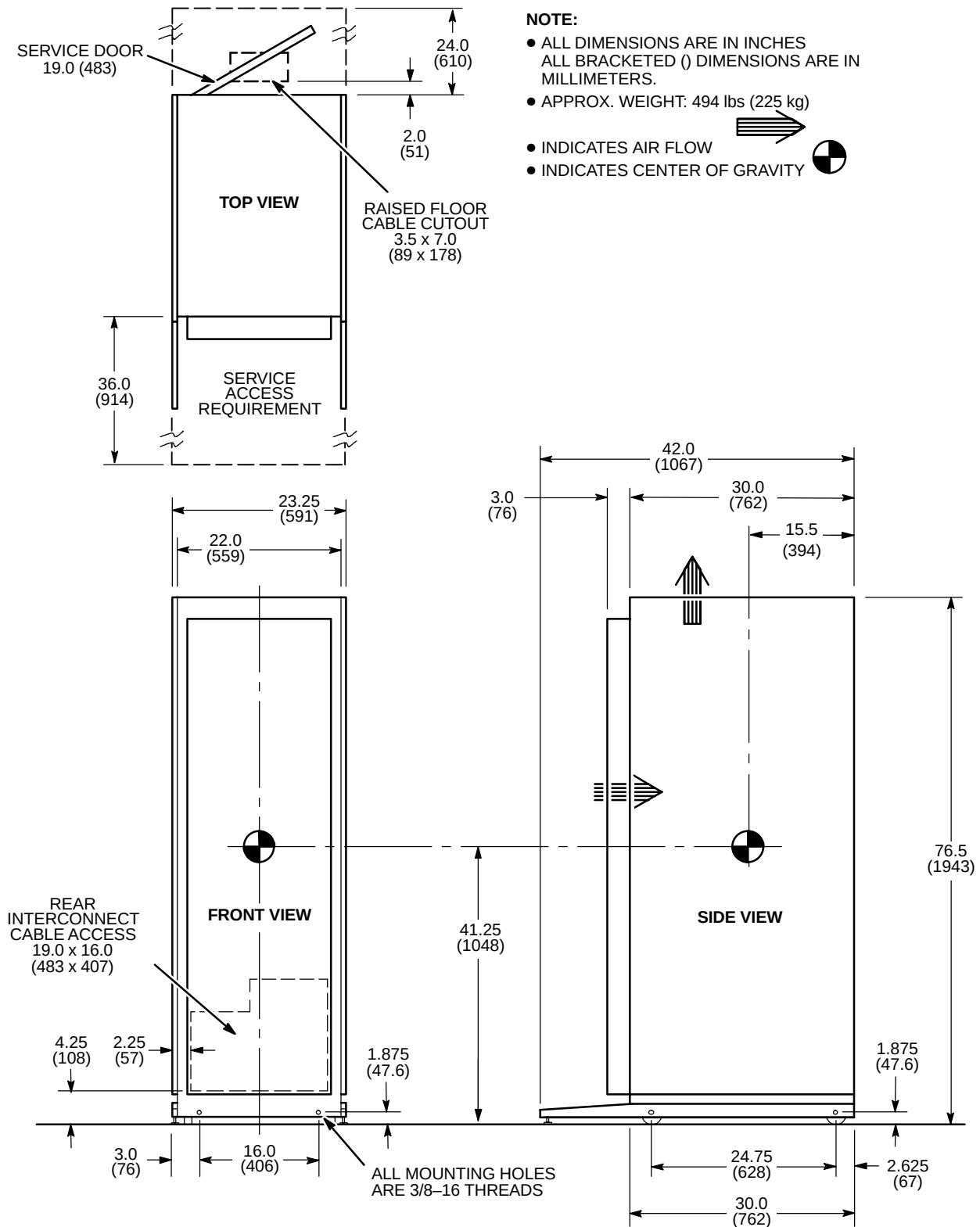


ACGD/PDU CABINET (MR3)

ILLUSTRATION 2-29

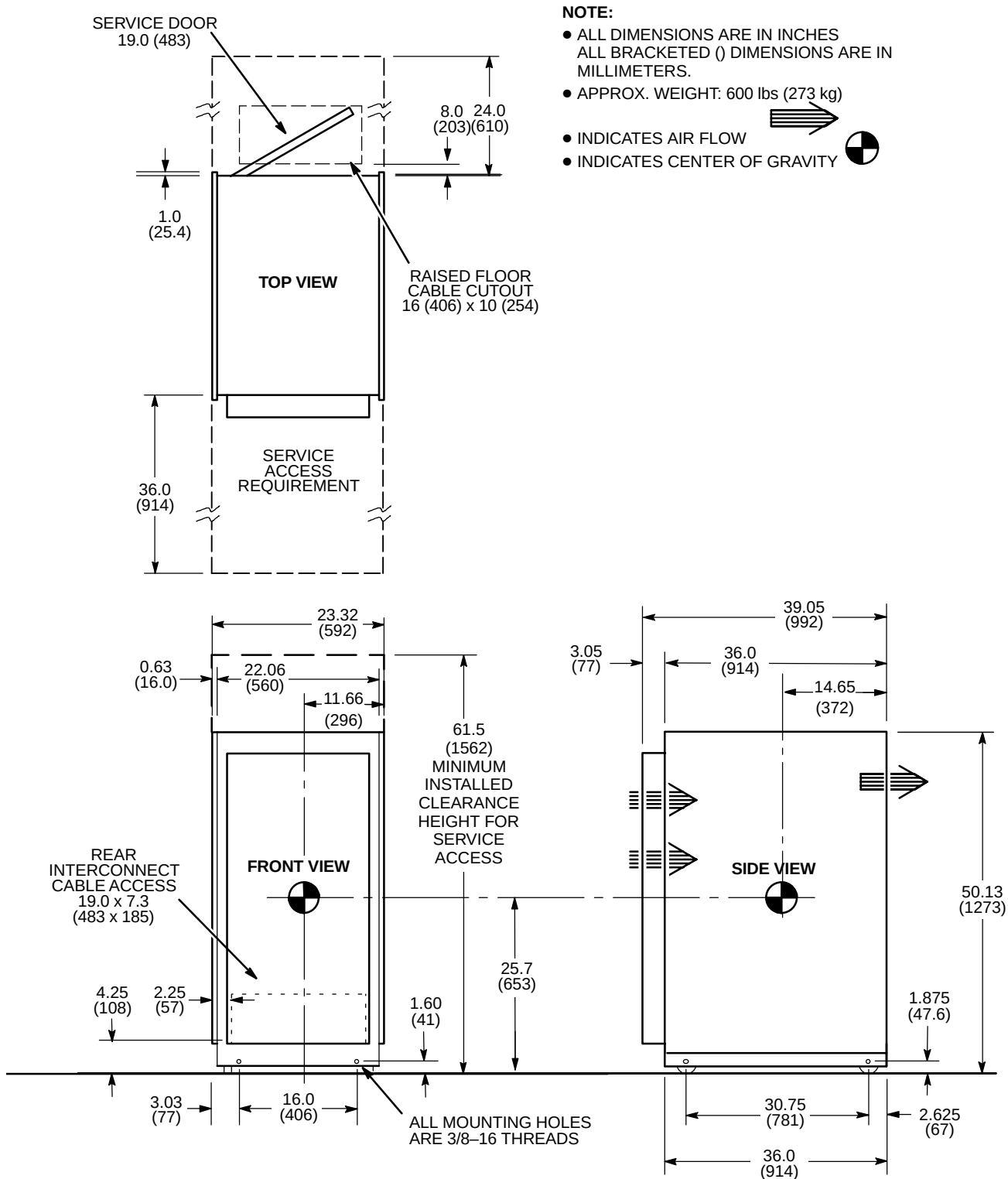


SYSTEM CABINET (MR2) FOR SIGNA TWINSPEED WITH EXCITE TECHNOLOGY
ILLUSTRATION 2-30



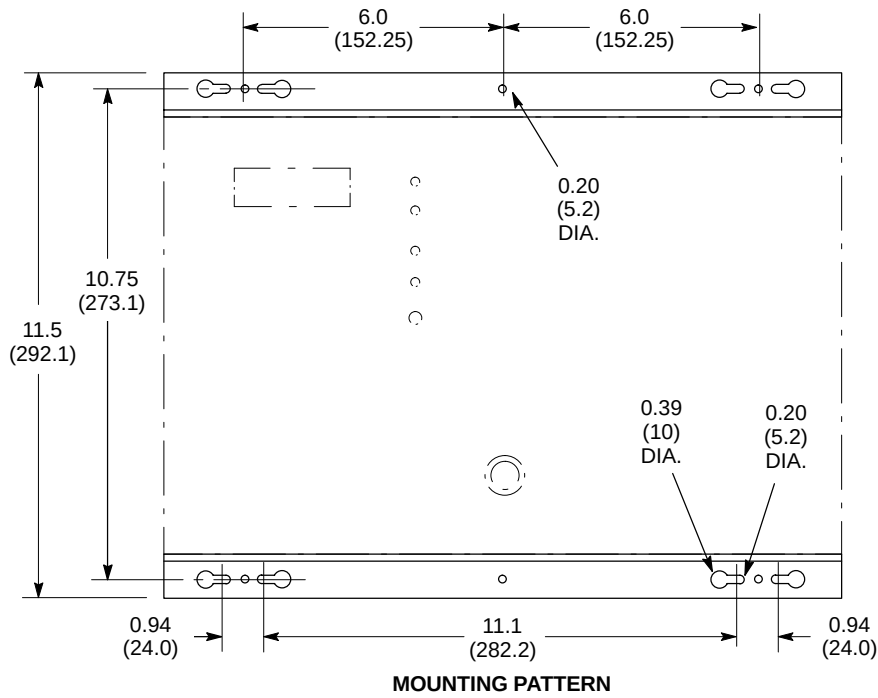
SYSTEM CABINET (MR2) FOR SIGNA TWINSPEED

ILLUSTRATION 2-31



TWIN ACCESSORY CABINET (TAC)

ILLUSTRATION 2-32



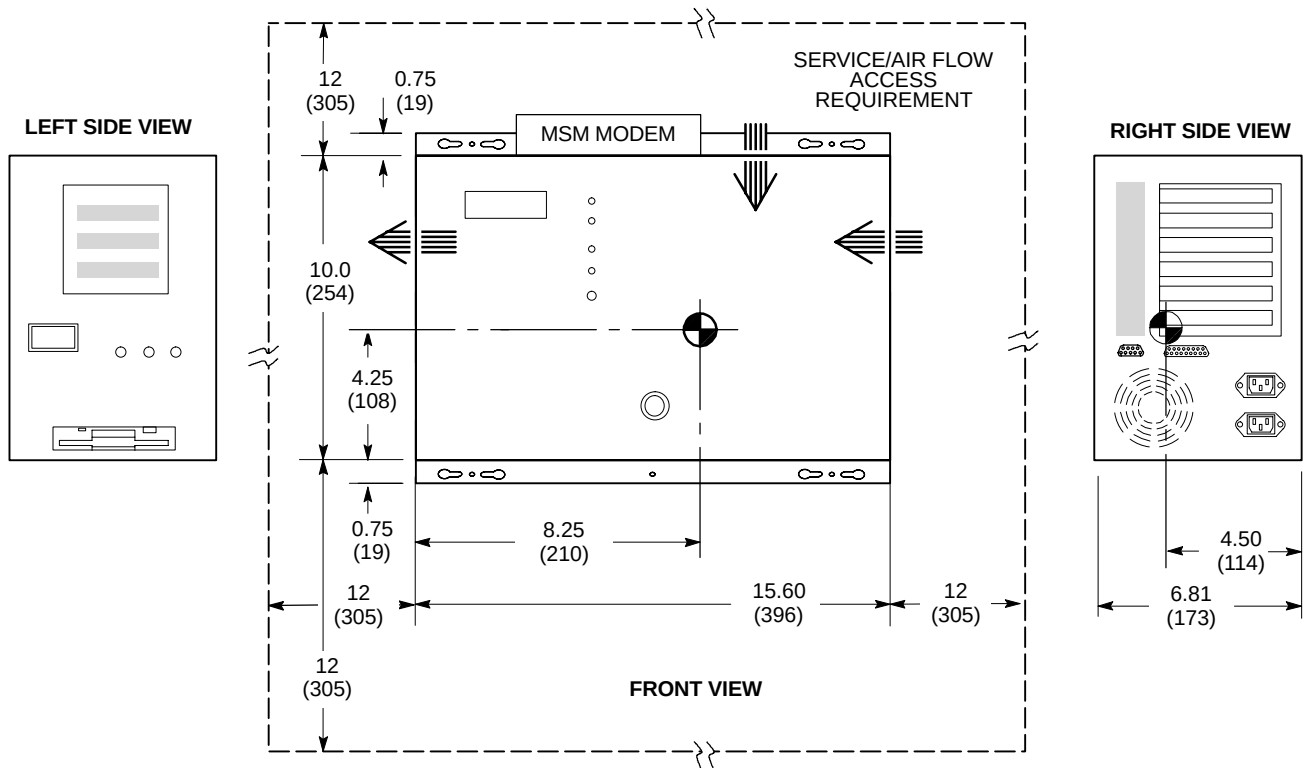
NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS
ARE IN MILLIMETERS.

- APPROX. WEIGHT: 22 lbs (10 kg)

- INDICATES AIR FLOW

- INDICATES CENTER OF GRAVITY

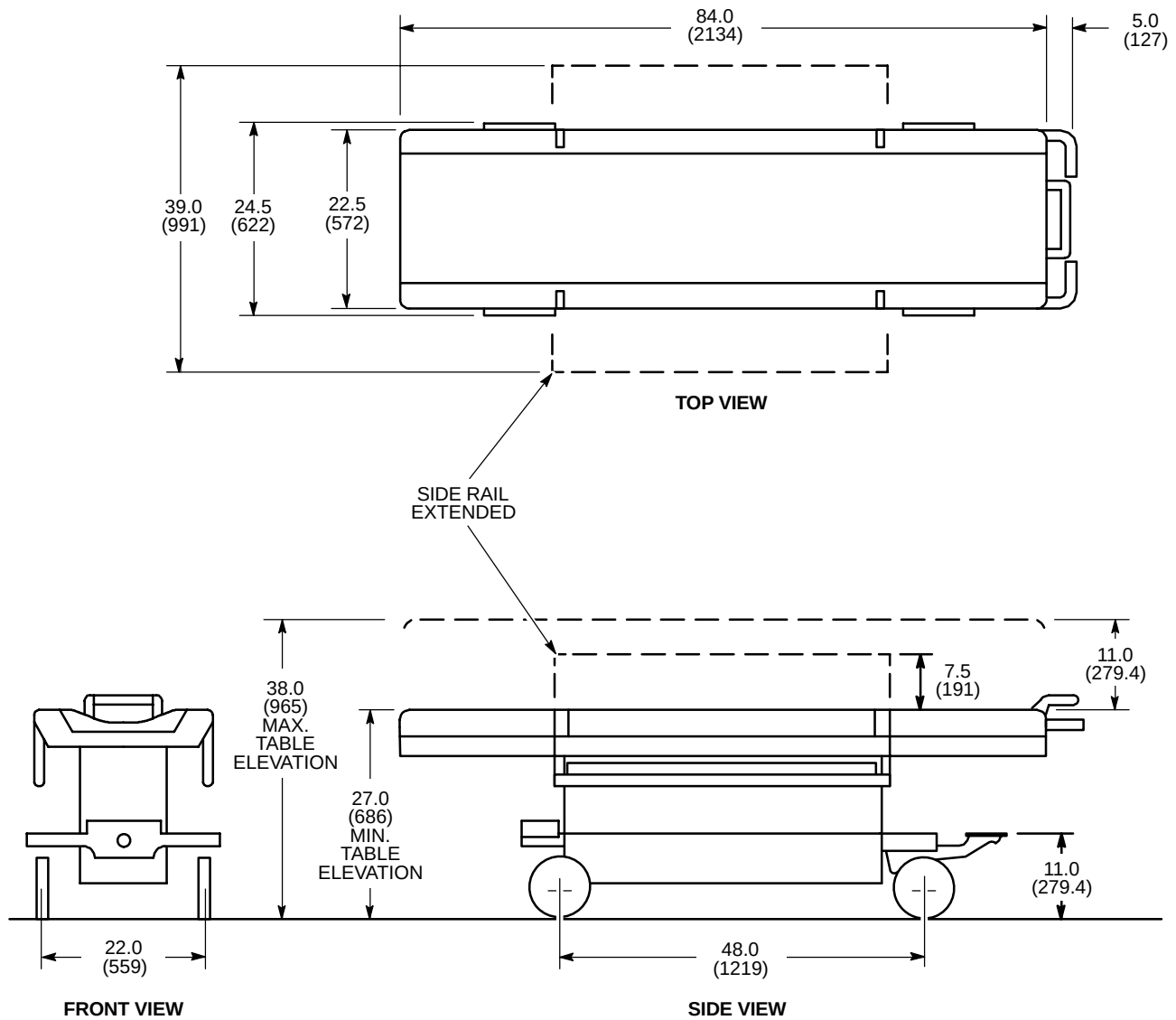


MAGNET MONITOR (MSM1)

ILLUSTRATION 2-33

NOTE:

- ALL DIMENSIONS ARE IN INCHES.
ALL BRACKETED () DIMENSIONS
ARE IN MILLIMETERS.
- WEIGHT: 280 lbs (127 kg)



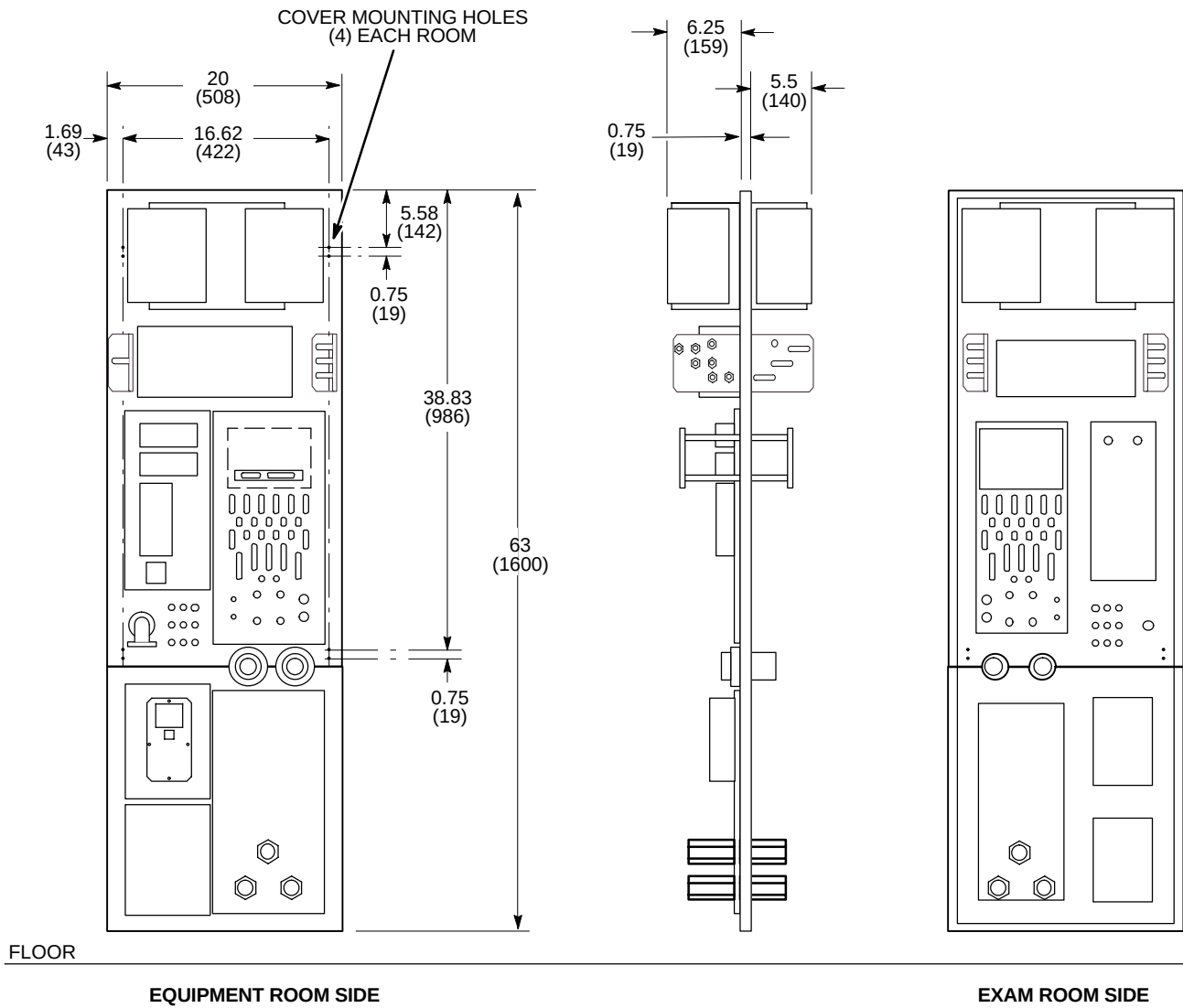
M1081A2

PATIENT TRANSPORT TABLE (PT1)

ILLUSTRATION 2-34

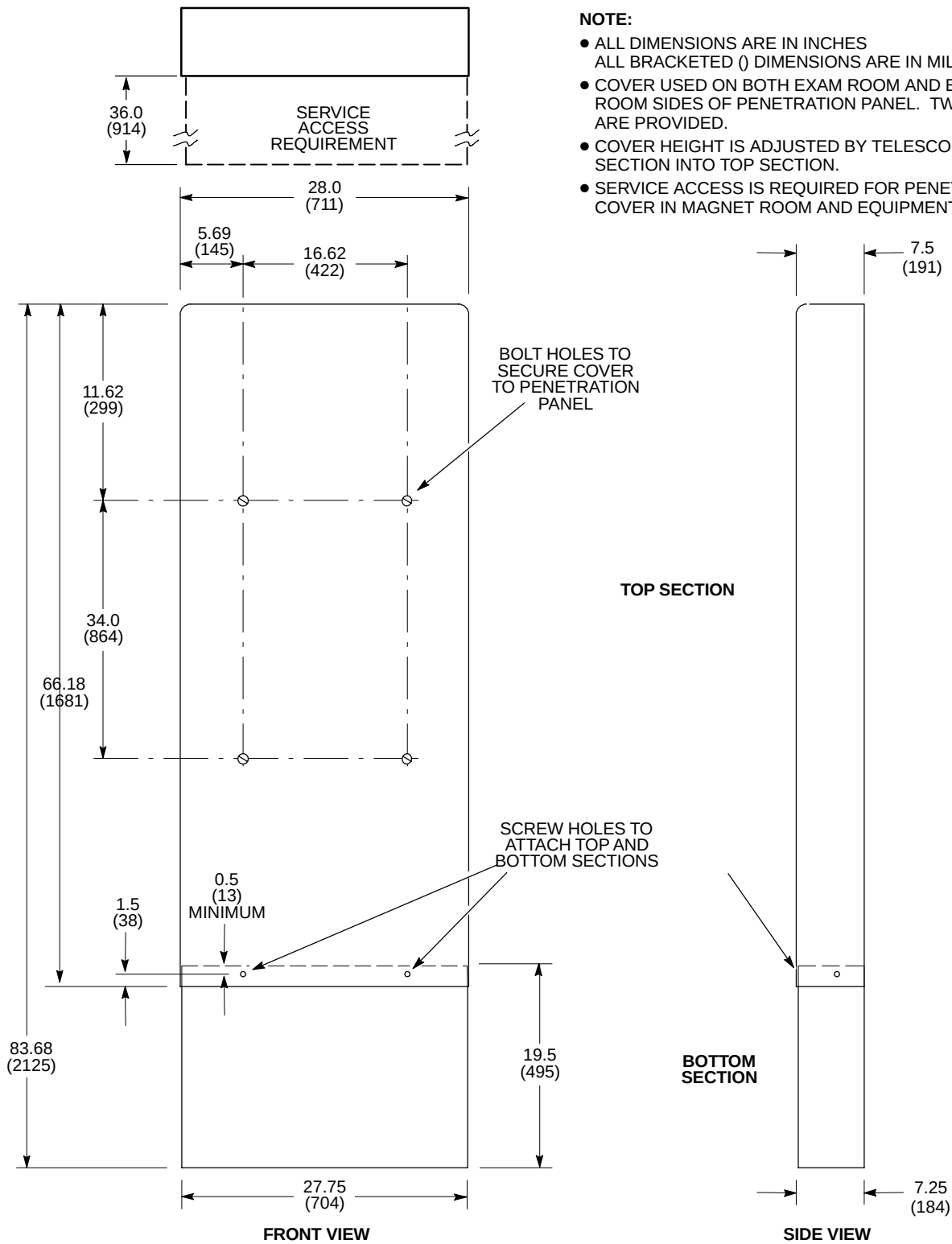
NOTE:

- ALL DIMENSIONS ARE IN INCHES.
ALL BRACKETED () DIMENSIONS
ARE IN MILLIMETERS.
- APPROX. WEIGHT: 52 lbs (23.6 kg)



M4494A

PENETRATION PANEL (PP1)
ILLUSTRATION 2-35



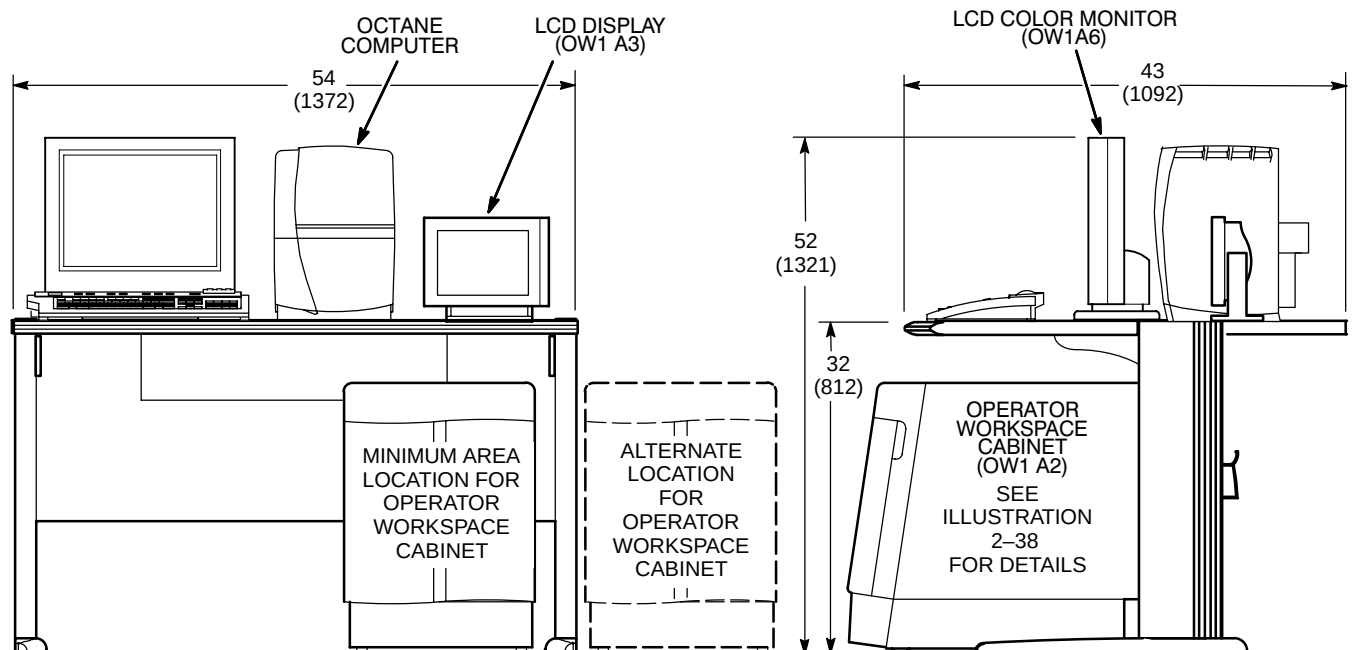
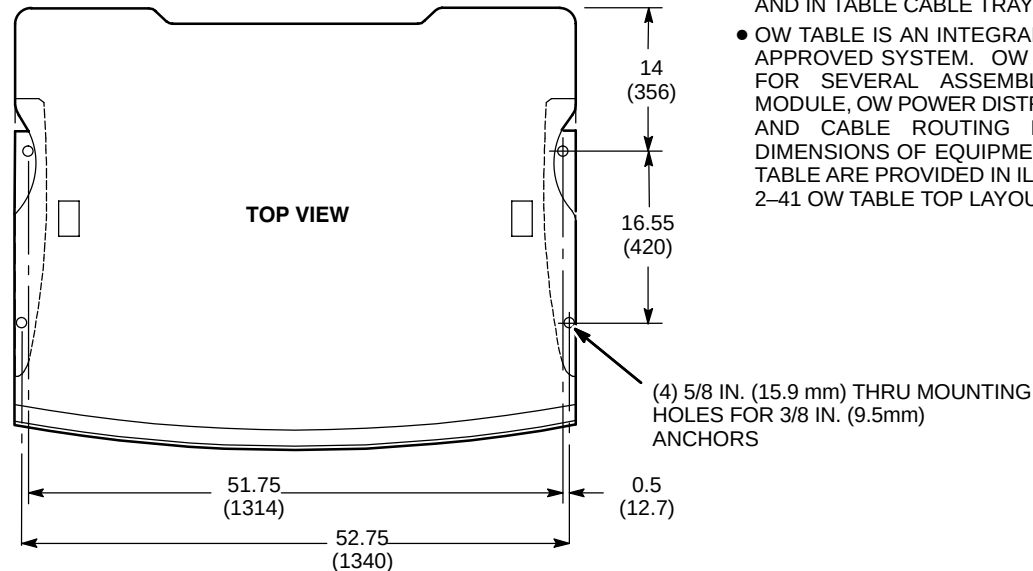
M4009A1M

PENETRATION PANEL COVER

ILLUSTRATION 2-36

NOTE:

- ALL DIMENSIONS ARE IN INCHES.
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- ASSEMBLIES WHICH MOUNT TO UNDERSIDE OF TABLE AND IN TABLE CABLE TRAY ARE NOT SHOWN.
- OW TABLE IS AN INTEGRAL PART OF THE REGULATORY APPROVED SYSTEM. OW TABLE PROVIDES MOUNTING FOR SEVERAL ASSEMBLIES (E.G. OW INTERFACE MODULE, OW POWER DISTRIBUTION BOX, MODEM, DASM) AND CABLE ROUTING FOR OW INTERCONNECTS. DIMENSIONS OF EQUIPMENT LOCATED ON TOP OF OW TABLE ARE PROVIDED IN ILLUSTRATIONS 2-39 THROUGH 2-41 OW TABLE TOP LAYOUT PLANNING PURPOSES.



OPERATOR WORKSPACE (OW1)

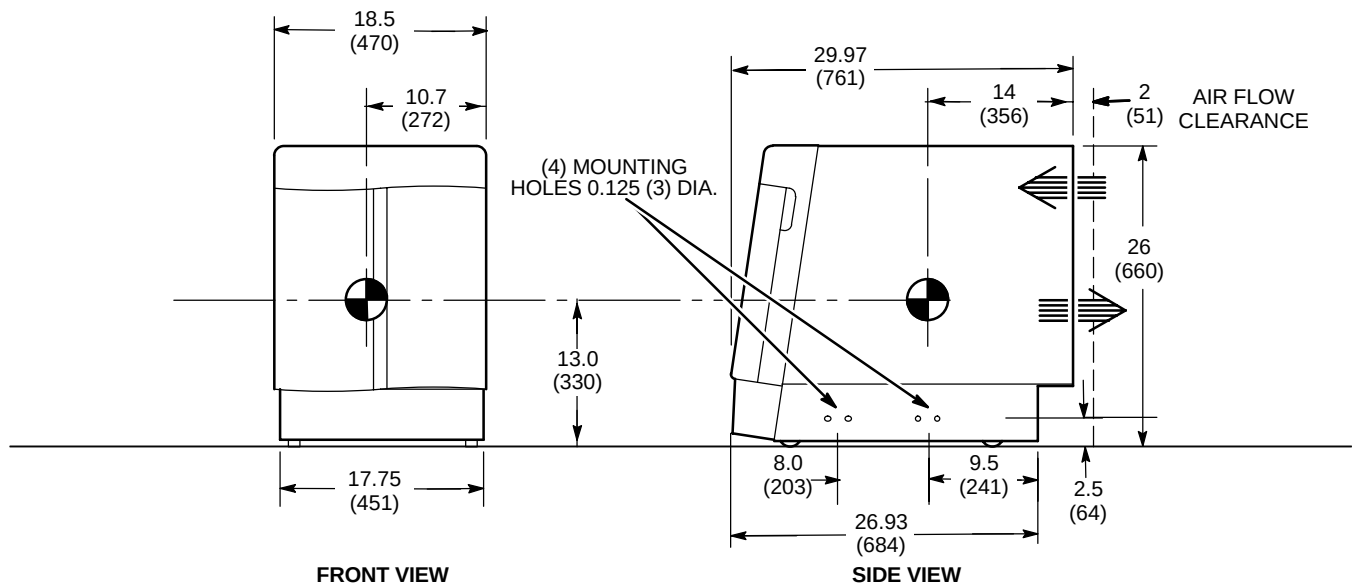
ILLUSTRATION 2-37

NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 192 lbs (87 kg)

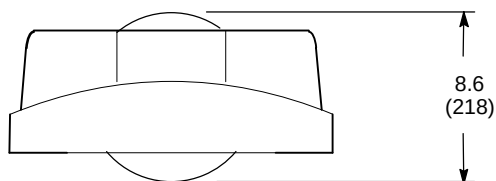
- INDICATES AIR FLOW 

- INDICATES CENTER OF GRAVITY 



OPERATOR WORKSPACE CABINET (OW1 A2)

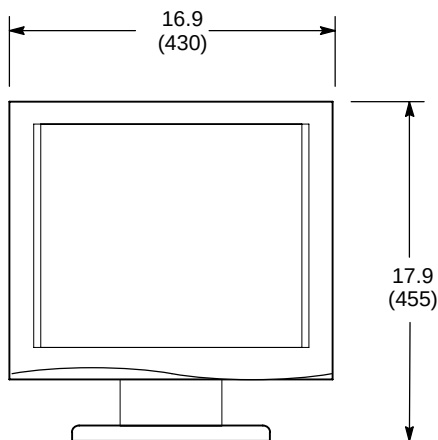
ILLUSTRATION 2-38



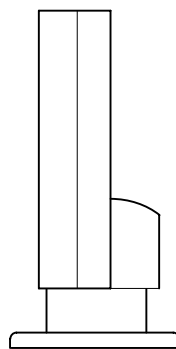
TOP VIEW

NOTE:

- ALL DIMENSIONS ARE IN INCHES
- ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.

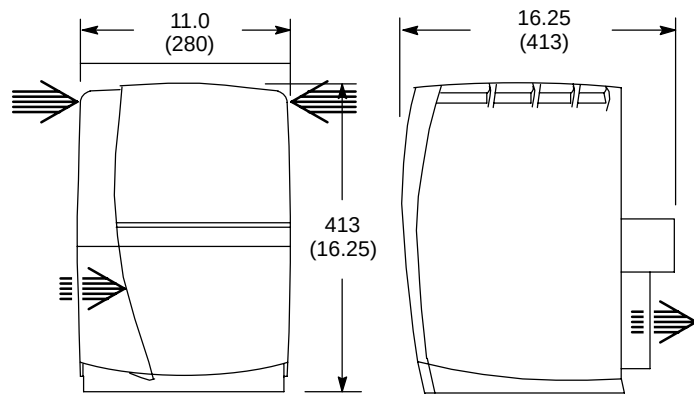


FRONT VIEW



SIDE VIEW

OPERATOR WORKSPACE COMPONENTS POSITIONED ON TABLE TOP – LCD COLOR MONITOR
ILLUSTRATION 2-39



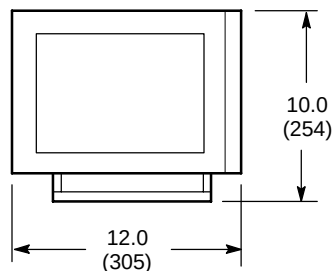
FRONT VIEW

SIDE VIEW

OCTANE COMPUTER

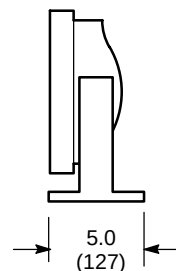
NOTE:

- ALL DIMENSIONS ARE IN INCHES
- ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- INDICATES AIR FLOW ➡



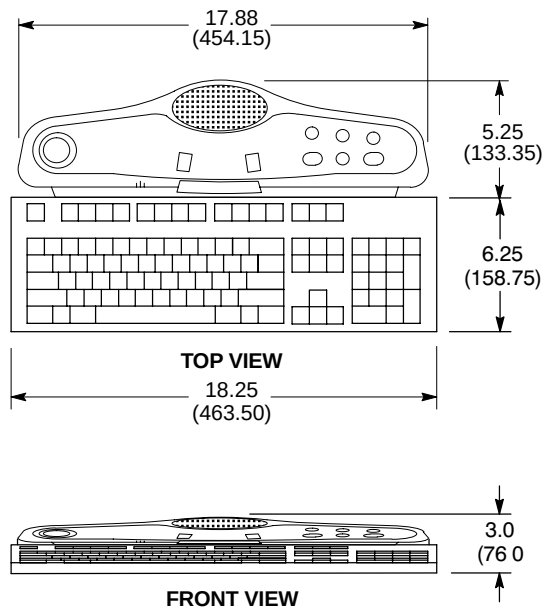
FRONT VIEW

LCD DISPLAY



SIDE VIEW

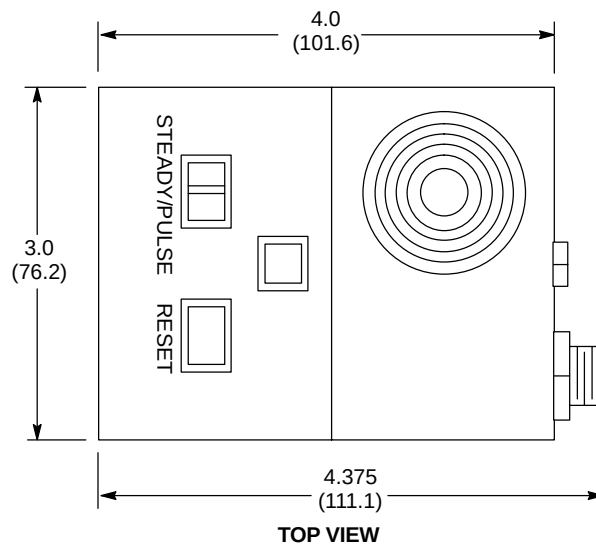
OPERATOR WORKSPACE COMPONENTS POSITIONED ON TABLE TOP – OCTANE COMPUTER
ILLUSTRATION 2-40



NOTE:

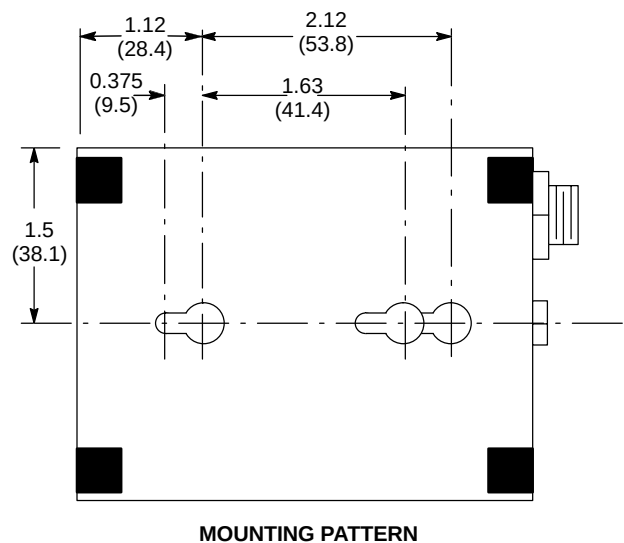
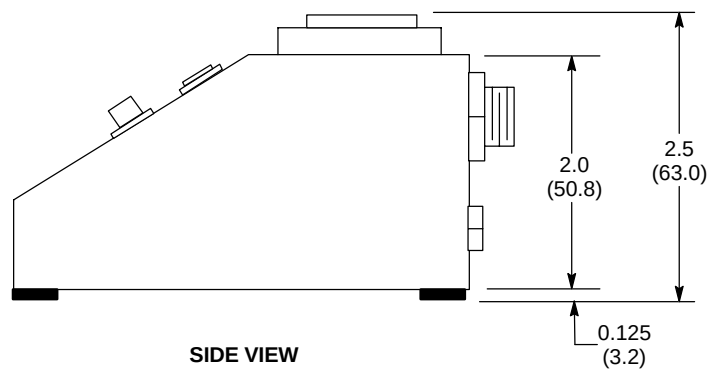
- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.

OPERATOR WORKSPACE COMPONENTS POSITIONED ON TABLE TOP – KEYBOARD
ILLUSTRATION 2-41



NOTE:

- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 0.5 lbs (0.2 kg)



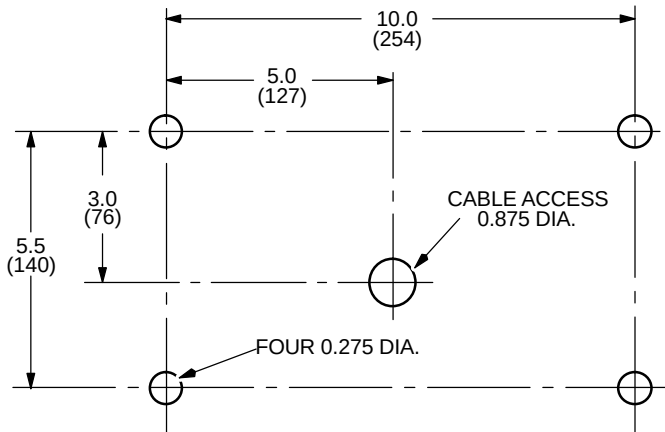
PNEUMATIC PATIENT ALERT CONTROL BOX (PA1)

ILLUSTRATION 2-42

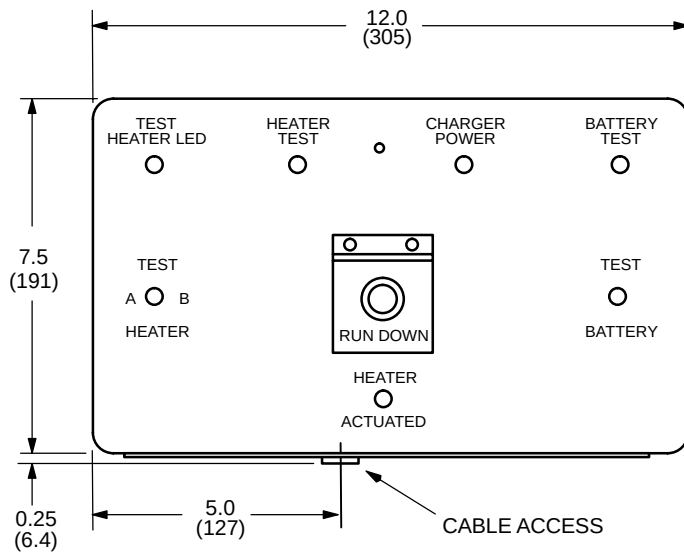
M4263A1

NOTE:

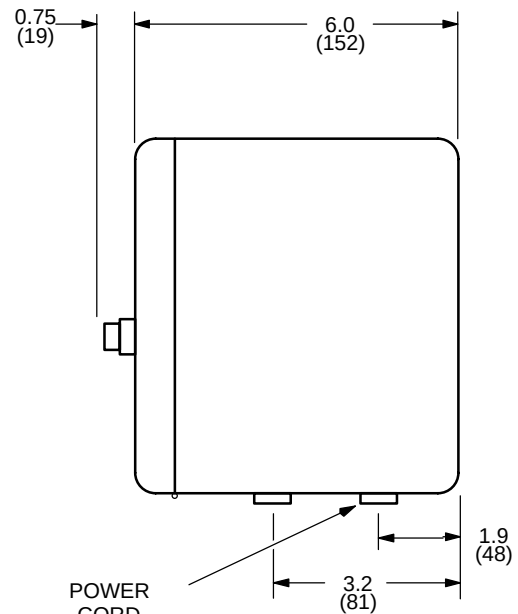
- ALL DIMENSIONS ARE IN INCHES.
ALL BRACKETED () DIMENSIONS
ARE IN MILLIMETERS.
- APPROX. WEIGHT: 8.8 lbs. (4kg)



MOUNTING PATTERN



FRONT VIEW



SIDE VIEW

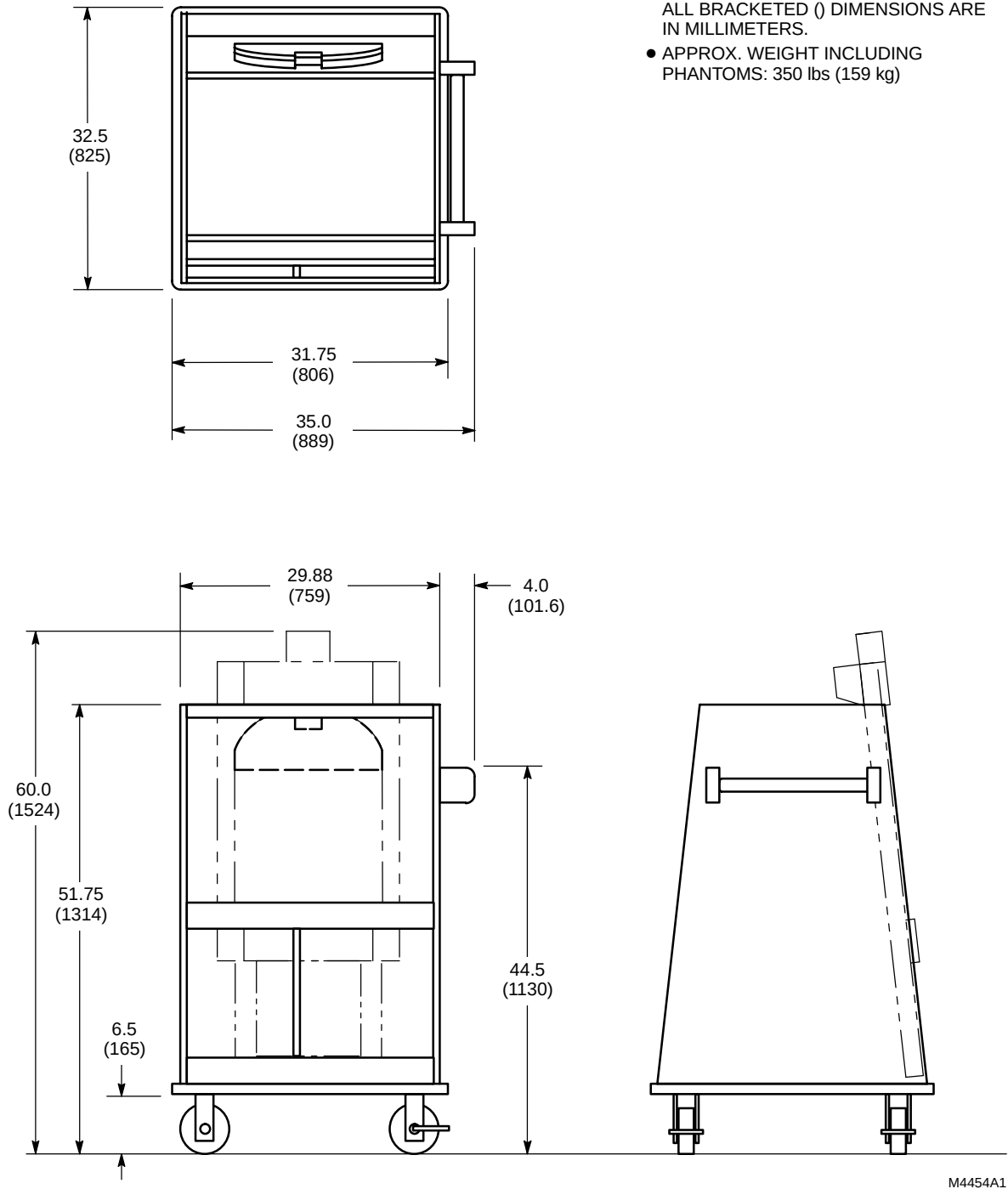
M1517A5

MAGNET RUNDOWN UNIT (MS4)

ILLUSTRATION 2-43

NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE
IN MILLIMETERS.
- APPROX. WEIGHT INCLUDING
PHANTOMS: 350 lbs (159 kg)



SPT PHANTOM SET SHIPPING/STORAGE CART
ILLUSTRATION 2-44

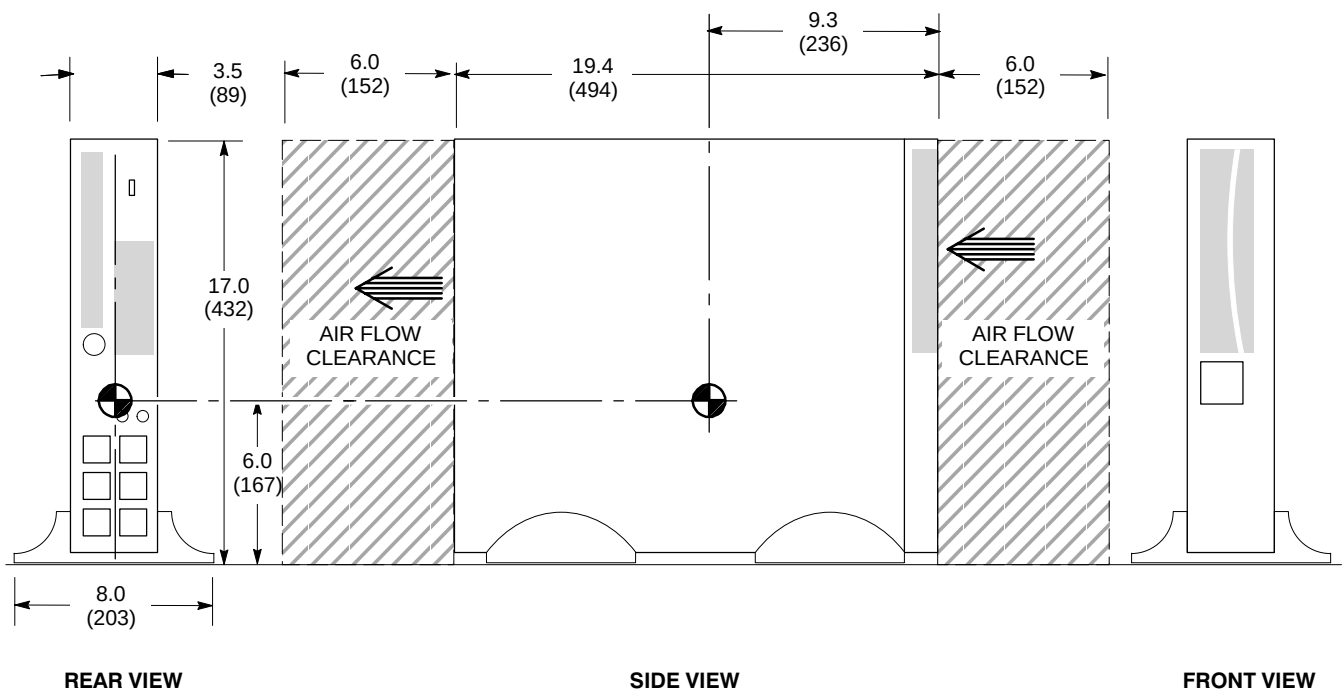
NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS
ARE IN MILLIMETERS.

- APPROX. WEIGHT: 50 lbs (23 kg)

- INDICATES AIR FLOW

- INDICATES CENTER OF GRAVITY

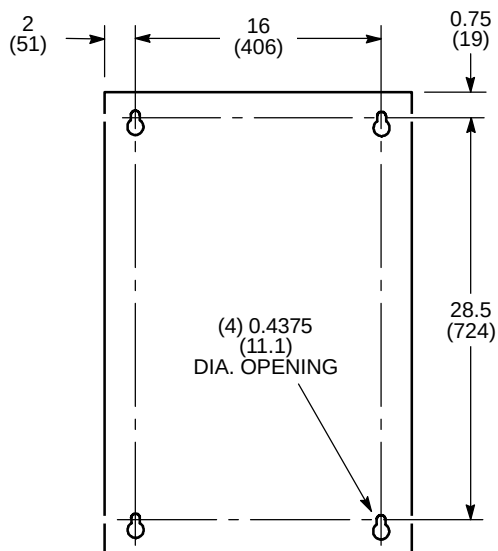


UPS FOR MAGNET MONITOR – STANDARD FOR TRUE MOBILE CONFIGURATION, OPTIONAL FOR OTHER CONFIGURATIONS

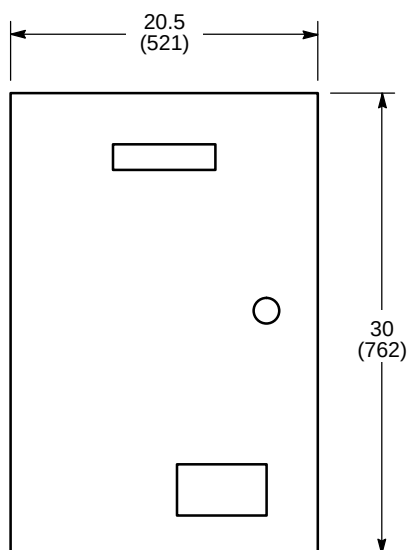
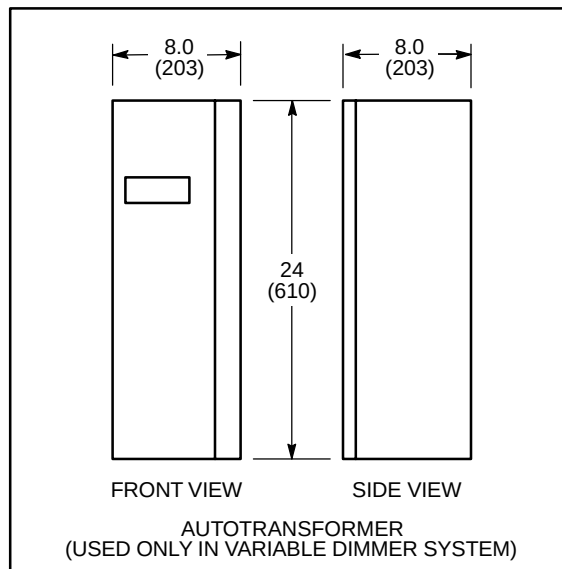
ILLUSTRATION 2-45

NOTE:

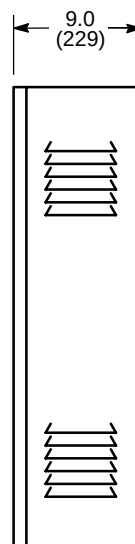
- ALL DIMENSIONS ARE IN INCHES.
ALL BRACKETED () DIMENSIONS
ARE IN MILLIMETERS.
- APPROX. WEIGHTS:
CONTROL PANEL: 155 lbs (70 kg)
AUTOTRANSFORMER: 60 lbs (27 kg)



MOUNTING PATTERN
(CONTROL PANEL)



FRONT VIEW
(CONTROL PANEL)



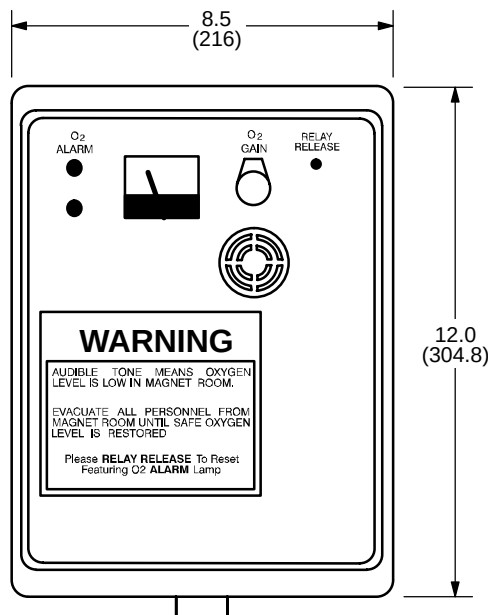
SIDE VIEW
(CONTROL PANEL)

M1519A3

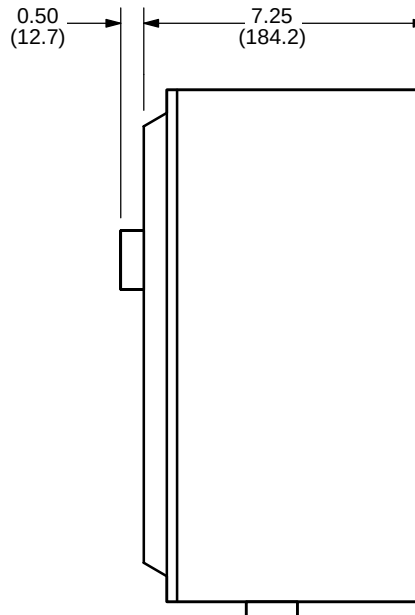
DC LIGHTING CONTROLLER – OPTIONAL
ILLUSTRATION 2-46

NOTE:

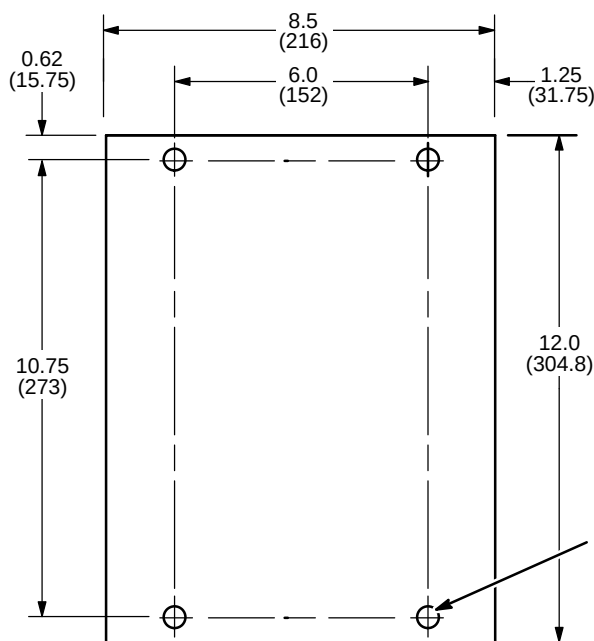
- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 9 lbs (4.1 kg)



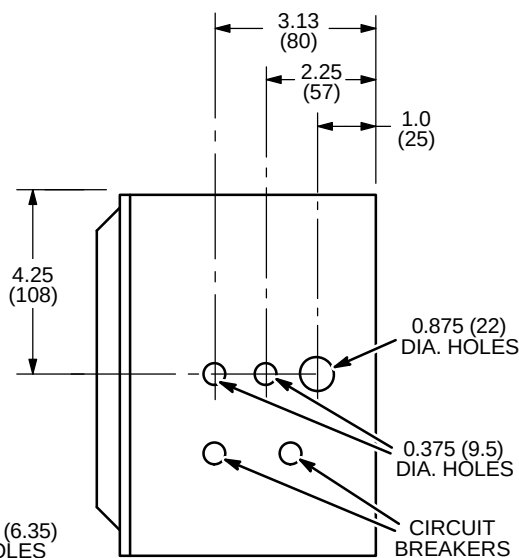
FRONT VIEW



SIDE VIEW



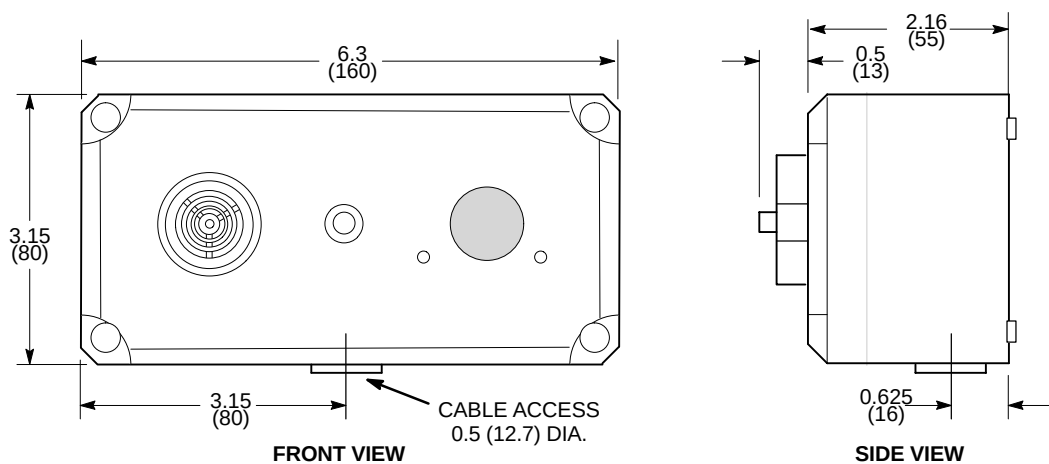
MOUNTING PATTERN



BOTTOM VIEW

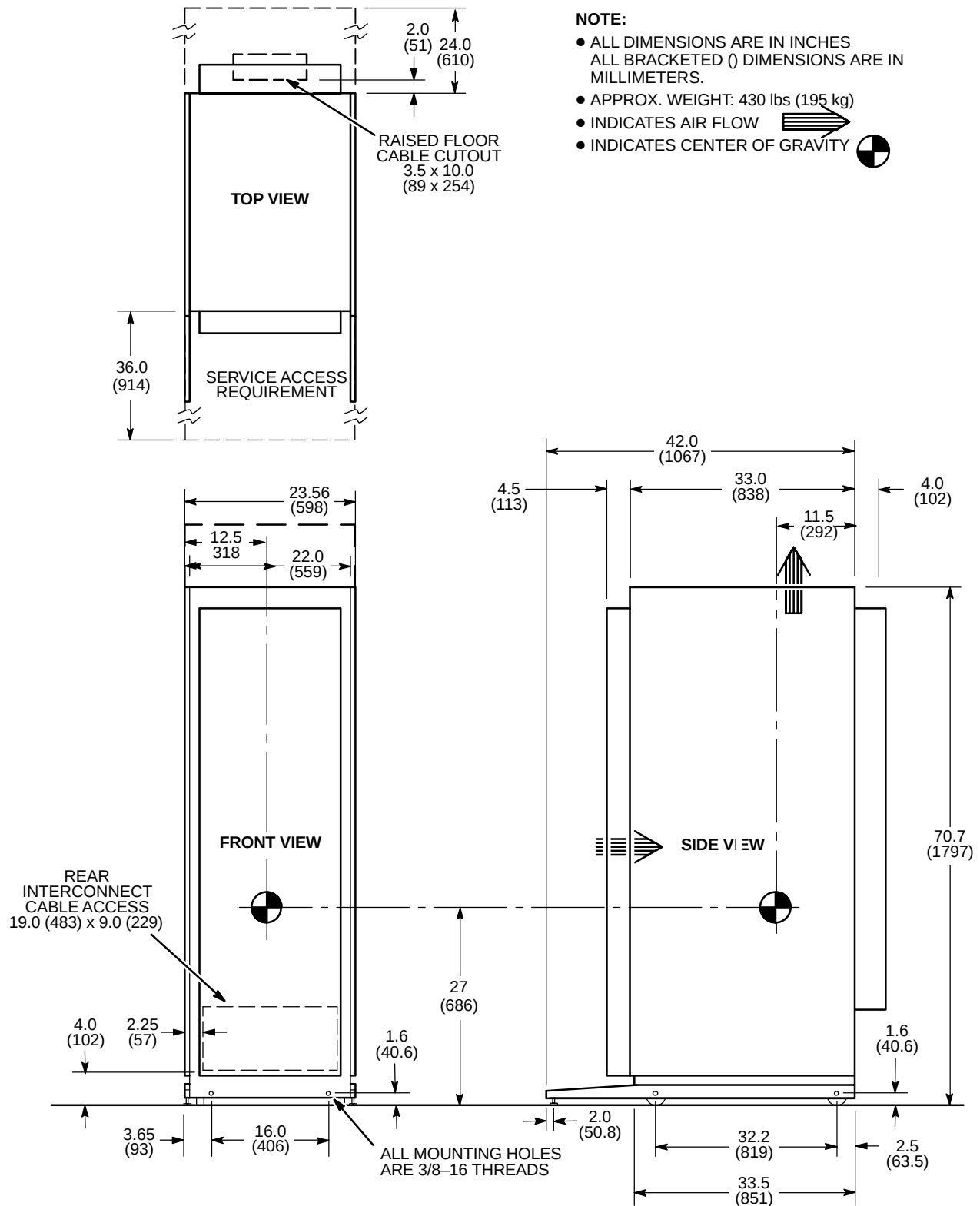
OXYGEN MONITOR (OM1) – OPTIONAL
ILLUSTRATION 2-47

M1516A4



REMOTE OXYGEN SENSOR MODULE (OM3) – OPTIONAL
ILLUSTRATION 2-48

M3346A1M

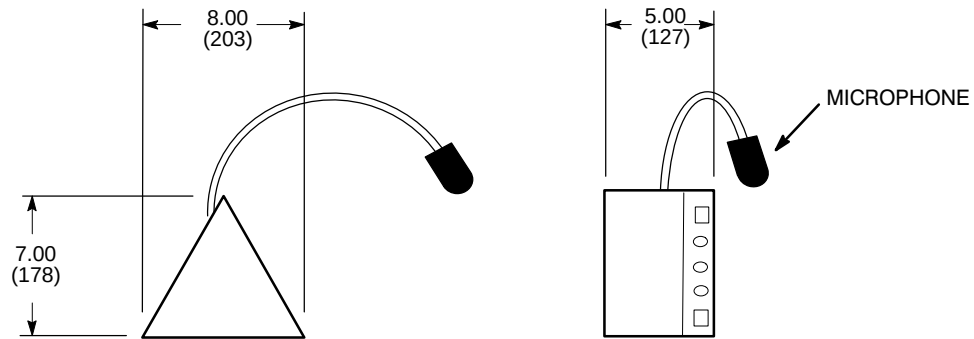


MNS CABINET (MNS) - 1.5T OPTION

ILLUSTRATION 2-49

NOTE:

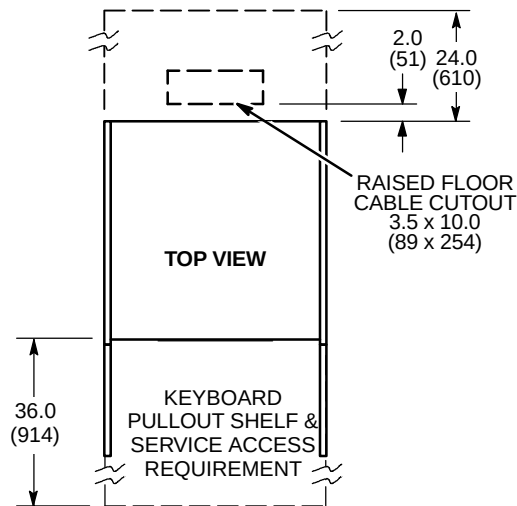
- ALL DIMENSIONS ARE IN INCHES
- ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- MICROPHONE POSITION IS ADJUSTABLE.



M4546A

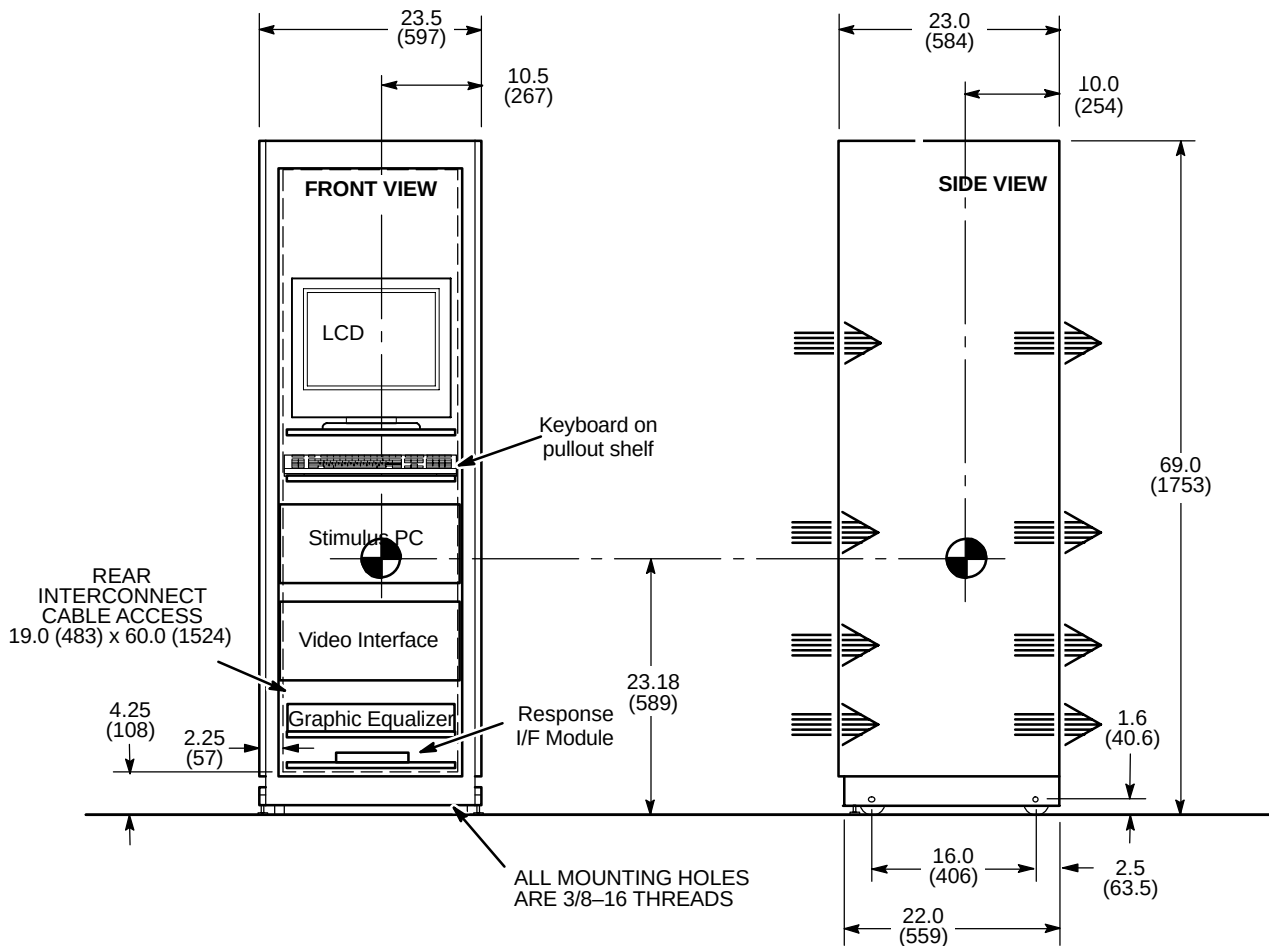
BRAINWAVE AUDIO CONSOLE – BRAINWAVE OPTION

ILLUSTRATION 2-50



NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 387 lbs (176 kg)
- INDICATES AIR FLOW
- INDICATES CENTER OF GRAVITY

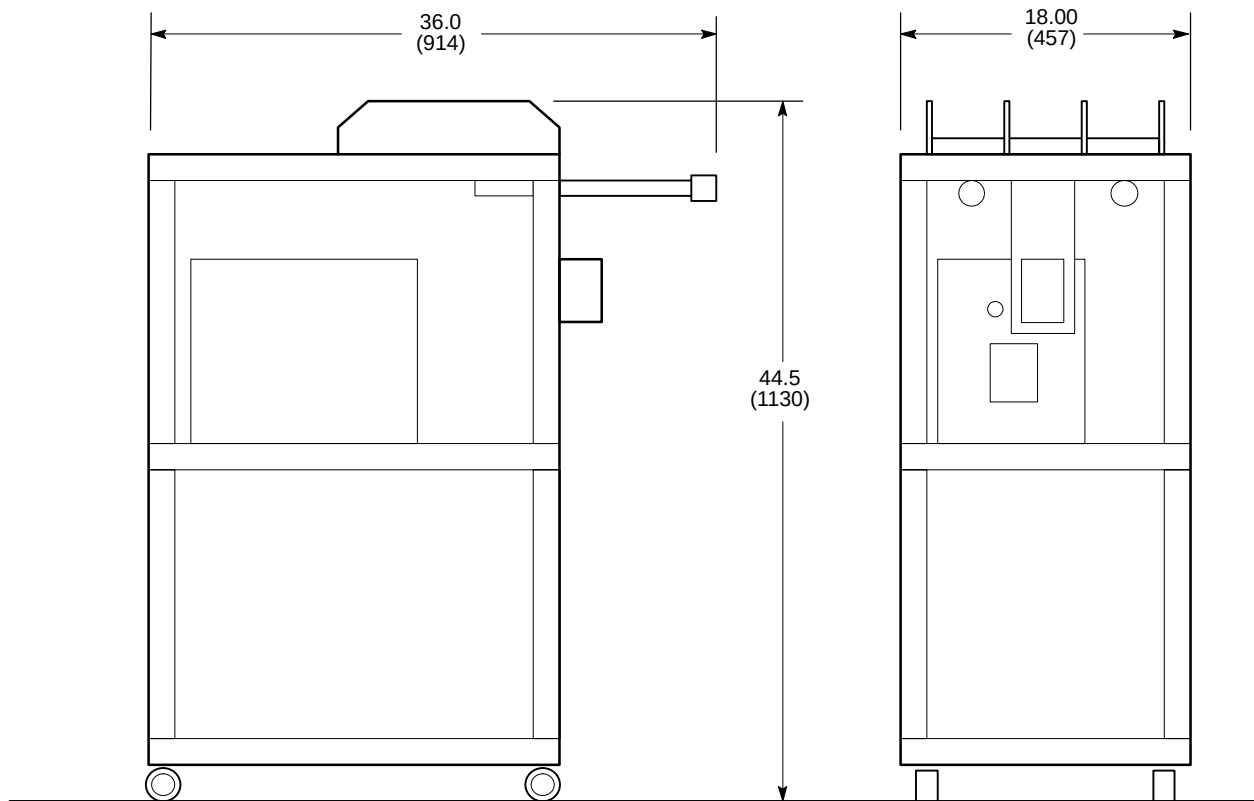


BRAINWAVE CABINET – BRAINWAVE OPTION

ILLUSTRATION 2-51

NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 71 lbs (32 kg)



M4546A

BRAINWAVE CART – BRAINWAVE OPTION

ILLUSTRATION 2-52